Rent Inflation and Monetary Policy

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Abstract

Rental prices enter the inflation indices that central banks target, yet rents behave unlike other consumer prices after a monetary policy shock. We build a Heterogeneous-Agent New Keynesian model where households rent, own, or become landlords, with endogenous house prices and rents, and sticky leases. A monetary tightening keeps would-be buyers renting, and rents rise until enough owners supply the extra rental housing. About two-thirds of the effect of higher interest rates on the rental market is supply-driven: higher financial returns and mortgage costs make supplying rental housing less attractive. We study how policy should treat rent inflation. A Taylor rule based on a price index that incorporates rents generates more consumption and inflation volatility than one that ignores rents. The welfare-optimal weight on rent inflation is negative under both general demand and supply shocks: central banks should react to rent increases with accommodative policy.

JEL codes: D31, E21, E31, E52, E58, G51, R21, R31.

Keywords: Rent inflation, monetary policy, heterogeneous-agent New Keynesian models, housing tenure choice, sticky rental leases, inflation targeting, rental market.

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1 Introduction

Shelter costs are a large part of the price indices that central banks target. In the euro area, the Harmonised Index of Consumer Prices (HICP) targeted by the European Central Bank (ECB) includes the actual rents paid by tenants, about 6 percent of the basket.[1] In the U.S., the weight of rents is larger because they are used to impute owner-occupied housing costs: they account for about 15 percent of headline PCE and 31 percent of headline CPI (Bolhuis et al., 2022). The post-pandemic episode showed how much these weights matter. Market rents surged in 2021 and 2022; the shelter components of the indices were still climbing when central banks tightened, and they kept rising long afterward, as existing leases caught up with market rents.

This paper studies how monetary policy should respond to rent inflation. Rents generate tradeoffs for monetary policy because they do not fall when policy tightens, unlike other consumer prices. In Spain, we estimate that a monetary tightening by the ECB lowers prices excluding rents within a year and raises real rents. We argue that understanding this empirical regularity requires considering the tenure choice and portfolios of different participants in rental markets. When a tightening lifts the return on financial assets, landlords require a higher rental yield. Higher rates also raise landlords' mortgage costs, a housing-specific form of the cost channel, and keep would-be buyers renting. We study these forces in a Heterogeneous-Agent New Keynesian (HANK) model in which households choose to rent, to own their home, or to become landlords, and can buy housing with a mortgage. Rental leases are sticky: only new contracts reprice at the market rent.

The model delivers three results. First, the supply side dominates: about two-thirds of the effect of higher interest rates on the rental market comes from higher returns on financial assets and higher mortgage costs for landlords. Second, a rule that targets a rent-inclusive index, as central banks effectively do, keeps reacting to rent pressure of its own making: the consumption contraction becomes deeper, and both nondurable and rent inflation become more volatile. Third, within simple Taylor rules, the welfare-optimal response to rent inflation is neither to tighten against it nor to look through it, but to ease: under both general demand and supply shocks, rent inflation should lower the policy rate relative to a rule that ignores rents. Accommodation lowers the cost of supplying rental housing and relieves poorer renters and mortgage borrowers, at the expense of unlevered owners and unlevered landlords.

In the model, households facing uninsurable earnings risk decide each period whether to rent, own their home, or own an additional housing unit that they supply to the rental market as landlords. House prices, market rents, residential construction, and the real interest rate are equilibrium objects. Because only new contracts reprice, the model distinguishes the market rent on new leases from the average rent households actually pay, which is the object measured in

[1] Following its 2021 strategy review, the ECB recommended a roadmap for incorporating owner-occupied housing costs into the index, on the grounds that doing so would better represent the inflation rate relevant for households (European Central Bank, 2021). The ECB's preferred approach is based on net acquisitions, in contrast to the rental-equivalence imputation used in the United States (Lane, 2022). The ECB's 2025 strategy assessment reiterates the Governing Council's support for including owner-occupied housing costs in the HICP (European Central Bank, 2025).

consumer price indices. On the nominal side, unions set wages subject to Rotemberg adjustment costs, yielding a New Keynesian wage Phillips curve, and the central bank sets the nominal rate with a Taylor rule. Our policy experiments vary whether, and with what weight, rent inflation enters the policy rule, either through the inflation index the rule targets or as a separate term.

We calibrate the model to Spain at quarterly frequency and validate it against two pieces of untargeted evidence: tenure transitions from the Spanish survey of household finances, and the responses of prices and rents to ECB monetary policy shocks, whose sign and timing the model reproduces. In Spain, although many new mortgages are now fixed-rate, most outstanding mortgages are still adjustable-rate (Banco de España, 2025), as in our model.

The rent channel of monetary policy. We first characterize how a monetary tightening reaches the rental market. A contractionary monetary policy shock lowers consumption, output, and nondurable inflation, and at the same time raises market rents. The market rent on new leases rises, while the average rent, the component that enters the price index, inherits the increase only as leases turn over: it peaks at 4 basis points above its steady-state level eight quarters after a 25-basis-point annualized innovation to the monetary policy rule, long after the shock itself has faded. We show that about two-thirds of the effect on the rental market driven by higher interest rates comes from the supply side: the rise in returns on alternative financial assets reduces the willingness of current landlords to invest in real estate, and the increase in mortgage costs discourages owners who would borrow to buy a rental unit from becoming landlords if rents do not increase. The demand side, would-be buyers who keep renting because mortgages cost more and savings pay more, accounts for the remaining third. The model generates an impact house-price decline of 11 basis points, consistent with the empirical evidence that monetary tightenings lower house prices (Corsetti et al., 2022). This price decline offsets more than half of the upward pressure on rents, and the rental market clears as higher rents draw wealthy owners into becoming landlords. On impact, the largest consumption losses fall on mortgagor-owners, while renters, whose rents rise as leases reset, recover more slowly than unlevered owners and landlords.

Rent in the inflation target. We next put measured rent inflation inside the policy rule. We give rent inflation its share of household spending in the inflation index the rule targets, and hold the economy and the monetary shock fixed, so the comparison with the rent-exclusive rule isolates the policy feedback from measured rent inflation to the path of interest rates. Because the rent component reaches the policy index only as leases turn over, the rule keeps responding to the second-round rent pressure of its own tightening long after the shock has faded; most of the additional contraction therefore accumulates after impact, while rents are still rising. Over the first two years, house prices fall, rents rise and consumption contracts by more, with no additional nondurable disinflation. How owner-occupied housing enters the index matters more than the weight on rents: rental equivalence more than triples the feedback relative to the model's rent share, and net acquisitions moves the additional tightening into the housing recovery.

The optimal response to rent inflation. We let the central bank choose the weight on rent inflation in an additive Taylor rule, which adds a term $\phi_r \pi_t^{\text{rent}}$ to a conventional rule that responds to nondurable inflation. We pick ϕ_r to maximize utilitarian welfare under two general demand shocks, to household patience and to government spending, and two supply shocks, to rental availability and to cost push. The optimal weight is negative under all four: $\phi_r^* = -0.50$ under the patience shock, -0.50 under the government-spending shock, -0.18 under the rental availability shock, and -0.48 under the cost-push shock. A negative weight means that rent inflation lowers the policy rate relative to a rule that ignores it. It works through the transmission mechanism: a lower policy rate reduces the cost of supplying rental housing and leans against the shock that raised rents, whether or not it is housing-specific. It also relieves mortgage borrowers and lowers the government's interest bill.

The poorest renters, who hold the least financial wealth and have the highest marginal utility of consumption, benefit most from the lower rents and higher transfers, while landlords lose rental income and indebted homeowners gain from lower mortgage costs. While tenure is a key determinant of the welfare impact of the optimal policy, net asset positions also matter: rich renters incur losses because of the lower returns on their savings, and so do unlevered homeowners, who are not affected by the reduction in mortgage rates.

These welfare gains may reflect both efficiency gains and redistributive gains, which arise whenever the (poorer) renters benefit from the monetary policy reaction. We use the Dávila and Schaab (2025) decomposition to establish that the redistributive motive is what pushes the monetary authority to respond negatively to rent inflation under the patience and rental availability shocks, and does most of the work under the government-spending shock, while under the cost-push shock both motives favor a negative response.

Such gains cannot be achieved with a Taylor rule that targets only nondurable inflation but is less hawkish overall. Under the patience and rental availability shocks, no such rule comes close to the gain from responding to rent inflation; under the government-spending and cost-push shocks, a less hawkish rule helps but falls short of it. The negative response also survives interest-rate smoothing, an output-gap term in the rule, a transfer proportional to labor productivity, and shorter leases. At a practical level, easing in response to rent inflation means that the part of measured inflation that comes from rents deserves less weight than other price increases: when rents drive a substantial share of inflation, central banks should tighten by less than they otherwise would.

Related literature. Our paper relates to three literatures. First, we build on general-equilibrium models of housing and rental markets with endogenous tenure choice (Sommer et al., 2013; Kaplan et al., 2020; Castellanos et al., 2024). Greenwald and Guren (2025) show that the slope of the tenure-supply curve, which reflects landlords' willingness to sell rental properties to owner-occupiers, governs how a credit-driven increase in the demand for homeownership moves the price–rent ratio and the homeownership rate; in our model, this slope is the endogenous landlord margin. These

models are real, whereas New Keynesian models with housing, starting with Iacoviello (2005), emphasize housing collateral and mortgage costs, whose role in monetary transmission depends on mortgage-market structure and is stronger where mortgages are adjustable-rate (Calza et al., 2013; Garriga et al., 2017). Empirically, the response of euro-area rents to a monetary tightening is small and imprecisely estimated (Corsetti et al., 2022), new-lease rents fall in Germany (Groiss and Syrichas, 2026), and rents rise in the United Kingdom (Lazarowicz and Richard, 2023) and the United States (Dias and Duarte, 2019; Abramson et al., 2025). We add evidence for Spain: after a monetary tightening, prices excluding rents fall and real rents rise. Tighter mortgage supply also raises rents (Gete and Reher, 2018). The consumption response to monetary policy differs between renters, mortgagors, and outright owners (Cloyne et al., 2020) and across household balance sheets in the euro area (Slacalek et al., 2020). Rents are sticky (Genesove, 2003; Baker and Wroblewski, 2026), and recorded shelter inflation lags market rents because lease turnover is slow (Bolhuis et al., 2022; Ball and Koh, 2025). We bring household landlords and sticky leases into a heterogeneous-agent New Keynesian economy, so that the real rate, inflation, and rents respond jointly to policy, and we show why rents rise when policy tightens: mainly because supplying rental housing becomes less attractive.

Second, the paper relates to work on which inflation central banks should stabilize. Since Aoki (2001) and Woodford (2003), the prescription has been to stabilize inflation in the sectors where prices are stickiest, with weights that also reflect each sector's cyclical sensitivity and shocks (Mankiw and Reis, 2003).[2] Rents are sticky, so a mechanical application of this logic would give rent inflation a large weight. The logic does not carry over to leases in our economy: a lease keeps its real rent until it turns over, so inflation creates no relative-price distortion in rents for policy to correct. A recent literature asks directly how monetary policy should treat housing inflation. Bianchi et al. (2024) argue that housing services are rationed through search and inelastic short-run supply, so that tightening against a housing-demand shock inefficiently depresses the goods sector, and optimal policy stabilizes non-housing inflation while disregarding housing inflation; Chodorow-Reich and Mehrotra (2026) apply this logic to U.S. shelter inflation and argue for placing less weight on shelter than its expenditure share. Dias and Duarte (2022) show in a two-agent model with tenure choice that a rent-inclusive index creates a policy feedback loop, in which the rule responds to rent movements that monetary policy itself induced. In an estimated New Keynesian model of the United States with a firm-operated rental stock, De Llanos (2026) finds that a tightening shifts constrained households toward renting and that, among the rules he considers, the shelter-inclusive rule has the largest welfare cost. Albuquerque et al. (2025) study a heterogeneous-agent economy with individual landlords and find that policy may do better by looking through a shock to rental supply. In our economy, households choose whether to become landlords, so a rule that responds to rents raises the very costs that make them supply less, and we find that the welfare-maximizing response to measured rent inflation is not to exclude or

[2] See also Benigno (2004) for currency areas, Eusepi et al. (2011) for a welfare-based price index, and Bodenstein et al. (2008) for core versus headline inflation.

down-weight it but to ease.

Third, our model builds on the Heterogeneous-Agent New Keynesian literature. Kaplan et al. (2018) study monetary transmission through household consumption under uninsurable risk and borrowing constraints, without a discrete tenure margin or rental market. On the normative side, optimal monetary policy in these economies trades off price stability against insurance and inequality (Bhandari et al., 2021; Acharya et al., 2023), and a utilitarian planner faces an incentive to redistribute toward indebted, high-marginal-utility households (Dávila and Schaab, 2023). Wealth-poor working households prefer a rule that stabilizes unemployment, while the wealth-rich prefer inflation targeting (Gornemann et al., 2021). Characterizing optimal rules in HANK, McKay and Wolf (2026) find that distributional concerns barely change optimal monetary policy, whose distributional effects are small. We add a discrete tenure choice and an endogenous landlord margin, so that a tightening transmits through tenure decisions and the rents that clear the rental market, and we characterize an optimal simple rule (Schmitt-Grohé and Uribe, 2007) rather than the full Ramsey optimum. In our economy, monetary policy moves a relative price between two groups of households, the rent that renters pay to landlords, and the redistribution it brings about shapes the optimal response to rent inflation.

We begin in Section 2 by developing the model. Two features of the housing block matter throughout: the endogenous tenure margin, which makes rents rise when policy tightens, and sticky leases, which separate the market rent on new contracts from the average rent that price indices measure. Section 3 calibrates the economy to Spain and validates it against untargeted evidence. Section 4 isolates the rent channel of a monetary tightening, decomposes the rent response into channels, and quantifies its consumption incidence across households. Section 5 puts rent inflation in the policy index and quantifies the cost of the resulting policy feedback. Section 6 characterizes the optimal weight on rent inflation and who gains from it. Section 7 concludes.

2 A HANK Model with Housing Tenure Choice

We study a discrete-time Heterogeneous-Agent New Keynesian economy with endogenous housing tenure choice, sticky rental leases, and housing supply. These margins determine how monetary shocks affect market rents, average rents, house prices, and households across tenure states. Time is indexed by $t \in \{0, 1, 2, \ldots\}$. The economy is populated by heterogeneous households, labor unions that introduce nominal rigidities through sticky wages, a representative consumption-good producer, a construction sector, a government that issues debt, and a central bank. The final good's price acts as the numeraire.

2.1 Households

We consider a continuum of infinitely lived, ex-ante identical households that become heterogeneous ex-post along four dimensions: financial wealth, housing tenure status, idiosyncratic labor

productivity, and a lease state. Each period, households consume a nondurable good, choose a financial asset position, and make a discrete housing tenure decision. The housing tenure choice determines whether a household rents, owns a single housing unit, or owns two housing units, in which case the household becomes a landlord and rents out one unit to the rental market, similarly to the framework in Castellanos et al. (2024).

Idiosyncratic risk. Households face two sources of idiosyncratic uncertainty. First, labor productivity follows a persistent stochastic process. Second, the discrete housing choice is subject to type-I extreme-value taste shocks, which smooth the tenure decision. There is no aggregate uncertainty. Aggregate shocks are modeled as one-time unanticipated MIT shocks: they are unexpected on impact, and the economy follows a deterministic transition thereafter.

Preferences. Households derive utility from nondurable consumption and housing services, and they dislike supplying labor. They value these flows according to the per-period utility function

$$\mathcal{U}(c,h,n) = u(c,h) - v(n) = \frac{(c\,f(h))^{1-\gamma}}{1-\gamma} - \psi_n \frac{n^{1+\eta}}{1+\eta}, \tag{1}$$

where γ is the coefficient of relative risk aversion, η is the inverse Frisch elasticity, ψ_n scales the disutility of labor, and $f(h)$ captures the housing services enjoyed by homeowners relative to renters,

$$f(h) = \mathbb{1}_{h=0} + \mathbb{1}_{h>0} \left(\theta\, \tilde{h}_2 / \tilde{h}_1\right)^{\alpha_h}. \tag{2}$$

Here $\tilde{h}_1 < \tilde{h}_2$ denote the two housing-quality (equivalently, size) tiers, with renters occupying the low tier and homeowners (owner-occupiers and landlords) the high tier, while $\theta > 1$ raises the service flow from ownership relative to renting and $\alpha_h < 1$ captures diminishing returns to housing quality. Housing consumption varies only through tenure: there is no intensive margin, so a renter cannot absorb higher rents by renting a smaller dwelling, and rents must move by more to clear the rental market. Labor supply n is set by labor unions, as described in Section 2.3.

The marginal utility of consumption, $u_c(c,h) = f(h)^{1-\gamma} c^{-\gamma}$, depends on the household's tenure state through the housing-services shifter $f(h)$. Because $\gamma > 1$, the term $f(h)^{1-\gamma}$ is decreasing in $f(h)$: at a given level of consumption, a homeowner, who enjoys more housing services ($f(h) > 1$), values an additional unit of consumption less than a renter does. Consumption and housing services are therefore substitutes: a household is willing to accept lower nondurable consumption in exchange for the housing services of ownership.

Budget constraint. Households allocate their savings across housing h and financial assets a, which can be negative, reflecting the net debtor position of households with mortgages. Households consume the nondurable good, and renters pay rent. Thus, their budget constraint reads:

$$c + a' + \psi_t(h,h') \leq \left(1 + r_t + \kappa\, \mathbb{1}_{a<0}\right) a + (1-\tau_y)\, w_t y n + T_t(y) + \widehat{p}^r_t(s)\,(h-1), \tag{3}$$

where a prime denotes a next-period variable, r_t is the real interest rate on balances carried from $t-1$ into t (Section 2.6), κ is a mortgage spread that makes the borrowing rate exceed the rate on savings by a constant margin, w_t is the real wage, τ_y is a flat tax on labor income, $T_t(y)$ is a government transfer (Section 2.7), and $\widehat{p}_t^r(s)$ is the rent paid or received under a lease in state s (see Section 2.2). The term $\widehat{p}_t^r(s)(h-1)$ captures the rental market: renters ($h=0$) pay rent, owner-occupiers ($h=1$) have no rental cash flow, and landlords ($h=2$) receive rent. The function $\psi_t(h,h')$ summarizes all housing costs, as detailed below.

Collateral constraints. Borrowing is subject to collateral and income constraints. A household that upgrades its housing position ($h' > h$) faces loan-to-value (LTV) and loan-to-income (LTI) limits. Let p_t^h denote the real price per unit of housing quality. Following Castellanos et al. (2024), houses of different qualities can be freely converted into one another, so a housing unit of quality $\tilde{h}_i$ is worth $p_t^h \tilde{h}_i$. Let $q_t(h,h')$ denote the quantity of housing quality acquired when moving from housing state h to h',

$$q_t(h,h') = \begin{cases} \tilde{h}_2, & \text{if } (h,h') = (0,1), \\ \tilde{h}_1 + \tilde{h}_2, & \text{if } (h,h') = (0,2), \\ \tilde{h}_1, & \text{if } (h,h') = (1,2), \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Thus, $p_t^h q_t(h,h')$ is the value of housing acquired in the transaction. An existing mortgage borrower who retains the same housing state must satisfy a minimum amortization requirement. Formally,

$$a' \geq \begin{cases} \max\left\{ -\lambda_{\text{LTV}} p_t^h q_t(h,h'),\ -\lambda_{\text{LTI}} w_t y n \right\}, & \text{if } h' > h, \\ (1-m)\, a, & \text{if } h' = h > 0 \text{ and } a < 0, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Households can borrow only when upgrading their housing position. The LTV constraint prevents them from borrowing more than a fraction λ_{LTV} of the value of the housing units purchased, while the LTI constraint caps borrowing at λ_{LTI} times current labor income. Existing borrowers who retain the same housing state may roll over their mortgage but must reduce its balance by at least a fraction m each quarter. Their minimum payment, $(r_t + \kappa + m)\,|a|$, therefore covers interest and amortization and moves one for one with the mortgage rate, as the installment of an adjustable-rate mortgage does when its rate resets. Households who become renters or move to a lower housing state must repay their debt in full. We abstract from refinancing and mortgage equity extraction.[3]

[3]Mortgages in the model reset automatically with the real rate, so refinancing is not needed for rate changes to pass through to debt service. We show using the Spanish Survey of Household Finances (*Encuesta Financiera de las Familias*, EFF), 2014–2022 waves, that only 3.3 percent of owner-occupiers hold debt secured on real estate for purposes other than buying property, compared with 37.9 percent with a loan for the main residence. International Monetary Fund (2008) likewise classifies mortgage equity withdrawal in Spain as limited.

Housing costs. The housing cost function $\psi_t(h, h')$ captures depreciation and the costs of buying and selling housing units. An owner-occupier holds the single unit $\tilde{h}_2$, a landlord holds both $\tilde{h}_1$ and $\tilde{h}_2$, and a renter holds none. Let $p_t(h)$ denote the market value of the housing position held in state h, $p_t(h) = p_t^h \tilde{h}_2 \mathbb{1}_{h=1} + p_t^h (\tilde{h}_1 + \tilde{h}_2) \mathbb{1}_{h=2}$, and recall that $q_t(h, h')$ is the quantity purchased and $q_t(h', h)$ the quantity sold when moving from h to h'. Housing costs are

$$\psi_t(h, h') = p_t(h') - (1 - \delta_h)\, p_t(h) + \tau_h\, p_t^h \big[q_t(h, h') + q_t(h', h)\big], \tag{6}$$

where δ_h is the depreciation rate and τ_h is a transaction cost, which the buyer pays on the units purchased and the seller on the units sold. The first two terms are the change in the value of the household's housing position net of depreciation; the last is the transaction cost, which accrues to the government. For example, an owner-occupier who sells and becomes a renter ($h = 1$, $h' = 0$) receives $(1 - \delta_h - \tau_h)\, p_t^h \tilde{h}_2$, and a renter who buys ($h = 0$, $h' = 1$) pays $(1 + \tau_h)\, p_t^h \tilde{h}_2$.

Recursive problem. We denote the individual state by (a, h, y, s). At the beginning of each period, the household observes a vector of choice-specific taste shocks $\varepsilon = \{\varepsilon(h')\}_{h'}$ and solves

$$V_t^{\varepsilon}(a, h, y, s) = \max_{c, a', h'} \mathcal{U}(c, h, n) + \sigma_\varepsilon \varepsilon(h') + \beta_t\, \mathbb{E}_t\big[V_{t+1}(a', h', y', s') \,\big|\, y, h, s, h'\big]. \tag{7}$$

The shocks $\varepsilon(h')$ are drawn independently across households, housing choices, and dates from a type-I extreme-value distribution with unit scale and zero mean, and $\sigma_\varepsilon > 0$ scales the taste shocks. The continuation value $V_t(a, h, y, s) = \mathbb{E}_\varepsilon[V_t^{\varepsilon}(a, h, y, s)]$ integrates over these shocks, yielding logit transition probabilities across housing states. Appendix A.1 provides the corresponding formulas. The discount factor β_t may vary over time.

Distribution. We denote the distribution over the idiosyncratic state space (a, h, y, s) by $g_t(a, h, y, s)$ and the stationary distribution by $g_{ss}(a, h, y, s)$. The share of households with housing state h and lease state s is $\mu_t^{h,s} = \iint g_t(a, h, y, s)\, da\, dy$, and the share with housing state h is $\mu_t^h = \sum_s \mu_t^{h,s}$.

2.2 Sticky Rental Leases

In practice, the rents paid by households do not adjust instantaneously to market conditions, both because existing contracts take time to be updated and because tenancy law limits when their rent may change. We capture these frictions with leases that retain their signing rent in real terms until they are renegotiated or either party changes tenure. Their nominal rent is therefore indexed to consumer-price inflation.

Leases as contracts. A lease is a contract between one renter and one landlord. New leases are signed at the market rent: households entering a rental tenure, and households whose previous

lease has ended, are matched at random in the market for new leases. A lease continues into the next quarter if it is not renegotiated, which occurs with probability φ, and if both parties remain in their rental tenure. A renegotiation resets the lease to the market rent. When either party changes tenure, the contract ends, and a partner who stays in its rental tenure signs a new lease at the market rent. Lease survival thus depends on both parties' endogenous tenure decisions. Households take aggregate tenure-retention rates as given; in equilibrium, these rates coincide with those generated by their choices – we provide more detail about these and their associated continuation values in Appendix A.1.

We track the lease with a state $s \in \{0, 1, \ldots, J\}$: $s < J$ denotes a lease signed s quarters ago, and $s = J$ pools leases signed J or more quarters ago, which allows us to keep the state space contained. A continuing lease ages to $\min\{s+1, J\}$, and a new lease starts at $s = 0$. Owner-occupiers hold no lease; their state is normalized to $s = 0$ and has no effect on payoffs. The rent paid by the renter and received by the landlord is

$$\widehat{p}_t^r(s) = \begin{cases} p_{t-s}^r, & s < J, \\ \widetilde{p}_t^r, & s = J, \end{cases} \tag{8}$$

where p_t^r is the market rent, determined by rental-market clearing, and $\widetilde{p}_t^r$ is the pooled rent of long-standing leases. We approximate the rents of these older contracts by their average, which is the same for renters and landlords. Appendix A.2 gives its updating rule and the evolution of lease masses. The two sides hold the same contracts, so renters' aggregate payments equal landlords' aggregate receipts at every date.

Average rent and rent inflation. The average rent is computed over the distribution of renters across lease vintages:

$$\overline{p}_t^r = \sum_{s=0}^{J-1} \frac{\mu_t^{0,s}}{\mu_t^0} p_{t-s}^r + \frac{\mu_t^{0,J}}{\mu_t^0} \widetilde{p}_t^r. \tag{9}$$

Rent inflation in the consumer price index is computed from the average rent $\overline{p}_t^r$, rather than the market rent p_t^r. This is the model counterpart of an index measured from outstanding leases and the relevant object for a central bank targeting an HICP-based index (Section 5). Lease turnover delays the transmission of market rents into this index.

Because $\overline{p}_t^r$ is expressed in units of the consumption good, the nominal average rent is $\overline{P}_t^r = P_t^c \overline{p}_t^r$, where P_t^c is the nominal price of the consumption good. With $\pi_t^c = P_t^c / P_{t-1}^c - 1$ denoting nondurable inflation, nominal rent inflation satisfies

$$1 + \pi_t^{\text{rent}} = (1 + \pi_t^c) \frac{\overline{p}_t^r}{\overline{p}_{t-1}^r}. \tag{10}$$

2.3 Labor Market

As is standard in New Keynesian models with sticky wages, both without household heterogeneity (Erceg et al., 2000) and with it (Auclert et al., 2024), a continuum of labor unions sets nominal wages and determines the hours households work.[4] Hours are rationed equally across households,

$$n_t = N_t, \tag{11}$$

where N_t is aggregate labor. Unions adjust nominal wages subject to a quadratic Rotemberg (Rotemberg, 1982) cost, which they pass through to their members as a utility loss. The flow payoff that the union internalizes augments the household utility (1) with this cost,

$$u(c,h) - v(n) - \frac{\theta_w}{2}(\pi_t^w)^2, \tag{12}$$

where $\pi_t^w = (W_t - W_{t-1})/W_{t-1}$ is nominal wage inflation and $\theta_w \geq 0$ governs the degree of wage rigidity.

Each union sets its wage to maximize the aggregate lifetime utility of the households it represents. Optimal nominal wage setting yields a nonlinear New Keynesian wage Phillips curve (see Appendix A.3 for the derivation),

$$\begin{aligned}\pi_t^w(1+\pi_t^w) = \frac{\epsilon_t}{\theta_w} N_t \left(v'(N_t) - (1+\tau_L)(1-\tau_y)\frac{\epsilon_t - 1}{\epsilon_t} w_t \int y_{it}\, u_c(c_{it}, h_{it})\, di \right) \\ + \beta_t \pi_{t+1}^w (1+\pi_{t+1}^w).\end{aligned} \tag{13}$$

Here c_{it} and h_{it} denote the consumption and housing of household i, and $\int(\cdot)\,di$ averages over the continuum of households. Equivalently, one can integrate with respect to the distribution $g_t(a,h,y,s)$. The parameter ϵ_t is the elasticity of substitution across labor varieties and τ_L is an employment subsidy. With a subsidy that offsets the union's market power in the steady state, namely $(1+\tau_L)(\epsilon-1)/\epsilon = 1$ at the steady-state elasticity ϵ, this equation shows that unions aim to equate the marginal disutility of labor to the marginal utility of the after-tax income earned in an additional hour of work.[5] The elasticity ϵ_t may vary over time.

Because goods prices are flexible, the model has no dispersion in relative prices across producers: inflation is costly through the wage-adjustment term in (12), and sticky leases matter for welfare through the rents households pay and the tenure choices they make.

[4] Placing the nominal rigidity in wages avoids the countercyclical profits that sticky-price heterogeneous-agent models generate and would have to assign to households (Broer et al., 2020; Auclert et al., 2025): with perfect competition and flexible prices, profits are zero at all dates. Symmetric rationing serves a separate purpose, shutting down the endogenous labor-supply margin so that labor income moves proportionally across households.

[5] We assume, as in Erceg et al. (2000), that employment in the steady state is subsidized to eliminate the monopolistic distortion associated with the wage-setting power of unions.

2.4 Consumption-Good Producer

A representative firm produces the final nondurable consumption good using only labor.[6] The technology is linear:

$$Y_t^c = A^c N_t, \tag{14}$$

where A^c is total factor productivity. Under perfect competition and flexible prices, profits from production are zero and the real wage equals the marginal product of labor,

$$w_t = A^c. \tag{15}$$

Nominal wage inflation π_t^w and nondurable inflation π_t^c are linked through the labor market. Since the real wage equals productivity, $w_t = A^c$, and the nominal wage is $W_t = w_t P_t^c$, nominal wage inflation satisfies $\pi_t^w = \pi_t^c$.

2.5 Construction Sector

The construction sector produces new housing units Y_t^h using land L_t and housing structures S_t (priced in units of the consumption good) according to a Cobb–Douglas technology,

$$Y_t^h = A^h L_t^{\alpha_L} S_t^{1-\alpha_L}, \tag{16}$$

where A^h is total factor productivity in construction and α_L is the land share. Under perfect competition, the construction firm solves

$$\max_{S_t, L_t} \; p_t^h A^h L_t^{\alpha_L} S_t^{1-\alpha_L} - p_t^L L_t - S_t, \tag{17}$$

where p_t^L is the price of land in consumption units. Land permits are in fixed supply, $L_t = \overline{L}$, and p_t^L adjusts to clear their market. The government issues the permits, and their revenue $p_t^L \overline{L}$ enters its budget (Section 2.7). Because houses of different qualities can be freely converted into one another, we do not need prices that clear the market for each individual housing type; it suffices to clear the market for the aggregate housing stock. Aggregate housing demand implied by the tenure distribution is $H_t = \mu_t^1 \tilde{h}_2 + \mu_t^2 (\tilde{h}_1 + \tilde{h}_2)$. The housing market clears when this demand equals the available aggregate stock. Since new construction augments the existing stock, which depreciates at rate δ_h, market clearing requires

$$H_{t+1} = (1 - \delta_h) H_t + Y_t^h. \tag{18}$$

[6] We adopt the simplest production structure, as in Dávila and Schaab (2023). An alternative is to assume firms produce differentiated varieties under monopolistic competition, as in Auclert et al. (2024). This generates positive profits that must be redistributed, adding a layer of complexity without affecting the main transmission channels we study.

Together with rental-market clearing, $\mu_t^0 = \mu_t^2$, and $\mu_t^0 + \mu_t^1 + \mu_t^2 = 1$, the definition of H_t ties the tenure distribution to the housing stock,

$$\mu_t^0 = \mu_t^2 = \frac{\tilde{h}_2 - H_t}{\tilde{h}_2 - \tilde{h}_1}. \tag{19}$$

When an owner-occupier becomes a renter and another becomes a landlord, the housing they demand falls from $2\tilde{h}_2$ to $\tilde{h}_1 + \tilde{h}_2$, so a higher renter share requires a smaller housing stock, and hence construction below depreciation. Because the stock adjusts slowly, the number of leases can change only gradually: in the short run, house prices and rents adjust so that households hold the existing stock in this tenure mix, and shifts in rental demand or in landlords' willingness to supply rental housing move rents rather than the number of rental units.

2.6 Financial Market

We consider a closed economy in which the wealth of savers ($a_t > 0$) finances the mortgages of borrowers ($a_t < 0$) and the public debt B_t. Aggregate net financial wealth is $A_t \equiv \sum_{h,s} \int\int a\, g_t(a, h, y, s)\, da\, dy$. In each period t, the real interest rate r_t adjusts so that the financial market clears,

$$A_t = B_t. \tag{20}$$

We adopt the following timing convention: the household's net position a_t is a one-period riskless claim rolled over each period. We fix the real payoff on the balance carried into t at the rate agreed when that balance was chosen, so monetary shocks realized at t affect the return on debt issued from t to $t+1$ but do not revalue the balance brought into t. Throughout, r_t denotes this real rate between $t-1$ and t.[7]

Real and nominal returns are linked by the Fisher equation. Along the perfect-foresight path after the shock, this claim is equivalent to a nominal bond with gross nominal rate $1 + i_t$, and the real gross return in consumption units is $1 + r_{t+1} = (1 + i_t)\, P_t^c / P_{t+1}^c$. The nominal interest rate i_t, the real interest rate r_{t+1}, and nondurable inflation π_{t+1}^c satisfy

$$1 + i_t = (1 + r_{t+1})(1 + \pi_{t+1}^c). \tag{21}$$

Mortgages are adjustable-rate: debt carried from t to $t+1$ bears the rate $r_{t+1} + \kappa$, where κ is the constant spread in (3).[8] Through the Fisher equation (21), the nominal policy rate i_t determines

[7] Because the payoff is fixed in real terms, surprise inflation does not redistribute wealth between nominal creditors and debtors, a channel emphasized by Doepke and Schneider (2006) and Auclert (2019); monetary policy redistributes in the model through the paths of real rates, rents, and house prices.

[8] Spanish variable-rate mortgages are indexed to Euribor plus a contractual margin. We represent this structure with a constant spread over the real rate and a quarterly reset, abstracting from the annual or semiannual reviews usual in Spanish contracts.

the real rollover rate r_{t+1} and hence mortgage financing costs and the minimum payment required by (5).

2.7 Government and Central Bank

Government. The government issues one-period bonds, finances purchases that may vary over time, pays an employment subsidy, and taxes labor income at the flat rate τ_y. It also collects the transaction costs on housing purchases and sales, the mortgage spread, and the revenue from land permits. Surpluses and deficits are financed with debt and returned to households through a lump-sum transfer that responds gradually to the level of debt. The fiscal authority pays an employment subsidy $\tau_L(1-\tau_y)w_t y_t n_t$ to each household with labor productivity y_t. To finance this subsidy, it raises a tax proportional to labor productivity, equal to $\tau_L(1-\tau_y)w_t y_t N_t$. In equilibrium this scheme does not redistribute resources across households, because hours are rationed symmetrically and $n_t = N_t$.[9]

Let B_t denote the stock of public debt carried into period t, G_t government purchases, $\overline{T}_t$ the lump-sum transfer paid to every household, and $M_t(h,h')$ the mass of households moving from housing state h to h'. Transaction-cost revenue $\mathcal{T}_t^h$ and aggregate mortgage debt A_t^- are

$$\mathcal{T}_t^h \equiv \tau_h\, p_t^h \sum_{h,h'} M_t(h,h')\,\big[q_t(h,h') + q_t(h',h)\big], \quad A_t^- \equiv \sum_{h,s} \iint a\, \mathbb{1}_{a<0}\, g_t(a,h,y,s)\, da\, dy < 0.$$

The government also collects the labor tax, $\tau_y w_t N_t$, and the spread on mortgage debt, $-\kappa A_t^- > 0$. Its budget constraint is

$$B_{t+1} = (1+r_t)\, B_t + G_t + \overline{T}_t - \tau_y w_t N_t - \mathcal{T}_t^h + \kappa A_t^- - p_t^L \overline{L}, \tag{22}$$

and the transfer follows the fiscal rule

$$\overline{T}_t = \overline{T}_{ss} - \Gamma_B\,\big(B_t - \overline{B}\big), \tag{23}$$

where $\overline{B}$ is steady-state debt and Γ_B governs how fast debt returns to it, as in the fiscal rules of Leeper (1991). In the stationary equilibrium, debt equals $\overline{B}$, purchases equal G_{ss}, and the transfer $\overline{T}_{ss}$ balances the budget. Along the transition, a change in the interest bill or in tax revenue first changes the stock of debt, and households receive it through the transfer over the following years rather than in the quarter in which it occurs.[10] The transfer schedule $T_t(y)$ combines the employment-subsidy scheme with the lump-sum transfer $\overline{T}_t$.

[9] Making both the subsidy and its financing proportional to labor productivity prevents this fiscal arrangement from redistributing income across households. This allows the employment subsidy to offset the steady-state wage markup without creating an additional fiscal transmission channel.

[10] Kaplan (2025) emphasizes that fiscal assumptions are not innocuous in HANK models: the way monetary-policy-induced fiscal effects are financed can shape aggregate and distributional transmission.

Central bank. Because the model contains only two priced components of household consumption, nondurable consumption and rental housing services, we define the model counterpart of HICP inflation, π_t^{hicp}, as a weighted average of nondurable and rent inflation. House purchases are asset transactions and do not enter the consumption price index. The central bank sets the nominal interest rate using a Taylor rule that targets this inflation measure,

$$i_t = r_{ss} + \phi_\pi \pi_t^{\text{hicp}} + \varepsilon_t^m, \qquad \pi_t^{\text{hicp}} = (1-\omega_r)\,\pi_t^c + \omega_r\,\pi_t^{\text{rent}}, \tag{24}$$

where r_{ss} is the steady-state real interest rate, ϕ_π is the inflation response coefficient, ε_t^m is a monetary policy shock, ω_r is the weight on rents in the index, and π_t^{rent} is the nominal inflation rate associated with the *average rent* $\overline{p}_t^r$ defined in (9). Interest and inflation rates in (24) are quarterly, and steady-state inflation is zero.[11]

2.8 Equilibrium

We now define the competitive equilibrium for this economy. Aggregate consumption is $C_t \equiv \int c_{it}\, di$. The equilibrium requires that household and firm choices are individually optimal, that all markets clear, and that the distribution of households evolves consistently with individual decisions and aggregate prices.

Definition 1 (Competitive Equilibrium)**.** *Given a sequence of shocks, a competitive equilibrium consists of household value and policy functions* $\{V_t, V_t^\varepsilon, c_t, a_{t+1}, h_{t+1}\}$*, prices* $\{w_t, r_t, p_t^h, p_t^r, \widetilde{p}_t^r, \overline{p}_t^r, p_t^L, i_t, \pi_t^c, \pi_t^{\text{hicp}}, \pi_t^w, \pi_t^{\text{rent}}\}$*, aggregate quantities* $\{N_t, C_t, Y_t^c, Y_t^h, H_t, A_t, L_t, S_t, B_t\}$*, lease-survival objects* $\{\xi_{h,t}^s, \lambda_t\}$*, a transfer schedule* T_t*, and a distribution* g_t *such that:*

(i) ***Household optimality.*** *Given prices and retention rates, household policy functions solve the Bellman equation (7), with continuation value (A.4), subject to the budget constraint (3) and the borrowing constraints (5).*

(ii) ***Consumption-good firm optimality.*** *The representative firm maximizes profits, yielding* $w_t = A^c$*.*

(iii) ***Construction firm optimality.*** *The construction firm maximizes profits as in (17).*

(iv) ***Labor union optimality.*** *Labor unions ration hours, so* $n_t = N_t$*, and nominal wage inflation* π_t^w *satisfies the New Keynesian wage Phillips curve (13).*

(v) ***Market clearing.*** *Four markets clear:*[12]

(a) Housing market: $H_{t+1} = (1-\delta_h)H_t + Y_t^h$.

[11] In Spain, the ECB sets the interest rate in response to euro-area inflation. Spanish inflation and inflation in the rest of the euro area share one cycle, with a correlation of 0.82 since 1999, so that, seen from Spain, the ECB's rule acts as a rule on Spanish inflation with a coefficient above one. Appendix D.10 derives this mapping.

[12] Goods-market clearing, $Y_t^c = C_t + S_t + G_t$, follows from Walras' law given the other clearing conditions (Appendix A.4).

(b) Rental market: *the total share of renters equals the total share of landlords,* $\mu_t^0 = \mu_t^2$*; by* (A.5)*, the two sides then coincide within each lease state.*

(c) Financial market: $A_t = B_t$.

(d) Land market: $L_t = \overline{L}$.

(vi) ***Government budget.*** *Public debt evolves according to* (22)*, and the lump-sum transfer follows the fiscal rule* (23)*.*

(vii) ***Law of motion for the distribution.*** *The distribution* g_t *evolves according to household policy functions, the idiosyncratic income transition probabilities, and the lease-state transition rules of Section 2.2.*

(viii) ***Lease consistency.*** *The retention rates households take as given equal those implied by their choices,* (A.3)*, for every lease state, and the pooled rent satisfies* (A.6)*.*

(ix) ***Monetary policy.*** *The nominal interest rate satisfies the Taylor rule and HICP-index definition* (24)*, and the Fisher equation* (21) *links the nominal and real interest rates.*

2.9 Timing and Rent Measurement

The lease structure of Section 2.2 determines when aggregate shocks can move rents and how rents enter measured inflation. We collect these implications here, since the quantitative sections use them repeatedly.

Predetermination. We index the initial stationary equilibrium by $t = 0$ and the impact period by $t = 1$. Aggregate shocks are realized after period-1 leases have been signed: leases are signed at the start of the quarter, before the shock is observed. Every incumbent renter and landlord enters the impact period at the steady-state rent: the market rent satisfies $p_1^r = p_{ss}^r$, every outstanding lease was signed at p_{ss}^r, and by (A.6) the pooled rent equals p_{ss}^r as well. The average rent is then predetermined, $\overline{p}_1^r = p_{ss}^r$, regardless of the distribution of households across lease states.

Remark 1 (Impact equivalence)**.** Since $\overline{p}_1^r = \overline{p}_0^r = p_{ss}^r$, the rent-inflation identity (10) implies that measured rent inflation coincides with nondurable inflation on impact,

$$\pi_1^{\text{rent}} = \pi_1^c \qquad \Longrightarrow \qquad \pi_1^{\text{hicp}} = \pi_1^c \quad \text{for any } \omega_r. \tag{25}$$

The rent weight in the Taylor rule (24) is therefore irrelevant for the mapping from inflation to the policy rate on impact. Its effects operate only from $t = 2$ onward and, at $t = 1$, through expectations of the future path.

Rent-inflation decomposition. Solving the rent-inflation identity (10) for π_t^{rent} and subtracting nondurable inflation gives

$$\pi_t^{\text{rent}} - \pi_t^c = \left(1 + \pi_t^c\right)\left(\frac{\overline{p}_t^r}{\overline{p}_{t-1}^r} - 1\right) \approx \text{ real-rent growth.} \tag{26}$$

Measured rent inflation exceeds nondurable inflation exactly when the real average rent grows. A rule that responds separately to π_t^{rent} and π_t^c therefore responds to real average rent growth over and above its response to nondurable inflation. Sections 5 and 6 use this decomposition.

Turnover lag. A lease ends when it is renegotiated, at rate $1 - \varphi$ per quarter, or when either party changes tenure. In the stationary equilibrium a fraction of about $\varphi\, \xi_{0,ss}\, \xi_{2,ss}$ of leases therefore survives each quarter, where $\xi_{h,ss}$ is the stationary tenure persistence of side h, and the average rent inherits movements in the market rent with a mean lag equal to the mean age of outstanding leases (Appendix A.2). Because lease survival responds to tenure decisions, a shock that leads more households to change tenure also shortens that lag. Market rents can move sharply while measured rent inflation responds only gradually.

3 Taking the Model to the Data

We take the model to Spanish data at quarterly frequency. Spain is particularly well suited to our analysis because it is an economy with a high homeownership rate, where the margin between renting and homeowning is particularly active, and because most outstanding mortgages and some newly originated ones are adjustable-rate, as in our model. We interpret the calibrated economy as a Spain-like economy whose central bank sets policy for it alone.

We divide the calibration between externally assigned parameters and seven internally calibrated parameters. We choose (A^h, δ_h, θ) to reproduce the Spanish house-value-to-income ratio,[13] residential investment over GDP,[14] and the renter share; the transaction cost τ_h and the taste-shock scale σ_ε to reproduce the quarterly flows from renting to owner-occupation and from landlord to owner-occupation; β to reproduce a 2 percent annual real interest rate; and the labor tax τ_y to reproduce the ratio of government consumption to GDP. The steady-state lump-sum transfer balances the government budget. All tenure moments refer to households that rent their main residence, own it, or own it and let another dwelling.[15] We then assess the model against

[13] The target is the mean value of homeowners' main residences divided by mean annual gross household income (ECB Household Finance and Consumption Survey, HFCS). Its model counterpart is the price of a reference dwelling of unit quality relative to annual mean income; see Appendix D.1.

[14] Gross fixed capital formation in dwellings divided by GDP at market prices (Eurostat national accounts). Its model counterpart is S_{ss}/Y_{ss}^c, the value of new housing net of land over consumption-good output, which equals model GDP; as in the national accounts, land is excluded from both.

[15] Authors' calculations using the EFF, 2014–2022 waves. Households that neither own their main residence, receive rental income, nor pay rent account for 7.2 percent of all households, and households that receive rental income without owning their main residence for 1.7 percent. The model has no counterpart for either group. Among the

Parameter	Description	Value	Source / Note
Housing and tenure			
$(\tilde{h}_1, \tilde{h}_2)$	Housing qualities	(0.905, 1.095)	Castellanos et al. (2024)
α_h	Housing preference parameter	0.5	Castellanos et al. (2024)
φ	Quarterly probability a lease is not renegotiated	0.99	Rental turnover (EFF)
J	Quarters a lease keeps its signing rent	20	Five-year minimum lease term
Mortgage and financial markets			
λ_{LTV}	Loan-to-value limit	0.8	20% down payment; Bover et al. (2019)
λ_{LTI}	Loan-to-income limit	24	Max. six times annual income
κ	Mortgage spread	0.00345	1.38 pp annual spread, as in the data
m	Minimum amortization	0.0190	90% repaid over 30 years
Preferences and idiosyncratic income			
γ	CRRA risk aversion	2	Standard
η	Inverse Frisch elasticity	2	Standard
ρ_y	Quarterly log-productivity persistence	0.95	Flodén and Lindé (2001)
σ_y	Std. of quarterly log-prod. innovation	0.104	Arellano et al. (2022)
Production and market structure			
α_L	Land share in housing production	0.33	Castellanos et al. (2024)
ϵ	Elasticity across labor varieties	10	Standard
τ_L	Employment subsidy	1/9	Offsets the steady-state wage markup
θ_w	Wage Rotemberg coefficient	778	Wage Phillips-curve slope of 0.02
Government and monetary policy			
ϕ_π	Inflation coefficient	1.5	Standard
$\overline{B}$	Government debt	4	Debt equal to 100% of annual GDP
Γ_B	Fiscal-rule coefficient	0.051	Burriel et al. (2010)

Table 1. Externally Calibrated Parameters

moments it does not target, among them the remaining entries of the quarterly tenure-transition matrix, estimated from the panel of the EFF, which govern mobility across tenure states. Section 4.2 compares the model's responses of prices and rents to a monetary tightening with our estimates for Spain.

3.1 Externally Calibrated Parameters

Table 1 reports the parameters assigned outside the internal calibration.

Housing and tenure. We adopt the housing qualities $(\tilde{h}_1, \tilde{h}_2) = (0.905, 1.095)$ and the housing preference parameter $\alpha_h = 0.5$ from Castellanos et al. (2024). We set $J = 20$ to reflect the five-year minimum term of residential leases for individual landlords under current Spanish tenancy law, and approximate rent updates by consumer-price indexation. We discipline φ using rental-turnover evidence. Our calculations using the EFF suggest that approximately 14 percent of market-rent tenants moved into their dwelling during the previous year.[16] With $\varphi = 0.99$, approximately 17

remaining households, 17.0 percent pay rent.

[16]Authors' calculations using the EFF, 2014–2022 waves; Appendix D.5 documents the turnover evidence and Spanish lease provisions.

percent of outstanding model leases were signed within the previous four quarters. In the model, a lease that is not ended by a change of tenure ends when the tenant moves to another rental dwelling or the two parties sign a new contract, and in either case the new lease is signed at the market rent. Neither φ nor J affects the stationary equilibrium, in which every lease pays the same rent, so both leave the calibration unchanged.

Mortgage and financial parameters. We set the loan-to-value limit to $\lambda_{\text{LTV}} = 0.8$, consistent with the Spanish evidence in Bover et al. (2019). The loan-to-income limit stands in for the affordability checks that lenders apply to new mortgages. Quantitative housing models usually state such checks as a limit on the ratio of payments to income (Kaplan et al., 2020; Greenwald and Guren, 2025); we use the loan-to-income form, which does not tighten mechanically when the interest rate rises. We set it to $\lambda_{\text{LTI}} = 24$ at quarterly frequency, six times annual income, the level above which International Monetary Fund (2026) classifies Spanish mortgages as high-LTI loans, so the limit binds only for the most leveraged buyers. We set the mortgage spread to $\kappa = 0.00345$, the sample average of the annualized spread, 1.38 percentage points.[17] We set the minimum amortization requirement to $m = 0.0190$, so that a borrower who retains the same housing state repays 90 percent of the initial balance over 30 years, $(1 - m)^{120} = 0.1$.

Preferences and earnings process. We set risk aversion to $\gamma = 2$ and the inverse Frisch elasticity to $\eta = 2$. Idiosyncratic labor productivity follows

$$z_{t+1} = \rho_y z_t + \sigma_y \zeta_{t+1}, \qquad z_t \equiv \log y_t, \qquad \zeta_{t+1} \sim \mathcal{N}(0, 1).$$

We set annual persistence to 0.8139, the estimate for Sweden in Flodén and Lindé (2001), which gives $\rho_y = 0.9498$ at quarterly frequency. We choose $\sigma_y = 0.104$ using the approximately 0.52 spread between the 90th and 10th percentiles of men's residualized annual log-earnings changes in 2005–2006 reported by Arellano et al. (2022). Appendix D.7 documents the frequency conversion and compares our persistence choice with the wage autocorrelations reported for Spain.

Production and wage rigidity. On the production side, we adopt a land share of $\alpha_L = 0.33$ from Castellanos et al. (2024). We set the elasticity across labor varieties to $\epsilon = 10$ and the employment subsidy to $\tau_L = 1/9$, which offsets the steady-state wage markup. We set the wage Rotemberg coefficient so that the slope of the wage Phillips curve with respect to the log labor wedge, $\epsilon\psi_n/\theta_w$, equals 0.02, which gives $\theta_w = 778$.[18] The slope, rather than θ_w itself, is the object that governs wage dynamics.

[17] Authors' calculations using the ECB Data Portal. The spread is the average, from May 2001 to April 2026, of the difference between the interest rate on new Spanish variable-rate mortgages with an initial rate fixation of up to one year and the one-year Euribor.

[18] The wage-setting parameters estimated by Galí et al. (2012), at the posterior mode of their specification including unemployment, imply a quarterly slope of approximately 0.022 on the log wage-markup gap.

Government, monetary policy, and normalizations. We set government debt to $\overline{B} = 4$, corresponding to 100 percent of annual GDP.[19] Steady-state government purchases equal the revenue from land permits, transaction costs, and the labor tax, $G_{ss} = p^L_{ss}\overline{L} + \mathcal{T}^h_{ss} + \tau_y w_{ss} N_{ss}$, and we calibrate the labor tax so that G_{ss} equals 19.0 percent of GDP.[20] We set the fiscal-rule coefficient to $\Gamma_B = 0.051$, the posterior mean that Burriel et al. (2010) estimate for Spain for the response of lump-sum transfers to public debt. Debt then closes half of any gap with its steady-state level in about four years. The steady-state transfer that balances the government budget is -1.8 percent of output: households pay a lump-sum tax, because the interest bill on public debt exceeds the revenue from the mortgage spread. We follow Auclert et al. (2025) in setting the inflation coefficient in the monetary policy rule to $\phi_\pi = 1.5$. We normalize consumption-sector productivity, land supply, and stationary mean labor productivity so that $A^c = 1$, $\overline{L} = 1$, and $\mathbb{E}[y] = 1$. The scale of labor disutility ψ_n is chosen so that aggregate employment also equals one in the stationary equilibrium. Mean labor income therefore equals one, and we use it as the unit of income throughout.

3.2 Internally Calibrated Parameters and Model Fit

Table 2 reports the seven internally calibrated parameters, chosen jointly to hit the targets listed above.

Parameter	Description	Value
A^h	Housing-sector productivity	0.0294
δ_h	Housing depreciation rate	0.0046
θ	Homeownership preference shifter	1.197
β	Discount factor	0.9881
σ_ε	Taste-shock scale	0.122
τ_h	Transaction cost	0.025
τ_y	Labor tax	0.150

Table 2. Internally Calibrated Parameters

The internally calibrated parameters pin down distinct margins of the stationary equilibrium. The housing parameters (A^h, δ_h, θ) jointly govern housing supply, the user cost of housing, and tenure choices. The discount factor β governs desired saving and is chosen to deliver a quarterly real interest rate of 0.5 percent, corresponding to 2 percent annualized. The transaction cost τ_h and the taste-shock scale σ_ε govern mobility across tenure states. The transaction cost dampens moves that require buying or selling a dwelling, while the taste shocks generate moves in every direction, regardless of prices. The flow from renting to owner-occupation responds mainly to the transaction cost, and the flow from landlord to owner-occupation mainly to the taste-shock scale. The calibrated transaction cost, 2.5 percent of the value of the units traded, is an average

[19] Spain's general government gross debt stood at 100.7 percent of GDP at the end of 2025, according to Eurostat.

[20] General-government final consumption expenditure over GDP at current prices (Eurostat, `nama_10_gdp`), averaged over 2014–2022 excluding 2020 and 2021, the period of the EFF tenure targets.

over tenure changes, many of which involve no market transaction.[21] The labor tax τ_y finances government consumption.

Table 3 compares the resulting targeted and untargeted moments with their Spanish counterparts; Appendix D documents the data. The calibrated model reproduces the targeted moments in Table 3 by construction.

Moment	Model	Spain
Targeted moments		
House value / annual income	5.25	5.25
Residential investment / GDP	6.9%	6.9%
Renter share	17.0%	17.0%
Government consumption / GDP	19.0%	19.0%
Renter → owner-occupier, quarterly	1.17%	1.17%
Landlord → owner-occupier, quarterly	2.55%	2.55%
Untargeted moments		
Rental burden (renters paying rent)	30.4%	35.0%
Gross rental yield (annual)	5.1%	5.6%
Household net worth / annual income	6.58	7.33
Owner-occupier share	66.0%	72.9%
Landlord share	17.0%	10.0%
Landlords with net debt	32.8%	34.0%
Owner-occupier → landlord, quarterly	0.66%	0.61%
Average quarterly tenure persistence	0.984	0.985
Other tenure transitions	Figure 1	

Table 3. Targeted and Untargeted Model Moments

Note. The 2 percent annual real interest rate is imposed as a calibration restriction. Spanish entries are sample averages. Tenure shares and transition probabilities are computed from the EFF for households that rent their main residence, own it, or own it and let another dwelling. In the model, the house value refers to a dwelling of unit quality. The rental burden is the mean across renters of rent relative to the household's own income, and the rental yield is the annual rent of a rental dwelling over its price. Landlords with net debt are landlords whose financial assets fall short of their debts. Sources and construction are documented in Appendix D.

Untargeted moments in Table 3. The level of rents is not targeted directly. The rental burden, the mean across renters of annual rent divided by the household's gross income, is 30.4 percent in the model, compared with 35 percent in Spain.[22] The gross rental yield, the annual rent of a rental dwelling over its price, is 5.1 percent, compared with the 5.6 percent that Khametshin et al. (2024) estimate for Spain from rents declared to the tax authority and registered sale prices, averaged over 2014–2022.

Turning to household balance sheets, the model generates household net worth equal to 6.58

[21] In the EFF panel, 88 percent of owner-occupiers who become landlords between two waves buy no dwelling in between: they let a dwelling they already owned or have inherited. Similarly, 53 percent of landlords who become owner-occupiers keep the dwelling they used to let. Authors' calculations using the EFF, 2014–2022 waves.

[22] Authors' calculations using the EFF, 2014–2022 waves, for households that pay rent; see Appendix D.1.

times annual income, compared with 7.33 in Spain.[23] We measure the model's ratio of net worth to annual income as $(A_{ss} + p^h_{ss} H_{ss})/(4\mathbb{E}[y])$, where A_{ss} denotes aggregate net financial wealth and H_{ss} aggregate housing in dwelling-size units.

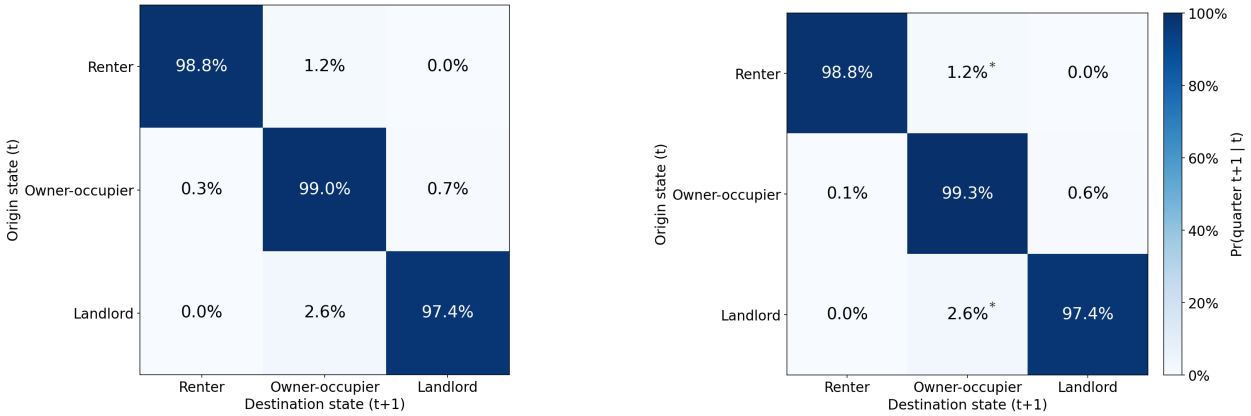
The model matches the overall share of property-owning households, although its composition differs from the data. Owner-occupiers account for 66.0 percent of households and landlords for 17.0 percent, compared with 72.9 and 10.0 percent in Spain. The landlord share equals the targeted renter share mechanically because each landlord supplies one rental unit and each renter occupies one unit. In the data, by contrast, a landlord household may supply several rental properties. The discrepancy concerns the number of landlord households; the size of the rental market itself is matched through the targeted renter share. The model also reproduces the share of landlords in debt, which governs how mortgage costs pass through to rental supply: 32.8 percent of landlords are net debtors in the model, against 34.0 percent in the EFF.[24] Owner-occupiers carry less debt than in the data, 9.9 against 40.7 percent net debtors, mainly because two thirds of new owner-occupiers in the model are landlords who sell their rental unit and repay their mortgage (Appendix D.6). Of the owner-occupiers who are net debtors and keep their home, 47 percent pay exactly the minimum installment, which moves one for one with the mortgage rate.

Wealth distribution. The upper tail of the model's wealth distribution is less concentrated than in the data, with 18.1 against 53.2 percent of net worth held by the top decile, because households share a common discount factor; heterogeneity in patience would bring it closer (Auclert et al., 2025). The model does reproduce who is hand-to-mouth in Spain, where such households are predominantly wealthy homeowners rather than poor (Kaplan et al., 2014): owner-occupiers and landlords with housing wealth but no positive net financial wealth are 16.0 percent of households, against 15.2 percent in Spain, although the model has fewer poor hand-to-mouth households, since renters cannot borrow. Landlords are the wealthiest tenure group in both, but the model compresses the distance between them and their tenants, so it understates the redistributive value of rent relief (Appendix D.3).

Tenure-transition fit. Figure 1 compares the quarterly tenure-transition matrix in the model with the one we estimate from the EFF panel (Appendix D.4). Two off-diagonal entries are calibration targets, and the rest test mobility across tenure states: the model reproduces every entry within sampling error except the flow from owner-occupation to renting, which it overstates, and hence the owner-occupier diagonal, which it understates. Two diagonal cells are load-bearing: the renter and landlord persistence probabilities, 0.9883 and 0.9745 in the model against 0.9880 and 0.9744 in the data, set the retention rates $\xi_{0,ss}$ and $\xi_{2,ss}$ that govern lease survival in (A.4), and hence the mean age of outstanding leases, 22 quarters, which is the mean lag of the measured rent index.

[23] Household net worth over income is financial assets plus housing wealth minus liabilities, divided by annual gross disposable household income (ECB Distributional Wealth Accounts and Quarterly Sector Accounts).

[24] Authors' calculations using the EFF, 2014–2022 waves, for landlords who own their main residence. A household is a net debtor when its financial assets fall short of its debts, the counterpart of $a < 0$ in the model.



(a) Model **(b)** Data

Figure 1. Quarterly Tenure-Transition Matrices

Note. Panel (b) excludes rent-free households and households that receive rental income without owning their main residence. Rows denote a household's current tenure state, columns denote its tenure state in the following quarter, and entries report the corresponding conditional transition probabilities. Asterisks mark the two transition probabilities used as calibration targets. Panel (a) reports the model matrix, computed from stationary joint tenure flows. Panel (b) reports the data matrix, constructed from households observed in consecutive EFF waves between 2014 and 2022. Each wave-pair matrix is converted to quarterly frequency, and the resulting matrices are pooled using origin-state household counts. Both panels use the same color scale, shown in Panel (b).

4 Monetary Tightening and the Rent Channel

We begin with the benchmark monetary experiment to highlight the transmission mechanism of monetary policy to rent inflation in our model. Throughout this section, the central bank responds only to nondurable inflation, so that $\omega_r = 0$ in the Taylor rule (24). In this environment, a monetary tightening raises the real rate and mortgage costs. Some households delay their move into ownership and keep renting, and supplying rental housing becomes less attractive, so market rents rise even as aggregate consumption and output fall; the second force is the larger one. We first describe the aggregate response to the monetary shock, then study the housing and rental market mechanism, decompose the rent response into its channels, and finally quantify the consumption incidence of the tightening across households.[25]

4.1 A Contractionary Monetary Policy Shock

The experiment starts from the calibrated stationary equilibrium and applies a 25-basis-point annualized innovation to ε_t^m in the Taylor rule (24). The shock follows an AR(1) process with

[25] Castellanos et al. (2024) establish this cross-sectional rent channel (displaced buyers crowding into the rental market and marginal landlords requiring higher compensation) in a life-cycle model with an exogenous real interest rate, and show that a rate rise lowers homeownership, raises rents, and shifts welfare from young renters toward landlords. We inherit this channel and contribute by studying how monetary policy propagates through it and by decomposing the rent response into the channels through which the tightening operates.

persistence $\rho_m = 0.7$, which implies a half-life of about two quarters. Figure 2 plots the policy shock, nondurable inflation, the real interest rate, and aggregate real activity.

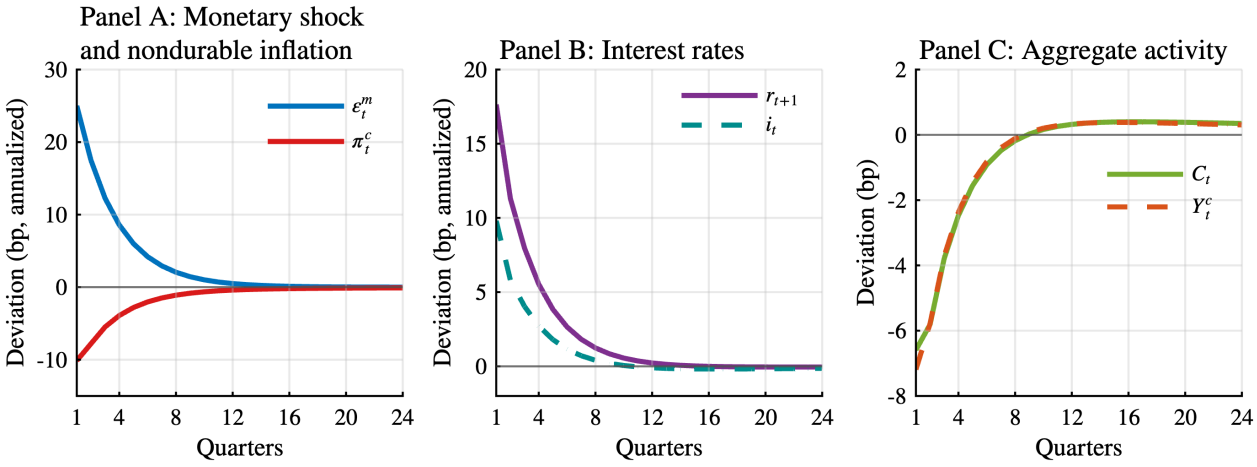


Figure 2. Aggregate Response to a Contractionary Monetary Shock

Note. All panels report deviations in basis points. Panels A and B annualize the monetary policy innovation, nondurable inflation, and interest rates. Panel C reports relative deviations of aggregate consumption and nondurable output from steady state.

The shock contracts real activity and lowers nondurable inflation. On impact, aggregate consumption falls by 6.6 basis points and nondurable output by 7.2 basis points, while nondurable inflation declines by about 10 annualized basis points. The real interest rate rises by 17.6 annualized basis points on impact, so it absorbs 71 percent of the monetary innovation, comparable to the roughly 60 percent in Kaplan et al. (2018). The policy rate itself rises by only 10 basis points, since the rule responds to the fall in inflation; the real rate rises by more because expected inflation falls.

4.2 The Housing and Rental Market Response

The housing-market response is concentrated in rents. Panel A in Figure 3 shows that the rent on new leases, the market rent p_t^r, rises as soon as leases are signed after the shock: it is 31 basis points above steady state in the second quarter and declines gradually afterward, without falling below its steady state. Because a new lease fixes its real rent for five years, the market rent is the price of a multi-year contract: it rises by less, and stays high for longer, than it would if leases reset every quarter. The real average rent responds gradually as leases reset (Section 2.9). Because only a fraction of leases reset each quarter, the increase reaches tenants with a delay: the average rent peaks at 4 basis points above steady state in the eighth quarter, when most of the innovation has already faded, and in the twelfth quarter it is still at 93 percent of its peak.[26]

[26]The length of the period during which a lease keeps its signing rent governs the shape of this response. Figure F.1 in Appendix F.2 repeats the experiment for J between one quarter and five years. With $J = 1$, a new lease pays the market rent for a single quarter, so the market rent must move by more to clear the rental market: it rises more than five times as much in the second quarter and falls below its steady state after a year and a half, and the average rent peaks in the fifth quarter at 15 basis points and is back to 43 percent of its peak in the twelfth quarter.

Panel B shows that the house price falls by 11 basis points on impact. With land and construction-sector productivity fixed, the housing-supply block maps this price decline into a fall of 22 basis points in new housing production.[27] The tenure distribution is predetermined on impact, so the house-price decline is forward-looking: it reflects the anticipated reallocation shown in Panel C. Because rental units are smaller than owner-occupied homes, the anticipated shift out of owner-occupation lowers desired holdings of the housing stock. With the stock itself nearly fixed in the short run, this shift appears almost entirely in the price: aggregate housing falls by less than 0.3 basis points over the transition, compared with a decline of 11 basis points in the house price. Section 4.3 shows that the drop in house prices mitigates the increase in rents.

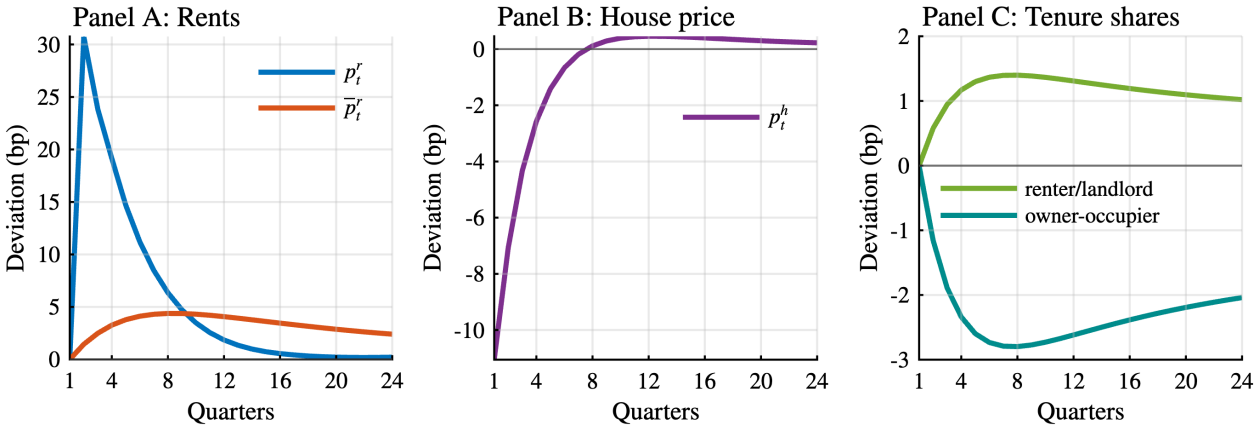


Figure 3. Rental and Housing Market Response to a Contractionary Monetary Shock

Note. All panels report deviations in basis points. Panel A reports relative deviations of the real market rent, at which new leases are signed, and the real average rent from steady state. Panel B reports the corresponding relative deviation of the real house price. Panel C reports absolute changes in tenure shares.

Panel C shows the reallocation that generates the rent response. Because mortgages reprice every quarter, the higher real rate raises the cost of servicing existing debt and financing new purchases in the budget constraint (3). The decline in house prices and labor income also tightens the collateral and income limits in (5). Households close to the ownership margin therefore become less likely to enter owner-occupation, raising the renter share and lowering the owner-occupier share. Rental-market clearing requires an equal increase in landlords. Because constrained households cannot take these positions, the market rent rises until households with sufficient balance-sheet capacity are willing to become landlords. The model also reproduces two qualitative features of the U.S. estimates in Dias and Duarte (2022), lower homeownership and a higher rent-to-price ratio after a monetary tightening, and its rental-supply elasticity lies in the range of available estimates (Appendices F.1 and F.4).

[27] Combining the construction technology (16) with the construction firm's problem (17) and fixed land supply gives

$$Y_t^h = (A^h)^{1/\alpha_L}\left((1-\alpha_L)p_t^h\right)^{(1-\alpha_L)/\alpha_L}\overline{L}.$$

Hence, conditional on construction-sector productivity, the elasticity of new housing production with respect to the house price is $(1-\alpha_L)/\alpha_L$.

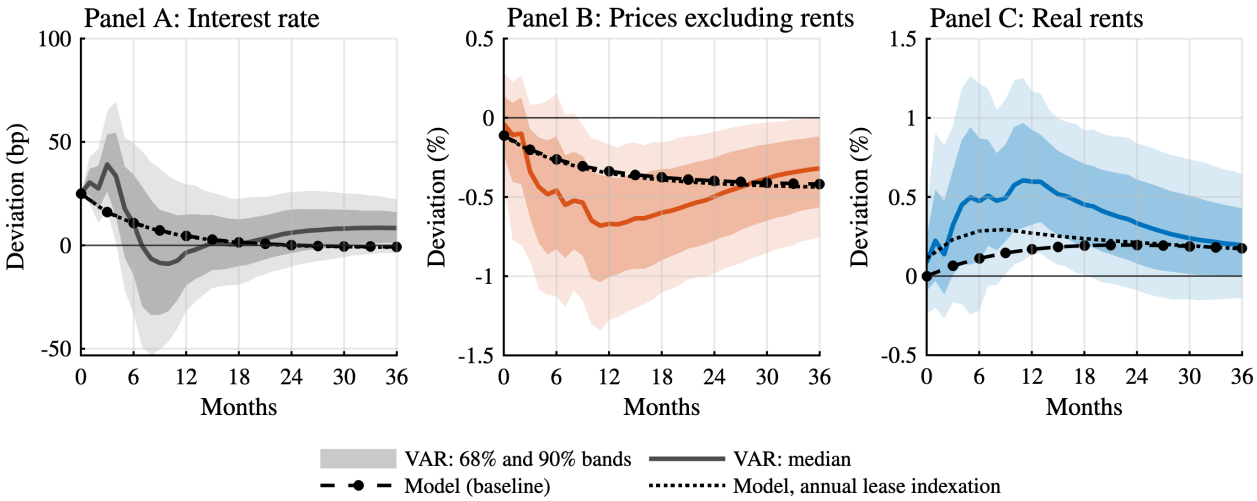


Figure 4. Monetary Tightening in Spain: Model and Evidence

Note. Responses to a monetary tightening that raises the one-year rate by 25 basis points on impact. VAR: Spain, monthly, 2005–2019; median of 2,000 bootstrap draws with 68 and 90 percent bands (Appendix E). Panel A reports the euro-area one-year rate and, for the model, the annualized deviation of the one-year nominal yield, $\sum_{k=0}^{3}(i_{t+k} - r_{ss})$. Panel B reports the HICP excluding rents and the model's price of the consumption good. Panel C reports real rents, the HICP index of actual rentals relative to the HICP excluding rents, and the model's real average rent $\overline{p}_t^r$. Model responses are scaled so that the one-year yield rises by 25 basis points in the impact quarter. The dotted line updates the rents of continuing leases once a year by the previous year's inflation (Appendix F.10).

To validate the model results against empirical evidence, we estimate a monthly vector autoregression for Spain over 2005–2019, in which the monetary shock is identified with high-frequency changes in the one-year swap rate around ECB announcements (Altavilla et al., 2019) and normalized to raise the one-year rate by 25 basis points on impact (Appendix E). We use the rental component of HICP as our measure of rents. Figure 4 compares the estimated responses with those of the model, scaled so that the model's one-year nominal yield also rises by 25 basis points on impact.

Although none of these responses is targeted in the calibration, the model reproduces their main features. In the data, consumer prices excluding rents fall by about 0.7 percent within a year of the tightening, while real rents rise by about 0.6 percent and remain above their pre-shock level for about two years. In the model, prices fall by about half as much, within the estimated 90 percent bands, and the real average rent rises gradually as leases reset, with the sign and persistence of the estimated response: it reaches about 30 percent of the estimated rise after a year and is close to the estimate by the third year. If anything, the model understates the rise in real rents after a monetary tightening. This is partially driven by the fact that, in the model, continuing leases keep the real rent at which they were signed, so the nominal rent falls with the price level, while in the data rents tend to be nominally stickier. When the rents of continuing leases are instead updated once a year by the previous year's inflation, as is common in Spanish contracts, nominal rents stay flat for the first three quarters and the real average rent peaks within the first year, close to the timing of

the estimated peak and at about half its size. We keep real indexation in the baseline because it isolates the rent channel: the average rent then moves only through the market rent on new leases. In Appendix F.10 we report further results for the case which is closer to the Spanish institutional arrangements.

4.3 Rent Decomposition

In our economy, a tightening affects rents through several channels. On the renter side, it makes it harder for households to get a mortgage, which increases their rental demand; on the landlord side, the increase in mortgage costs reduces rental supply, which is potentially exacerbated by the increase in the returns on financial assets, which are alternative investments for potential property owners. A tightening also reduces labor income through the standard HANK general-equilibrium channels, which may also impact rental demand. Lower house prices push the other way, making housing cheaper to buy.

To separate these channels, we compute excess rental supply in a set of counterfactual scenarios in which we keep all prices constant at their steady-state levels after the monetary shock, with the exception of one, which we vary in turn. For example, in one of the experiments, we change the mortgage rate paid by indebted landlords, while keeping the mortgage rate paid by owner-occupiers, the return on financial savings, labor income, taxes and transfers, the house price, and rents constant. By inputting the mortgage rate's equilibrium path with all other prices unchanged, we measure the contribution of this channel to rental-market excess supply, i.e., the difference between the amount of housing that landlords are willing to rent out and the amount of rental housing that renters demand. We then do the same with all other prices.[28] Although there are interactions between channels, to first order these single-input responses sum to the equilibrium response. We also report the direct effect of rents. Rents adjust to clear the disequilibrium in the rental market generated by the combined pressure of all the other channels, so that the total effect reported in the decomposition adds up to zero excess supply and the rental market clears. A channel that contracts excess supply therefore contributes to raising rents.

Figure 5, panel A, shows that the contribution of reduced rental supply to the rise in rents is around twice the size of that of increased demand, thus accounting for around two-thirds of the cumulative contraction generated by movements in the rental market over the first 24 quarters. The fall in house prices partially offsets these two effects. Housing becomes cheaper to buy for prospective landlords and homebuyers alike, encouraging rental supply and easing access to homeownership. Over the same horizon, this channel offsets more than half of the upward rent pressure generated by the other channels.[29]

[28] In equilibrium the three interest-rate inputs comove through the real rate r_t and the mortgage wedge κ in the budget constraint (3); the decomposition perturbs them separately to attribute the rent response across the savings, owner-mortgage, and landlord-mortgage margins. Appendix B states the decomposition formally, defines each input, and reports the resulting contributions to excess supply and to the market rent. The equilibrium paths fed into the household problem are those of Figures 2 and 3.

[29] The sign of the rent response depends on the rental yield, which the calibration ties to depreciation through the

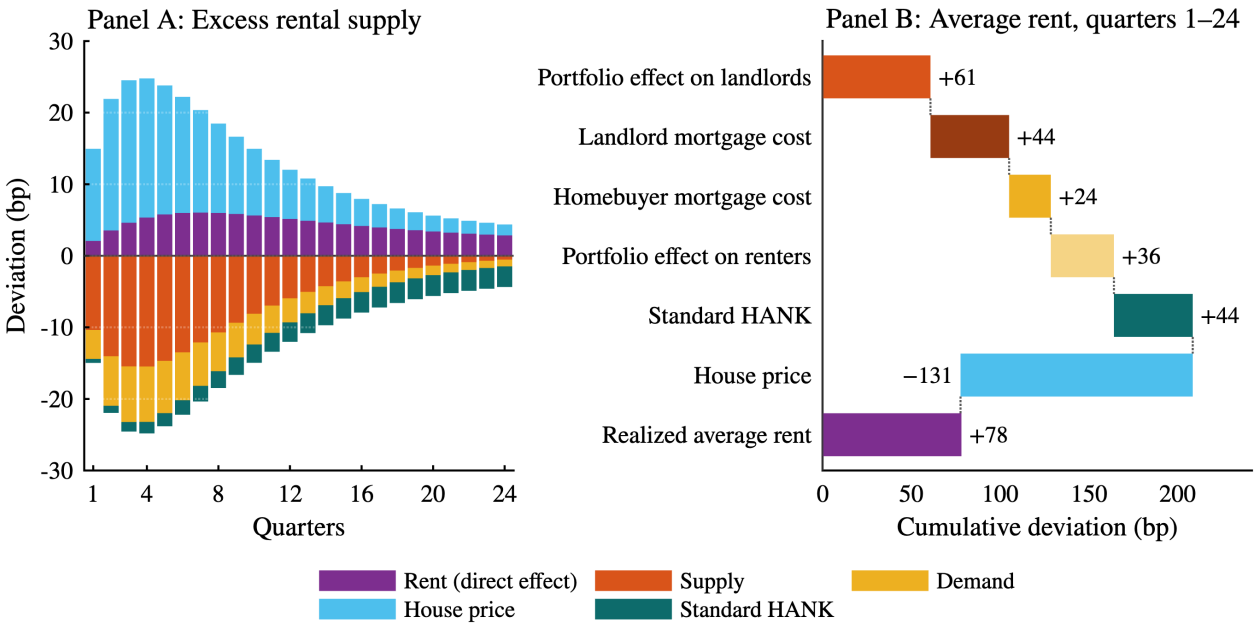


Figure 5. Decomposition of the Rent Response

Note. Panel A reports each group's contribution to rental-market excess supply, the landlord share minus the renter share, in basis points. Panel B reports each channel's contribution to the relative deviation of the real average rent from steady state, summed over quarters 1–24, in basis points; the portfolio effect on landlords includes that on owner-occupiers. Each channel feeds one equilibrium input path through the household problem, holding the remaining inputs at their steady-state values. Supply groups the mortgage rate on landlord debt and the return on financial savings of landlords and owner-occupiers; demand groups the mortgage rate on owner-occupier debt and the return on savings of renters; standard HANK groups labor income and taxes. Table B.1 reports each channel's contribution to excess supply and to the market rent.

The rental market clears through a rise in rents that induces additional supply. On impact, owners become more likely to enter landlordship, more than offsetting a slight increase in exits by incumbent landlords (Appendix F.5). The additional entrants are high-asset households, with net financial assets more than three times those of the average owner-occupier. However, most (86 percent) borrow to buy the rental unit. Expected higher future rents compensate them for the forgone return on financial savings and for the higher cost of that mortgage.

Figure 5, panel B, digs deeper into the specific channels determining rental supply and demand, and translates their contributions into the average rent paid by renters. Two separate channels determine the drop in rental supply. First, at unchanged rents, a higher return on financial savings raises the opportunity cost of holding rental property (portfolio effect on landlords), so incumbent landlords lean toward exit. This channel accounts for a rise in rents of 61bp. Second, higher mortgage costs make it more expensive to buy and hold a rental unit with debt (landlord mortgage cost), so owners who would need a mortgage to become landlords stay out; the channel

residential-investment target. If we instead calibrate to lower yields, the house price absorbs more of the tightening. Its impact response is approximately inversely proportional to the yield, as for an asset whose duration is the price–rent ratio, and real rents fall once the yield is below about 3.8 percent, against 5.1 percent in our calibration and 5.6 percent in Spanish data (Appendix F.3).

works through entry rather than by pushing levered incumbents out and would therefore also be present in a regime with fixed-rate mortgages in which there is significant pass-through of policy rate shocks to mortgage debt.[30] This channel is smaller than the former, but still relevant, rising rents by 44bp.

On the demand side, a higher rate on owner-occupier mortgage debt raises the cost of being a levered owner and lowers the value of becoming one (homebuyer mortgage cost), so would-be buyers stay renters and the pool of renters grows. A higher return on savings also keeps renters renting (portfolio effect on renters), and the standard HANK channel of lower labor income and higher taxes adds to this pressure.[31]

4.4 Consumption Incidence across Households

The consumption effects of the tightening depend on both mortgage debt and rental exposure. We measure them by following cohorts defined by tenure and net financial assets at the date of the shock, relative to the same initial cohorts in a no-shock counterfactual.

Mortgagor-owners experience the largest impact contraction as they anticipate higher debt service; levered landlords also cut consumption more than their unlevered counterparts. Renters face an additional housing-cost burden that reduces the resources available for nondurable consumption—the renter cohort's consumption falls more on impact and recovers less strongly than that of unlevered owners and landlords. Sticky leases delay the increase in rental payments, but households already adjust consumption in anticipation of rising rents. We report detailed consumption responses in Appendix F.6.

5 Rent in the Inflation Target

So far, we have focused on the case in which the central bank does not react to rent inflation. We now move to the case in which the policy index in rule (24) includes rent inflation with weight ω_r. Rent inflation enters with its share of household spending on consumption and rent in the steady state, 5.3 percent.[32] The rent component is nominal average rent inflation, π_t^{rent}, which reflects the rents paid on leases of every vintage. Because the experiment shocks the rule itself, any term that

[30] Holding all other inputs at their steady-state paths, the higher rate on landlords' debt lowers entry by 1.9 basis points of the population on impact, all of it among owners who would have been net debtors after entry. Over quarters 1–24, lower entry accounts for more than the entire contribution of the channel to rental supply: exits by levered incumbents rise on impact but fall afterward, as fewer levered landlords enter, and on net they slightly raise supply.

[31] The portfolio effect on renters is the demand counterpart of the effect on landlords: a higher return on savings makes remaining a renter, with wealth in the liquid asset, more attractive than buying a house. We group the portfolio effect on owners with supply because both of its margins concern becoming a landlord: the opportunity cost that deters owners from becoming landlords, and the wealth effect of higher interest income, which over time makes becoming a landlord more affordable. The two nearly offset.

[32] The share exceeds Spain's HICP weight on actual rentals, 3.1 percent over 2014–2022, because the model compresses the dispersion of rental burdens: aggregate rent is 28 percent of aggregate renter income in the model and 20 percent in the EFF, and at the data's aggregate burden the model's share would be 3.9 percent. Appendix D.8 documents the weights and their construction, and Appendix F.7 reports the responses at Spain's weight.

rises after a tightening amplifies it. The rent term is informative because of when it rises and where the additional tightening lands. The experiment therefore describes how the rent term changes the transmission of a tightening, not whether the rule is desirable; Section 6 studies the optimal response to rent inflation under structural shocks.

Three lessons emerge. First, a rule that responds to rent inflation keeps tightening after the innovation has faded, because measured rents rise only as leases turn over; at HICP weights it adds a fraction of a basis point to the policy rate. Second, the additional contraction operates mainly through lower house prices and higher rents. Responding to rent inflation raises the cost of supplying rental housing and lowers house prices; over the first two years, these housing channels account for the additional fall in consumption. Third, how owner-occupied housing enters the index matters more than the weight on rents: rental equivalence more than triples the feedback relative to the model's rent share, and net acquisitions moves the tightening into the housing recovery.

5.1 Policy Feedback under a Rent-Inclusive Rule

A rule that includes rent inflation responds to the lagged consequences of its own tightening: the additional tightening follows the rise in rents rather than the monetary innovation, and it persists after the innovation has faded. Table 4 summarizes the responses, and Appendix F.7 plots them and reports Spain's weight.

The feedback operates through the gradual transmission of market rents into rental payments. On impact, the average rent is predetermined, so the policy index coincides with nondurable inflation for any rent weight (Remark 1).[33] Once leases reset at the higher market rent, measured rent inflation exceeds nondurable inflation, so the rule keeps the policy rate above its benchmark path. The higher rate raises landlords' financing costs and the return on their savings, the supply channels of Section 4.3, so rents must rise further to keep rental housing supplied: the rule reinforces the rent pressure of its own tightening.

How long the reinforcement lasts depends on the path of real rents. Summing (26) over time, and using that the average rent is predetermined on impact, the rent term accumulated in the policy index up to quarter t is

$$\sum_{k=1}^{t} \left(\pi_k^{\text{hicp}} - \pi_k^{c} \right) \;\approx\; \omega_r \, \ln\!\left(\frac{\overline{p}_t^{\,r}}{p_{ss}^{r}} \right), \tag{27}$$

the rent weight times the log deviation of the real average rent from its steady state. The rent term adds to the policy index while the real average rent rises and subtracts from it only as the

[33] If continuing leases are indexed to past rather than current inflation, as Spanish leases are updated once a year by the previous year's change in consumer prices, measured rent inflation is close to zero on impact while nondurable inflation falls, so the rent term raises the policy rate already on impact, by 0.6 to 0.7 basis points under the model's rent share. The additional tightening stays below one basis point, still lowers house prices, and buys no additional disinflation (Appendix F.10).

			Owner-occupied housing	
	Benchmark (no rents)	Actual rentals	Rental equivalence	Net acquisitions
Real interest rate	+	+2.3	+8.3	−0.2
Real house price	−	−3.1	−11.5	−8.6
Real average rent	+	+3.4	+12.5	+14.4
Consumption	−	−2.7	−9.9	−9.4
Nondurable inflation	−	0.0	+0.3	−5.6

Table 4. Rent-Inclusive Targets: Responses Relative to the Benchmark Rule

Note. Responses to the monetary policy shock of Section 4.1, cumulated over quarters 1–8. The first column gives the direction of the cumulative deviation from steady state under the benchmark rule, whose index excludes rents ($\omega_r = 0$). Each other entry is the difference between the cumulative deviation under the rule in the column and under the benchmark rule, in percent of the benchmark's absolute cumulative deviation; a negative entry means the variable is lower than under the benchmark rule. Actual rentals enter the index at the model's rent share, 5.3 percent. Rental equivalence weighs owner-occupied housing by 13 percent and net acquisitions by 9 percent, with actual rentals at the model's share scaled by $1 - \omega_o$. Appendix F.7 reports Spain's weight and the first five years.

average rent falls back toward its steady state. The average rent rises for eight quarters, as long as new leases are signed above it, and returns slowly because leases last about five years. The real interest rate accordingly stays above its benchmark path from the second to the thirteenth quarter. The additional tightening reaches at most 0.3 basis points on the policy rate after a 25-basis-point innovation. Lease length decides this timing: when a new lease pays the market rent for a single quarter ($J = 1$), real rents peak and return sooner, and the rule front-loads the tightening instead, gives it back from the third year, and reduces the cumulative fall in consumption over five years (Appendix F.2).

Households anticipate the higher path of rates, so the rent term affects the economy before the rent increase reaches the index. Consumption falls by more on impact under the rent-inclusive rule, although the impact real interest rate is slightly lower.[34]

5.2 The Costs of Responding to Rent Inflation

Responding to rent inflation deepens the consumption contraction without delivering additional nondurable disinflation over the first two years, and the additional contraction runs through the housing market. The differences grow with the weight on rents: under the model's rent share they are nearly twice as large as under Spain's HICP weight (Table F.3), and they keep growing almost in proportion to the weight up to $\omega_r = 0.15$ (Appendix F.8). The higher path of real rates lowers house prices further, and lower house prices account for almost the entire additional fall in consumption; higher rents and lower transfers add to it, while the higher real rates themselves

[34] In a first-order decomposition of the impact consumption difference by household input, the anticipated real rates account for more than a third of the additional fall, and lower house prices, labor income, and transfers and higher rents for the rest.

lower consumption only on impact.[35] Nondurable inflation, in contrast, falls by the same amount over two years as under the benchmark rule.[36] Appendix F.9 reports how the additional contraction is distributed across households.

Over five years, the later easing of real interest rates relative to the benchmark is too small to offset the earlier additional tightening. The real interest rate falls below its benchmark path only from the fourteenth quarter, and then by about a hundredth of a basis point, so over five years, under the model's rent share, the cumulative fall in consumption is still 1.4 percent larger than under the benchmark rule, house prices fall by more, and the cumulative rise in the rent tenants pay is 3.1 percent larger. Cumulative nondurable disinflation is larger than under the benchmark rule only until the eighth quarter, while households anticipate the higher rates; by the end of the second year the rent term has bought no additional disinflation, and over five years nondurable inflation falls by 2.7 percent less.

De Llanos (2026) shows, in an estimated model of the United States, that rents rise after a tightening because constrained households shift toward renting and housing is costly to convert to rental use, and that tightening in response to rising rents reinforces that shift. Our economy adds the supply side of the rental market: because landlords' cost of holding a rental unit rises with the policy rate, the rule's additional tightening raises rents further, and it reaches households mainly through house prices. The feedback scales with the weight of housing in the index, which is several times larger in the U.S. consumer price index, where rents are imputed to owners, than in Spain's HICP.

5.3 Owner-Occupied Housing in the Index

How owner-occupied housing enters the index matters more for the feedback than the weight on actual rentals. The HICP excludes the cost of owner-occupied housing; the ECB's 2021 strategy review concluded that including it, through the net acquisitions approach, would better represent the inflation rate relevant for households (European Central Bank, 2021; Eiglsperger et al., 2022), and the 2025 assessment reaffirmed the case (European Central Bank, 2025). The alternative, rental equivalence, imputes to owners the rent their home would fetch, as the U.S. CPI does. With owner-occupied housing, the index becomes $(1-\omega_o)\left[(1-\omega_r)\pi_t^c + \omega_r \pi_t^{\text{rent}}\right] + \omega_o \pi_t^o$, where π_t^o is nominal average-rent inflation under rental equivalence and nominal house-price inflation under our net-acquisitions approximation.[37] The weight ω_o is 13 percent under rental equivalence, the

[35] In a first-order decomposition of the consumption difference by household input, cumulative consumption over quarters 1–8 under the model's rent share is lower than under the benchmark rule by 0.0058 percent of steady-state consumption. House prices contribute -0.0057, rents -0.0022, transfers -0.0016, labor income -0.0006, and real interest rates $+0.0043$.

[36] With interest-rate smoothing of 0.8 and a 25-basis-point annualized innovation, responding to rent inflation still lowers house prices, raises rents, and deepens the consumption contraction relative to the inertial benchmark. These additional effects are smaller than without smoothing. The rule delivers 0.5 percent more nondurable disinflation than the inertial benchmark over two years and 1.3 percent less over five years, under the model's rent share.

[37] We proxy net-acquisitions inflation by nominal house-price inflation; the official index also includes acquisition and ownership costs and restricts dwelling purchases to those entering the household sector.

imputed-rent share of Spain's household consumption, and 9 percent under net acquisitions, the euro-area estimate.[38]

Under rental equivalence, the composite raises the weight on rent inflation to 18 percent, and every response in Table 4 is between three and four times as large as under the model's rent share. Under net acquisitions, the index incorporates house prices, which fall on impact and recover while rents are still rising. The rule therefore eases on impact and tightens during the recovery, when it lowers house prices again and reinforces the rise in rents. Only an index that excludes housing costs avoids the feedback altogether: how owner-occupied housing is measured decides not only how much the rule responds to housing, but when. Appendix F.7 reports the responses. Under the structural shocks of Section 6, rental equivalence also costs the most welfare (Table 5).

6 The Optimal Response to Rent Inflation

The previous section showed that putting rent inflation directly into the policy index changes the timing and incidence of monetary transmission. We now turn to the normative question: how should a central bank that maximizes welfare treat rent inflation?

To do so, we let the central bank choose the weight on nominal rent inflation in an additive Taylor rule,

$$i_t = r_{ss} + \phi_\pi \pi_t^c + \phi_r \pi_t^{\text{rent}}, \tag{28}$$

and report gains relative to $\phi_r = 0$, the rule that responds only to nondurable inflation. The rule incorporates nominal average rent inflation π_t^{rent}, the object observed in a consumer price index.[39]

We evaluate welfare after four shocks, two related to demand and two related to supply, that raise rents. For all four, the welfare-optimal response to rent inflation is negative: rent inflation should lower the policy rate relative to a rule that ignores rents. We first introduce the structural shocks used to evaluate policy, then characterize optimal simple rules, and finally decompose their welfare gains across cohorts and channels.

6.1 Structural Shocks

We evaluate policy under four shocks that raise rents through different mechanisms. Two are general demand shocks, to household patience and to government spending. They expand activity and raise nondurable inflation, and rents rise because the rule tightens in response. Two are supply shocks. The rental supply shock raises rents at the source, by making it costlier to finance a rental unit, with little nondurable inflation or monetary response. The cost-push shock raises

[38] The imputed-rent share averages 13.5 percent in Spain over 2014–2022; the 9 percent net-acquisitions weight follows Eiglsperger et al. (2022). All remaining HICP weights are scaled by $1 - \omega_o$.

[39] A rule on the gap of the real average rent from its steady state, $(\overline{p}_t^r - p_{ss}^r)/p_{ss}^r$, also calls for easing under the government-spending, rental supply and cost-push shocks. Under the rental supply shock it gains 0.68 basis points, against 0.11 for the inflation rule, because the rent gap there is created by the shock itself and the level term keeps easing while rents stay high. Under the patience shock, real rents peak after the expansion, and the optimal coefficient is slightly positive. Appendix F.16 reports the results.

nondurable inflation outside housing; rents again rise through the rule's tightening, but with activity contracting rather than expanding. Together, the four shocks separate whether rent pressure originates in the rental market or in the policy response to inflation elsewhere, and whether it comes with a boom or a contraction. Each shock is a one-standard-deviation innovation with the persistence estimated on previous work, as we detail in Appendix D.9. Appendices F.11, F.12, F.13, and F.14 report the impulse responses.

6.1.1 Demand Shocks

A patience shock lowers the discount factor β_t by 0.0016 on impact, and a government-spending shock raises purchases G_t by 0.72 percent, financed by debt and, through the fiscal rule (23), by lower transfers over the following years. In both cases, as in a standard demand expansion, output, labor, and nondurable inflation rise (Clarida et al., 1999; Galí, 2015), and the benchmark Taylor rule raises the policy rate, transmitting the expansion to housing decisions through borrowing costs and returns on savings.

Much of the rise in rents comes from the tightening rather than from the expansion itself. Higher mortgage rates keep would-be buyers renting, and higher mortgage rates and returns on savings make supplying rental housing less attractive, through the channels of Section 4.3. Higher labor income works in the opposite direction by relaxing the loan-to-income limit in (5). Under the patience shock, lower patience also reduces the incentive to save for a down payment and lowers the discounted benefits of owning in the future, which adds to the shift toward renting. Under the benchmark rule, the real average rent peaks at about 1.1 percent above steady state after the patience shock and 0.13 percent after the government-spending shock.

6.1.2 Supply Shocks

The rental supply shock originates in the rental market. Households in the landlord state who carry mortgage debt pay a spread κ_t^L in place of κ, 22 annualized basis points higher on impact, without any change in owner-occupiers' borrowing costs or in the return on savings; the additional spread revenue accrues to the government (Appendix A.4). At unchanged rents, higher financing costs discourage owners from becoming landlords and encourage indebted landlords to exit. The resulting contraction in rental supply raises rents and shifts households toward owner-occupation. Under the benchmark rule, the real average rent peaks at about 0.39 percent above steady state, while nondurable inflation and the policy rate move little: the shock generates rent pressure with almost no monetary response.

The cost-push shock originates outside the housing market. A transitory fall in the elasticity across labor varieties in the wage Phillips curve (13) raises the desired wage markup by 0.9 percent on impact, and with it wage and nondurable inflation, without directly changing housing costs or borrowing conditions. Under the benchmark rule, nondurable inflation rises by about 0.15 annualized percentage points on impact, the real interest rate rises, and consumption and the

house price fall. As after a monetary tightening, higher rates raise rents through the financing and portfolio margins of Section 4.3: the real average rent peaks at about 0.03 percent above steady state. The rent pressure therefore originates, as under the demand shocks, in the rule's own response to inflation, but here inflation rises while activity contracts.

6.2 Optimal Simple Rules

We evaluate welfare conditional on a one-time, unanticipated MIT innovation to each structural shock and choose ϕ_r to maximize welfare over the resulting transition, in the spirit of the optimal simple rules of Schmitt-Grohé and Uribe (2007),

$$\phi_r^* \in \arg\max_{\phi_r} \mathcal{W}(\phi_r), \qquad \mathcal{W}(\phi_r) = \sum_{t=1}^{\infty} \widetilde{\beta}_t \left[\int u(c_{it}, h_{it})\, di - v(N_t) - \frac{\theta_w}{2} (\pi_t^w)^2 \right]. \tag{29}$$

Whenever we evaluate welfare, we adopt an equal-weighted utilitarian criterion.[40] The discount factor satisfies $\widetilde{\beta}_1 = 1$ and $\widetilde{\beta}_t = \prod_{\ell=1}^{t-1} \beta_\ell$.[41]

The benchmark is $\phi_r = 0$, under which policy responds only to nondurable inflation. Thus, the exercise asks whether nominal rent inflation should raise or lower the policy rate relative to that benchmark, holding fixed the response to nondurable inflation. We compute the optimum separately for each of the four shocks[42] and report welfare differences as consumption equivalents, in basis points of permanent consumption. Finally, we also report welfare changes from the benchmark to the rent-inclusive (HICP-style) rule of Section 5.

6.2.1 The Optimal Weight on Rent Inflation

The welfare-optimal response to rent inflation is negative under all four shocks. The optimal coefficient is -0.50 under both demand shocks, -0.18 under the rental supply shock, and -0.48 under the cost-push shock, and a positive coefficient lowers welfare under every shock (Table 5 and Figure C.1). The welfare gains range from 0.09 to 1.24 basis points of permanent consumption, after a single unanticipated shock.[43] They offset between 7 and 42 percent of the welfare cost of the shock itself.[44]

[40] The objective excludes the type-I extreme-value taste shocks $\sigma_\varepsilon \varepsilon(h')$ that smooth the discrete tenure choice (Iskhakov et al., 2017). Their scale is calibrated to tenure flows, not to any preference moment; we treat them as a smoothing device of the solution method, with no welfare content.

[41] The planner discounts with the households' realized discount-factor path, so, under the patience shock, welfare inherits the shocked path—a form of time-dependent discounting in the spirit of Dávila and Schaab (2023).

[42] We do not optimize ϕ_r against the monetary shock itself. Optimizing a rule's response to a shock that originates in the rule conflates responding to rent inflation with reversing the policy innovation.

[43] Welfare gains of this size are typical of stabilization policy against a single unanticipated shock in this class of models. In our replication of the one-asset HANK model of Dávila and Schaab (2023), even replacing a Taylor rule with fully optimal Ramsey policy raises welfare by between 0.01 and 0.2 basis points of permanent consumption, using the same consumption-equivalent definition. Computing Ramsey policy in our model is left for future work.

[44] The welfare cost of a shock is the consumption equivalent that makes the benchmark transition as good as remaining in the stationary equilibrium: 4.5, 4.9, 0.5 and 0.2 basis points for the patience, government-spending, rental supply and cost-push shocks, of which the optimized rule offsets 28, 7, 20 and 42 percent. The patience shock changes

	Demand		Supply	
	Patience	Government spending	Rental supply	Cost push
Rule on nominal average rent inflation, π_t^{rent}				
Optimal weight, ϕ_r^*	-0.50	-0.50	-0.18	-0.48
Welfare gain at ϕ_r^*	1.242	0.366	0.106	0.087
Rent-inclusive targets of Section 5, welfare change				
Actual rentals, $\omega_r = 0.053$	-0.176	-0.022	-0.081	-0.003
Rental equivalence, $\omega_o = 0.13$	-0.724	-0.086	-0.327	-0.013
Net acquisitions, $\omega_o = 0.09$	-0.172	-0.018	-0.075	-0.011
Cumulative deviation of the real average rent				
Benchmark, $\phi_r = 0$	41.77	8.40	2.10	1.00
Optimized rule, ϕ_r^*	20.78	4.17	1.45	0.60

Table 5. The Optimal Response to Rent Inflation

Note. Each column optimizes rule (28) against a one-standard-deviation shock, holding $\phi_\pi = 1.5$. Welfare changes are consumption equivalents relative to the $\phi_r = 0$ benchmark under the same shock, in basis points of permanent consumption. Rent-inclusive targets weigh actual rentals by the model's rent share; rental equivalence and net acquisitions are the composites of Section 5.3, with actual rentals scaled by $1 - \omega_o$. The lower block reports the sum of percent deviations of the real average rent from steady state over the transition.

The rent-inclusive rules of Section 5 do the opposite: they lower welfare and raise the rent burden under all four shocks. The welfare criterion therefore favors a negative response to rent inflation over both excluding rents and including them with their expenditure weight.

How owner-occupied housing enters the index does not change this ranking, but it decides how much welfare the rule gives up. Rental equivalence, which raises the weight on rent inflation to 18 percent, loses about four times as much as actual rentals alone under every shock. Net acquisitions loses about as much as actual rentals under the demand and rental supply shocks, more under the cost-push shock, and less than rental equivalence under all four. The ECB favors net acquisitions on measurement grounds (European Central Bank, 2021; Eiglsperger et al., 2022); the welfare criterion ranks the two approaches the same way, although neither improves on an index without housing costs.

A negative coefficient tells the central bank to lower the policy rate when measured rent inflation rises. The lower rate works on the margins that raised rents in Section 4.3: it reduces landlords' mortgage costs and the return on the financial assets they could hold instead of a rental unit. Supplying rental housing becomes more attractive, so rents need to rise by less to clear the market. This holds whatever raised rents in the first place, whether the rule's own tightening, as under the demand and cost-push shocks, or landlords' higher financing costs, as under the rental supply shock.

the households' own discount factor, so for this shock we measure the cost and the gain at the steady-state discount factor.

The relief reaches the rent that tenants pay. A lower rent on new leases passes into the average rent as leases turn over, and the optimized rule cuts the cumulative rise in the real average rent by between 31 and 50 percent (Table 5).

Accommodation also has costs. After a demand shock, the economy is already expanding, and a lower rate pushes consumption, hours and inflation higher still: under the patience shock, nondurable inflation on impact is 1.6 annualized percentage points under the optimized rule, against 0.7 under the benchmark. The welfare criterion counts these costs through labor disutility and wage-adjustment costs, so under the demand shocks the case for a negative coefficient rests largely on who gains from it, which Section 6.3 examines. After a cost-push shock, the economy is contracting, and accommodation softens the contraction: consumption falls by half as much on impact. Appendices F.11 to F.14 report the responses under both rules.

In practice, it is hard for a central bank to decompose the relative contribution of different shocks in real time. However, the four welfare curves overlap, so any coefficient between -0.25 and 0 improves welfare under each of the four shocks (Figure C.1).[45] A single coefficient of -0.30 maximizes the sum of the four gains. It keeps about half of the gain under the demand shocks and two-thirds under the cost-push shock, but little under the rental supply shock, whose welfare falls quickly once the coefficient goes beyond its own optimum.

6.2.2 Is This Just a Less Hawkish Rule?

This negative weight admits two readings. The additive rule (28) can be rewritten identically as

$$i_t = r_{ss} + (\phi_\pi + \phi_r)\,\pi_t^c + \phi_r\big(\pi_t^{\text{rent}} - \pi_t^c\big), \tag{30}$$

a lower coefficient on nondurable inflation plus a negative response to real rent growth (26). The gain could come entirely from the first term, that is, from a less hawkish rule. To check, we compare the optimized rule with rules that have no rent term but a different coefficient on nondurable inflation.

Figure 6 shows that, independently of the shock under consideration and across a wide range of potential ϕ_π, the welfare gain implied by including a rent term is higher than that deriving from an optimized rule on nondurable inflation. However, whether this alternative rule can approximate the optimal rule, and how, varies across shocks. Under the government-spending and cost-push shocks, reducing ϕ_π reduces the tightening reaction of the central bank. The effect is similar to what $\phi_r < 0$ generates, and therefore recovers 44 and 80 percent, respectively, of the gain from the rent term. Under the patience shock, a similar argument is true, but the best rule without a rent term only generates 10% of the welfare gains from our optimal policy. In the case of the rental availability shock, given that it generates almost no nondurable inflation, the rule that ignores rents hardly improves welfare compared to the baseline.

[45] The same holds when welfare includes the expected taste-shock utility of tenure choices, which also leaves the four optimal coefficients unchanged. Under the patience shock, evaluating welfare at the steady-state discount factor gives the same optimal coefficient, with a gain of 1.26 basis points instead of 1.24.

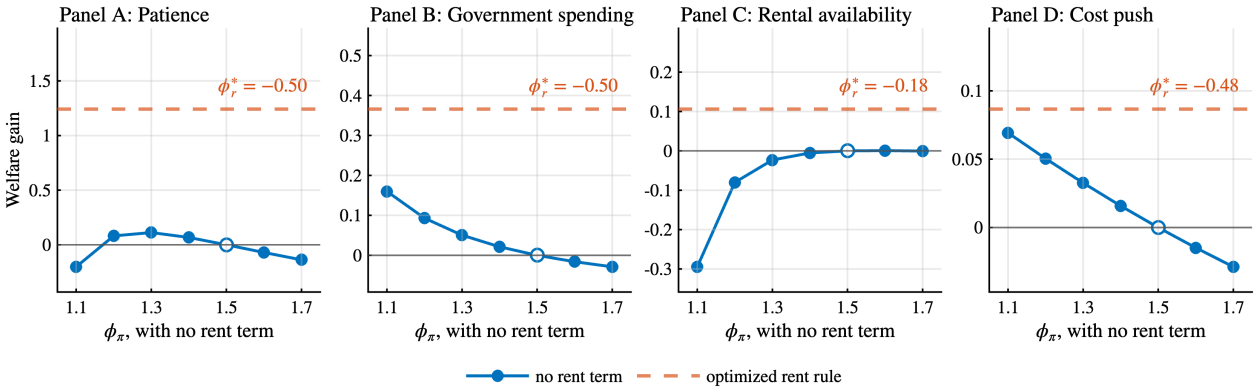


Figure 6. Welfare under Alternative Monetary Policy Rules

Note. The solid lines report welfare relative to the shock-specific benchmark $(\phi_\pi, \phi_r) = (1.5, 0)$ for rules without a rent term, varying ϕ_π over $[1.1, 1.7]$ with $\phi_r = 0$. The dashed lines report the welfare gain from optimizing the rent coefficient while holding $\phi_\pi = 1.5$. Both are consumption-equivalent gains, in basis points of permanent consumption.

Alternatively, in Table F.8, we consider the case in which we keep $\phi_\pi + \phi_r$ constant and change only the response to real rent growth. For effective coefficients of 1.2, 1.3 and 1.5, welfare rises as ϕ_r falls from 0 to -0.4 under all four shocks. Under the rental supply shock, the best of these rules keeps the benchmark's effective coefficient of 1.5 and eases with rents, with $\phi_\pi = 1.9$ and $\phi_r = -0.4$. The rent term therefore does more than lower the overall response to inflation: it lowers the policy rate when real rents are rising, which allows it to respond to different types of shocks in a more targeted way by design.

The negative response to rent inflation does not depend on the rule's other features or on the lease block. With interest-rate smoothing, an output-gap term, a transfer proportional to labor productivity, shorter leases, or more frequent lease turnover, moving ϕ_r from 0 to -0.20 raises welfare under each of the four shocks (Table F.7 in Appendix F.15). When leases reprice more often, the gain is larger than in the baseline.

6.3 Distributional Decomposition across Cohorts and Channels

We now ask who gains from the optimal response, and through which channels. We focus on the patience shock, under which rents rise the most, and report the other shocks in Appendix F. We follow cohorts defined at the date of the shock and decompose their welfare changes with the inputs of Section 4.3: each channel replaces one input path under the benchmark rule with its path under the optimized rule. Figure 7 reports the channels and the net change for each cohort, in utility units, which is how the planner adds them up.

Lower rents (purple) move resources from landlords to tenants: the rent channel raises welfare in every renter cohort and lowers it for landlords. A lower rent is worth most to tenants with low consumption and high marginal utility, which gives a utilitarian planner a distributional reason to value rent relief.

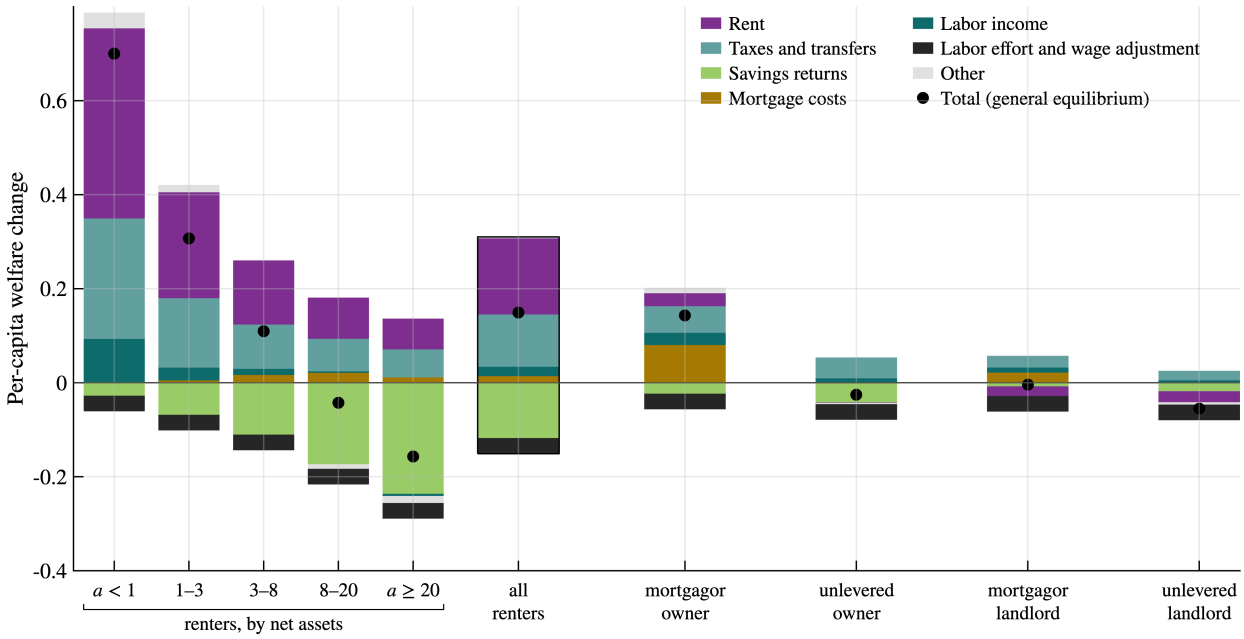


Figure 7. Distributional Decomposition of the Welfare Gain

Note. Per-capita welfare change from moving ϕ_r from 0 to -0.50 under the patience shock, by date-of-shock cohort, in utility units. Net financial assets a are measured in units of quarterly steady-state income. Each channel holds the benchmark equilibrium and replaces one input with its path under the optimized rule; the rent channel moves the rent tenants pay and the rent landlords receive together. Savings returns groups the portfolio effects on renters, owners and landlords of Figure 5; mortgage costs group the homebuyer and landlord mortgage rates; other groups the house price and lease survival. Labor effort and wage adjustment is the change in labor disutility and wage-adjustment costs, common to all households. The all-renters bar aggregates the five renter cohorts. Black dots are the general-equilibrium total.

The net gain is progressive within renters. Poor renters gain the most from lower housing costs and hold little wealth exposed to lower returns on savings. As wealth rises, the loss of interest income (light green) grows and eventually outweighs the rent relief, so the wealthiest renters lose. Renters as a group still gain. For the poorest renters, rent relief is the largest gain, about one and a half times the gain from higher transfers (gray-blue).

Mortgagor-owners gain mainly through lower mortgage costs (dark yellow). Unlevered owners and landlords lose: they earn less on their savings, landlords collect lower rents, and every household bears the cost of higher hours and inflation (black).

Higher transfers come from the government's lower interest bill, which first reduces public debt and reaches households over the following years through the fiscal rule (23) (Figure F.16 in Appendix F.15). Because households hold the public debt, the higher transfers are largely paid for by the interest income that savers lose. Taken together, the two channels account for a third of the aggregate gain under the patience shock, half under the government-spending shock, and almost none under the two supply shocks. The negative coefficient does not rest on the progressivity of the transfer: when the transfer is distributed in proportion to labor productivity, moving ϕ_r from 0

to -0.20 still raises welfare under every shock, by between 53 and 119 percent of the baseline gain.

The other shocks share this pattern, with two differences. Under the supply shocks, higher transfers matter as much as rent relief for the poorest renters, and mortgagor-landlords gain, because lower mortgage costs outweigh the rent they lose. Measured in consumption equivalents, renters and mortgagor-owners gain under all four shocks, while unlevered owners and unlevered landlords lose (Table F.9).

Efficiency versus redistribution. Following Dávila and Schaab (2025), we split the marginal welfare effect of the rent coefficient into efficiency, the households' total willingness to pay, and redistribution, which is positive when the change favors the households the planner weighs more (Figure 8). The planner weighs renters most, then owners, then landlords, and, among owners and landlords, those with a mortgage.

Under the patience shock, efficiency favors a less negative coefficient at every point: accommodation during a boom raises hours and inflation, and these costs outweigh what households would pay for lower rates. The negative optimum rests on redistribution. Under the government-spending and rental supply shocks, efficiency alone would call for a less negative coefficient, about -0.18 and close to zero respectively, and redistribution accounts for the rest. Under the cost-push shock, both components favor a more negative coefficient: accommodation softens a contraction, which most households value.

The rental market shapes this redistribution. Monetary policy moves the rent, a price that renters pay to landlords, two groups with very different marginal utilities. Leaning against rent increases moves resources toward renters, and lower debt service moves them toward mortgagors.

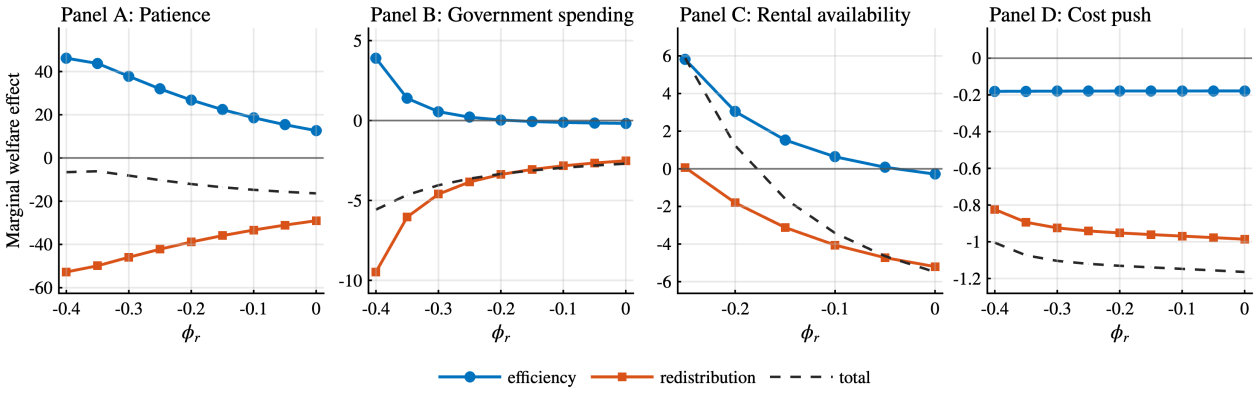


Figure 8. Efficiency and Redistribution

Note. Marginal welfare effect of the coefficient on rent inflation in rule (28), $\mathrm{d}\mathcal{W}/\mathrm{d}\phi_r$, holding $\phi_\pi = 1.5$, split into efficiency and redistribution components following Dávila and Schaab (2025). The dashed line is their sum. Positive values favor a higher coefficient. Effects are measured in 10^{-5} units of perpetual consumption, a unit of consumption to every household at every date. Appendix C.1 contains the derivations.

7 Conclusion

Rent inflation poses a distinct problem for monetary policy because interest rates affect both the demand for rental housing and the return required to supply it. In Spain, an ECB tightening lowers prices excluding rents and raises real rents, and the model reproduces the sign and timing of this response without targeting it. In our model, a monetary tightening keeps would-be buyers renting while raising landlords' financing and opportunity costs. Rents rise until households are willing to supply the additional rental housing. About two-thirds of the effect of higher interest rates on the rental market comes from landlords' financing and portfolio incentives. Sticky leases delay the response of measured rents.

This combination changes the consequences of including rents in the inflation target. A rule that responds positively to measured rent inflation keeps reacting to rental pressure generated by its own earlier tightening. The additional response raises the cost of supplying rental housing and prolongs the consumption contraction. Over the first two years after a monetary tightening, consumption falls by more than under a rule that ignores rents, with no additional nondurable disinflation. Under each of the four structural shocks we consider, the rent-inclusive rule also lowers welfare.

The welfare-optimal response is accommodative under the two general demand shocks and the two supply shocks we consider: rent inflation should lower the policy rate relative to a rule that ignores it. Accommodation reduces the financing and opportunity costs of supplying rental housing, relieves mortgage borrowers, and lowers the government's interest bill. Returning these interest savings to households changes the distribution of financial income, while savers receive lower returns. The welfare gain reflects these household exposures together with changes in employment and inflation costs. Within the simple rules we study, a less hawkish response to nondurable inflation recovers at most part of the gain, and the negative response survives interest-rate smoothing, an output-gap term, a transfer proportional to labor productivity, and shorter leases.

Accommodation also reaches the rents that tenants pay: the optimized rule lowers the cumulative rise in the average rent by between a third and a half. This rent relief has a redistributive effect: landlords receive less rental income, while tenants face lower housing costs. It particularly benefits renters with little financial wealth and high marginal utility of consumption. Wealthier renters can nevertheless lose overall because lower interest income outweighs their rent relief, while indebted homeowners benefit through lower mortgage payments. The appropriate treatment of rent inflation therefore depends on rental-supply incentives and households' exposure to changes in rents and interest rates.

Our framework abstracts from institutional features that shape monetary transmission in housing and rental markets. Mortgages in our model are effectively adjustable-rate. With fixed-rate debt, a tightening would not immediately reprice existing mortgages, weakening the direct mortgage-cost channel. Higher rates would still affect new borrowing and raise the return on

alternative financial investments, so the landlord portfolio channel would remain operative. Nor does our economy have a fiscal instrument targeted at renters, such as a housing allowance, which could take over part of the redistribution that monetary policy achieves here.

In our welfare analysis, we optimize a simple rule against one unanticipated shock at a time and stop short of solving the full Ramsey problem. Our results should therefore be read as conditional on the shocks we study, although any coefficient on rent inflation between -0.25 and 0 raises welfare under all four of them. Both solving the full Ramsey problem and incorporating our framework in a macro model with aggregate risk are natural, but computationally intensive, avenues for future research.

In our economy, a central bank that eases when measured rent inflation rises does better than one that looks through it, which in turn does better than one that gives rents their expenditure weight in the policy index. How owner-occupied housing would enter the index matters too: imputing rents to owners, as the U.S. CPI does, costs about four times as much welfare as actual rentals alone, while the net-acquisitions approach favored by the ECB costs less.

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APPENDIX

A Derivations and Model Details

A.1 Tenure Choice and Lease Survival

At the beginning of each period, the household observes the vector of choice-specific taste shocks $\varepsilon = \{\varepsilon(h')\}_{h'}$ in the household problem (7) and then chooses its next-period housing state. The formulas below are ex-ante with respect to these shocks: they condition only on the state (a, h, y, s).

For each next-period housing state $h' \in \{0, 1, 2\}$, we define the deterministic choice-specific value

$$\widetilde{V}_t(a, h, y, s; h') = \max_{c, a'} \mathcal{U}(c, h, n) + \beta_t \mathbb{E}_t \left[V_{t+1}(a', h', y', s') \mid y, h, s, h' \right],$$

where the maximization is subject to the budget constraint (3) and the borrowing constraints (5), and the expectation over the next lease state s' follows (A.4). The value $\widetilde{V}_t$ collects everything the household controls once the tenure decision is fixed: it is the value of choosing consumption and savings optimally, conditional on moving to housing state h'.

After the taste shocks are observed, the household problem (7) reduces to a deterministic discrete choice across the three shock-augmented values,

$$V_t^{\varepsilon}(a, h, y, s) = \max_{h' \in \{0,1,2\}} \widetilde{V}_t(a, h, y, s; h') + \sigma_\varepsilon \, \varepsilon(h').$$

Because the shocks $\varepsilon(h')$ are drawn independently from a type-I extreme-value distribution with unit scale and zero mean, integrating over them yields closed-form expressions (Iskhakov et al., 2017). The ex-ante probability that a household in state (a, h, y, s) chooses housing state h' is

$$\mathbb{P}_t(h' \mid a, h, y, s) = \frac{\exp\left(\widetilde{V}_t(a, h, y, s; h')/\sigma_\varepsilon\right)}{\sum_{j \in \{0,1,2\}} \exp\left(\widetilde{V}_t(a, h, y, s; j)/\sigma_\varepsilon\right)}, \tag{A.1}$$

and the expected value before the shocks are realized is the log-sum

$$V_t(a, h, y, s) = \mathbb{E}_\varepsilon\left[V_t^{\varepsilon}(a, h, y, s)\right] = \sigma_\varepsilon \log\left[\sum_{j \in \{0,1,2\}} \exp\left(\widetilde{V}_t(a, h, y, s; j)/\sigma_\varepsilon\right)\right]. \tag{A.2}$$

Equation (A.2) is the continuation value that enters the Bellman equation (7) one period earlier, and the probabilities in equation (A.1) govern the evolution of the distribution $g_t(a, h, y, s)$ across housing states and, through (A.3), the retention rates that determine lease survival. Castellanos et al. (2024) and Albuquerque et al. (2025) use the same extreme-value structure for the housing

tenure decision.

The scale parameter σ_ε governs how sensitive tenure choices are to differences in the choice-specific values. As $\sigma_\varepsilon \to 0$, the probabilities in equation (A.1) concentrate on the housing state with the highest deterministic value, and the tenure decision reduces to choosing the h' that maximizes $\widetilde{V}_t$. A larger σ_ε smooths the transition probabilities across the renter, owner-occupier, and landlord states: households whose choice-specific values are close transition gradually rather than all at once. The shocks are idiosyncratic taste shocks over tenure states, capturing motives for moving that are orthogonal to the household's financial state.

Lease survival and continuation value. Lease survival depends on the decisions of both sides of the market. Let $\mathbb{P}_t(h' \mid a, h, y, s)$ denote the probability that a household in state (a, h, y, s) chooses housing state h', given in (A.1), and let $\xi^s_{h,t}$ denote the tenure-retention rate of households in rental tenure $h \in \{0, 2\}$ and lease state s, the share of them that keep their tenure between t and $t+1$,

$$\xi^s_{h,t} = \frac{1}{\mu^{h,s}_t} \iint \mathbb{P}_t(h' = h \mid a, h, y, s)\, g_t(a, h, y, s)\, da\, dy. \tag{A.3}$$

Because partners are matched at random within each lease state, a renter with lease state s who remains a renter keeps its lease with probability $\varphi\, \xi^s_{2,t}$, and a landlord who remains a landlord keeps its lease with probability $\varphi\, \xi^s_{0,t}$. Writing $2 - h$ for the rental tenure of the partner, the continuation value in (7) is

$$\begin{aligned} &\mathbb{E}_t\left[V_{t+1}(a', h', y', s') \,\middle|\, y, h, s, h'\right] \\ &= \begin{cases} \mathbb{E}_{y'|y}\Big[\varphi\, \xi^s_{2-h,t}\, V_{t+1}(a', h', y', s^+) \\ \qquad + \left(1 - \varphi\, \xi^s_{2-h,t}\right) V_{t+1}(a', h', y', 0)\Big], & h' = h \in \{0, 2\}, \\ \mathbb{E}_{y'|y}\left[V_{t+1}(a', h', y', 0)\right], & h' = 1 \text{ or } h' \neq h. \end{cases} \end{aligned} \tag{A.4}$$

where $s^+ = \min\{s+1, J\}$ is the state of a lease that continues. Households take the paths of retention rates $\{\xi^s_{h,t}\}$ as given, and in equilibrium these paths coincide with the ones their choices generate.

A.2 Evolution of Average Rents

Lease masses. Random matching implies that the two sides of the market hold the same contracts. The masses of leases of each age on each side evolve as

$$\begin{aligned} \mu^{h,s+1}_{t+1} &= \varphi\, \mu^{h,s}_t\, \xi^s_{0,t}\, \xi^s_{2,t}, && s < J - 1, \\ \mu^{h,J}_{t+1} &= \varphi \sum_{s \in \{J-1, J\}} \mu^{h,s}_t\, \xi^s_{0,t}\, \xi^s_{2,t}, && h \in \{0, 2\}. \end{aligned} \tag{A.5}$$

and new leases account for the remainder, $\mu^{h,0}_{t+1} = \mu^{h}_{t+1} - \sum_{s\geq 1} \mu^{h,s}_{t+1}$. Whenever renters and landlords coincide within each lease state at t, they coincide at $t+1$ as well. Rental-market clearing, $\mu^{0}_{t} = \mu^{2}_{t}$, holds lease state by lease state, and the rent paid by renters equals the rent received by landlords in every period.

Pooled rent. The pool accumulates the leases that reach age J, which carry the rent at which they were signed, and the pooled rent is the average over its contracts. In this way, it reflects the persistence of existing rental contracts, but also updates with past market conditions, because every period a cohort of leases enters the pool. Formally,

$$\begin{aligned} \widetilde{p}^{r}_{t} &= \lambda_{t-1}\, \widetilde{p}^{r}_{t-1} + (1-\lambda_{t-1})\, p^{r}_{t-J}, \\ \lambda_{t} &= \frac{\mu^{0,J}_{t}\, \xi^{J}_{0,t}\, \xi^{J}_{2,t}}{\mu^{0,J}_{t}\, \xi^{J}_{0,t}\, \xi^{J}_{2,t} + \mu^{0,J-1}_{t}\, \xi^{J-1}_{0,t}\, \xi^{J-1}_{2,t}}. \end{aligned} \tag{A.6}$$

where λ_t is the share of the leases in the pool at $t+1$ that were already in it at t, and $1-\lambda_t$ is the share of leases that reach age J.

Average-rent weights. Iterating the pooled-rent recursion (A.6) backward expresses the pooled rent as a weighted average of the market rents at which its leases were signed,

$$\widetilde{p}^{r}_{t} = \sum_{j=0}^{\infty} \left(\prod_{i=1}^{j} \lambda_{t-i} \right) (1-\lambda_{t-j-1})\, p^{r}_{t-J-j},$$

with $p^{r}_{t} = p^{r}_{ss}$ for $t \leq 1$ and $\lambda_t = \lambda_{ss}$ for $t \leq 0$. Substituting this expression into (9) writes the average rent as a weighted average of current and past market rents,

$$\overline{p}^{r}_{t} = \sum_{k=0}^{\infty} \varpi_{t,k}\, p^{r}_{t-k}, \qquad \varpi_{t,k} = \begin{cases} \dfrac{\mu^{0,k}_{t}}{\mu^{0}_{t}}, & k < J, \\ \dfrac{\mu^{0,J}_{t}}{\mu^{0}_{t}} \left(\displaystyle\prod_{i=1}^{k-J} \lambda_{t-i} \right) (1-\lambda_{t-k+J-1}), & k \geq J. \end{cases}$$

For $k < J$, the weight on the market rent of k quarters ago is the share of renters holding a lease signed k quarters ago; the pool spreads its share over older vintages according to the survival of pooled leases. The weights are nonnegative and sum to one at every date, so a lasting movement in market rents passes into the average rent one for one: lease turnover delays the movement without shrinking it. Because a lease is a contract between one renter and one landlord, the same weights describe the rent that landlords receive.

The weights move along the transition only through the composition of leases by age and the survival of pooled leases, and both move little. At their stationary values, the average rent is a

fixed linear filter of market rents,

$$\overline{p}^r_t \approx \sum_{k=0}^{\infty} \omega_k \, p^r_{t-k}, \qquad \omega_k = \begin{cases} \mu^{0,k}_{ss} / \mu^0_{ss}, & k < J, \\ \left(\mu^{0,J}_{ss} / \mu^0_{ss}\right)(1-\lambda_{ss})\lambda^{k-J}_{ss}, & k \geq J, \end{cases} \tag{A.7}$$

which tracks the solved average rent to within 0.7 percent of its peak deviation after the monetary shock of Section 4.1, under the benchmark rule and under both rent-inclusive rules of Section 5. Leases younger than $J = 20$ quarters carry 59 percent of the weight, and their weights decline by about 5 percent per quarter, the rate at which renegotiations and changes of tenure end leases of every age. The pool carries the remaining 41 percent, spread geometrically at its survival rate $\lambda_{ss} = 0.96$. The index therefore lags the market rent by a mean of $\sum_k k\,\omega_k = 22$ quarters, the mean age of outstanding leases. Figure A.1 illustrates both properties.

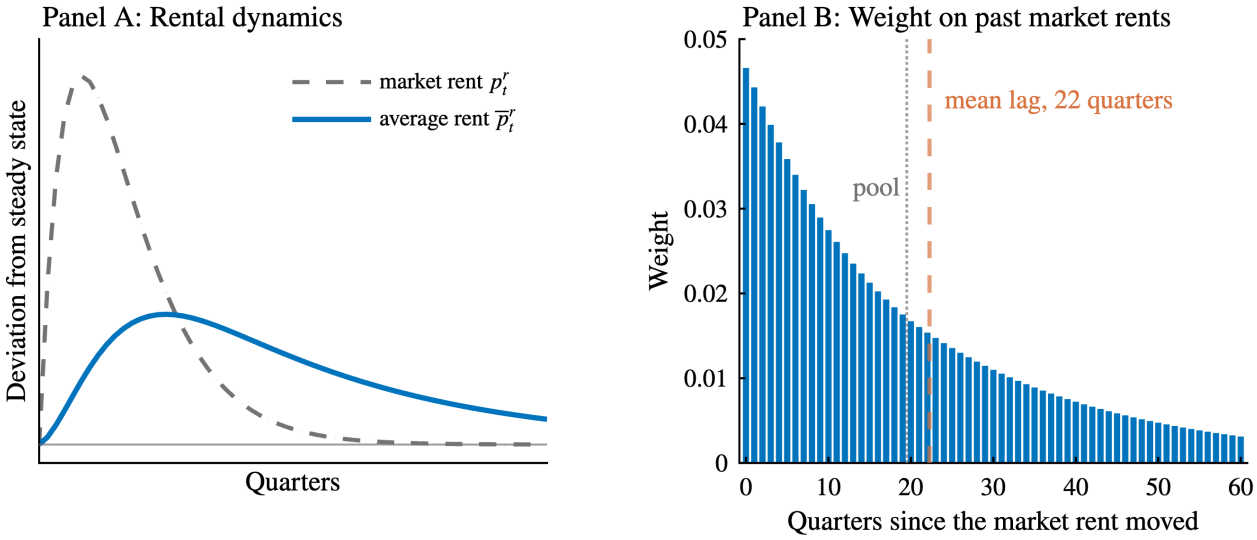


Figure A.1. Adjustment of Average Rents

Note. Panel A plots illustrative rental dynamics: a market rent that rises and returns to its steady state, and the average rent implied by the stationary weights in (A.7). Panel B shows the weight ω_k that the index places on the market rent of k quarters ago, with the implied mean lag marked; leases aged $J = 20$ quarters or more belong to the pool.

Nominal indexation. Let P^c_t denote the nominal price of the consumption good and $\pi^c_t = P^c_t / P^c_{t-1} - 1$ nondurable inflation. In nominal terms, a lease signed in $t - s$, with $s < J$, pays

$$\widehat{P}^r_t(s) = P^c_t \, p^r_{t-s} = \widehat{P}^r_{t-s}(0) \prod_{k=t-s+1}^{t} (1 + \pi^c_k),$$

where $\widehat{P}^r_{t-s}(0) = P^c_{t-s} p^r_{t-s}$ is its nominal rent at signing. Thus, its nominal rent is updated every quarter by accumulated nondurable inflation. The pooled rent is indexed in the same way.

A.3 Derivation of the Nonlinear New Keynesian Wage Phillips Curve

Each union introduced in Section 2.3 chooses its nominal wage to maximize the discounted utility of its members, internalizing the effect of its wage on their labor income and hours, and passing the Rotemberg adjustment cost through to them as a utility loss. Optimal wage setting delivers equation (13).

Labor packer and union labor demand. A continuum of unions indexed by $k \in [0,1]$ transforms household hours into differentiated labor services. Each household supplies hours to every union, so a household's total hours aggregate the services supplied through all unions, and each union rations hours equally across its members. A competitive labor packer combines the union services N_{kt} into aggregate labor according to the CES technology

$$N_t = \left(\int_0^1 N_{kt}^{\frac{\epsilon_t - 1}{\epsilon_t}} \, dk \right)^{\frac{\epsilon_t}{\epsilon_t - 1}},$$

where ϵ_t is the elasticity of substitution across labor varieties, and sells the bundle to the consumption-good producer at the nominal wage W_t. Cost minimization by the packer yields the demand for union k's service and the wage index,

$$N_{kt} = \left(\frac{W_{kt}}{W_t} \right)^{-\epsilon_t} N_t, \qquad W_t = \left(\int_0^1 W_{kt}^{1-\epsilon_t} \, dk \right)^{\frac{1}{1-\epsilon_t}},$$

where W_{kt} is the nominal wage set by union k, so that $\partial N_{kt} / \partial W_{kt} = -\epsilon_t N_{kt} / W_{kt}$. Consistent with the definition of aggregate wage inflation in Section 2.3, union-specific gross wage inflation is $1 + \pi_{kt}^w = W_{kt} / W_{k,t-1}$. When varying its own wage, the union takes all aggregate objects as given, including W_t, N_t, prices, and the household distribution.

Union's wage-setting problem. Because the discount factor may vary over time, we define the compound discount $D_{t,j} = \prod_{\ell=t}^{j-1} \beta_\ell$, with $D_{t,t} = 1$. At date t, union k chooses the sequence of its wages $\{W_{kj}\}_{j \geq t}$ to maximize the discounted utility of its members, augmented with the Rotemberg cost as in (12),

$$\sum_{j=t}^{\infty} D_{t,j} \left[\int \Big(u(c_{ij}, h_{ij}) - v(n_{ij}) \Big) \, di - \frac{\theta_w}{2} \left(\pi_{kj}^w \right)^2 \right], \tag{A.8}$$

where i indexes households. Union k internalizes that its wage changes both its members' labor income and, through its labor demand, their total hours: $\partial n_{ij} / \partial W_{kj} = \partial N_{kj} / \partial W_{kj}$. In a symmetric equilibrium, hours rationing delivers $n_{ij} = N_j$, as in equation (11).

A member with productivity y_{it} receives from union k the after-tax labor income $(1 - \tau_y) \, W_{kt} \, y_{it} \, N_{kt} / P_t^c$, where P_t^c is the nominal price of the consumption good, together with the

proportional employment subsidy $\tau_L(1-\tau_y)\,W_{kt}\,y_{it}\,N_{kt}/P_t^c$, which the union internalizes when setting its wage. The subsidy and the tax that finances it, $\tau_L(1-\tau_y)\,w_t\,y_{it}\,N_t$, both enter the transfer $T_t(y)$ described in Section 2.7; because the tax depends on aggregate employment, the individual union takes it as given. The derivative of the wage bill with respect to the union's wage is

$$\frac{\partial(W_{kt}N_{kt})}{\partial W_{kt}} = N_{kt} + W_{kt}\frac{\partial N_{kt}}{\partial W_{kt}} = (1-\epsilon_t)\,N_{kt},$$

so the direct change in member i's resources is

$$\frac{\partial}{\partial W_{kt}}\left[(1+\tau_L)(1-\tau_y)\frac{W_{kt}\,y_{it}\,N_{kt}}{P_t^c}\right] = (1+\tau_L)(1-\tau_y)(1-\epsilon_t)\,y_{it}\,\frac{N_{kt}}{P_t^c}.$$

By the envelope theorem, the effect on member i's value is its marginal utility of consumption times this change in resources, $u_c(c_{it},h_{it})\,(1+\tau_L)(1-\tau_y)(1-\epsilon_t)\,y_{it}\,N_{kt}/P_t^c$.

First-order condition. The date-t wage enters the objective (A.8) through members' date-t resources, their date-t hours, and the wage-adjustment costs at t and $t+1$. From the definition of union wage inflation,

$$\frac{\partial \pi_{kt}^w}{\partial W_{kt}} = \frac{1}{W_{k,t-1}}, \qquad \frac{\partial \pi_{k,t+1}^w}{\partial W_{kt}} = -\frac{W_{k,t+1}}{W_{kt}^2},$$

so the first-order condition with respect to W_{kt} is

$$\int u_c(c_{it},h_{it})(1+\tau_L)(1-\tau_y)(1-\epsilon_t)y_{it}\frac{N_{kt}}{P_t^c}\,di - \nu'(n_t)\frac{\partial N_{kt}}{\partial W_{kt}} - \theta_w\pi_{kt}^w\frac{1}{W_{k,t-1}} + \beta_t\theta_w\pi_{k,t+1}^w\frac{W_{k,t+1}}{W_{kt}^2} = 0. \quad \text{(A.9)}$$

Symmetry and the aggregate equation. All unions face the same problem, so we impose the symmetric equilibrium $W_{kt}=W_t$, $N_{kt}=N_t$, and $\pi_{kt}^w=\pi_t^w$. Multiplying equation (A.9) by W_t, using $W_t/W_{t-1}=1+\pi_t^w$, $W_{t+1}/W_t=1+\pi_{t+1}^w$, the real wage $w_t=W_t/P_t^c$, and $\partial N_{kt}/\partial W_{kt} = -\epsilon_t N_t/W_t$, and rearranging yields

$$\pi_t^w(1+\pi_t^w) = \frac{\epsilon_t}{\theta_w}N_t\left(\nu'(N_t) - (1+\tau_L)(1-\tau_y)\frac{\epsilon_t-1}{\epsilon_t}\,w_t\int y_{it}\,u_c(c_{it},h_{it})\,di\right) + \beta_t\,\pi_{t+1}^w(1+\pi_{t+1}^w),$$

which is equation (13). Because consumption is conditional on the realized housing tenure choice, the integral is shorthand for the probability-weighted cross-sectional object

$$\int y_{it}\,u_c(c_{it},h_{it})\,di = \sum_s\sum_{h',h}\iint y\,u_c\big(c_t(a,h,y,s;h'),\,h\big)\,\mathbb{P}_t(h'\mid a,h,y,s)\,g_t(a,h,y,s)\,da\,dy,$$

where $c_t(a,h,y,s;h')$ is the consumption policy conditional on choosing housing state h' and $\mathbb{P}_t(h'\mid a,h,y,s)$ is the logit probability (A.1) of Appendix A.1.

A.4 Aggregation, Government Budget, and Goods-Market Clearing

We aggregate the household budget constraint and combine it with the government budget constraint to show that goods-market clearing follows from the other equilibrium conditions.

Aggregate objects. Aggregate mortgage debt $A_t^- < 0$, transaction-cost revenue $\mathcal{T}_t^h$, and the masses of households moving across housing states, $M_t(h, h')$, are defined in Section 2.7. Because the employment subsidy and its financing are both proportional to productivity and $n_t = N_t$, they cancel household by household, so the aggregate transfer $\int T_t(y)\, dg_t$ equals the lump-sum transfer $\overline{T}_t$.

Aggregate housing outlays. Each entry of the housing-cost function (6) is a change in the value of the housing position and a transaction cost on the units transacted. Summing over the transition masses, using $\sum_{h'} M_t(h, h') = \mu_t^h$ and $\sum_h M_t(h, h') = \mu_{t+1}^{h'}$, and recalling that the aggregate stock satisfies $H_t = \mu_t^1 \tilde{h}_2 + \mu_t^2 (\tilde{h}_1 + \tilde{h}_2)$, we can write aggregate housing outlays as

$$\sum_{h,h'} M_t(h, h')\, \psi_t(h, h') = p_t^h \left[H_{t+1} - (1 - \delta_h) H_t \right] + \mathcal{T}_t^h = p_t^h Y_t^h + \mathcal{T}_t^h, \tag{A.10}$$

where the second equality uses the housing law of motion (18).

Rental cash flows. From the budget constraint (3), the rental cash flow of a household in state (h, s) is $\widehat{p}_t^r(s)(h - 1)$. Because a lease is a contract between one renter and one landlord, renters and landlords coincide within each lease state, $\mu_t^{0,s} = \mu_t^{2,s}$ (Section 2.2), so aggregate rent received by landlords equals aggregate rent paid by renters,

$$\sum_s \left(\mu_t^{2,s} - \mu_t^{0,s} \right) \widehat{p}_t^r(s) = 0.$$

Aggregate budget. Integrating the budget constraint (3), which holds with equality at the optimum, using the housing-outlay identity (A.10), the unit mean of labor productivity, and $w_t N_t = Y_t^c$ from equations (14) and (15), gives

$$C_t + A_{t+1} + p_t^h Y_t^h + \mathcal{T}_t^h = (1 + r_t) A_t + \kappa A_t^- + (1 - \tau_y)\, Y_t^c + \overline{T}_t.$$

The government budget constraint (22) gives the transfer,

$$\overline{T}_t = B_{t+1} - (1 + r_t)\, B_t - G_t + \tau_y Y_t^c + \mathcal{T}_t^h - \kappa A_t^- + p_t^L \overline{L}.$$

Substituting it and imposing financial-market clearing (20) at t and $t+1$, the interest payments, the mortgage spread, the transaction costs, and the labor tax cancel, leaving

$$C_t + p_t^h Y_t^h = Y_t^c - G_t + p_t^L \overline{L}.$$

Construction input and land. The first-order conditions of the construction problem (17) imply $S_t = (1-\alpha_L)\, p_t^h Y_t^h$ and $p_t^L \overline{L} = \alpha_L\, p_t^h Y_t^h$, so that payments to the construction input and to land exhaust construction revenue,

$$p_t^h Y_t^h = S_t + p_t^L \overline{L}.$$

The land revenue accrues to the government, which returns it to households through the transfer. Substituting gives the goods-market clearing condition

$$Y_t^c = C_t + S_t + G_t. \tag{A.11}$$

Goods-market clearing therefore holds whenever households and the government satisfy their budget constraints and the housing and financial markets clear. In the stationary equilibrium, government purchases equal the revenue from land permits, transaction costs, and the labor tax (Section 3.1), so the transfer that balances the budget is $\overline{T}_{ss} = -r_{ss}\overline{B} - \kappa A_{ss}^-$: households receive the revenue from the mortgage spread and pay the interest bill on public debt. In the monetary experiments, $G_t = G_{ss}$, and τ_y, τ_h, and κ remain fixed, and the transfer follows the fiscal rule (23). Under the rental availability shock of Section 6.1.2, which raises the spread on landlord debt to κ_t^L, replace κA_t^- throughout by $\kappa A_t^- + (\kappa_t^L - \kappa) A_{t,L}^-$, where $A_{t,L}^- = \sum_s \iint a\, \mathbb{1}_{a<0}\, g_t(a,2,y,s)\, da\, dy$ is aggregate landlord mortgage debt; the landlord-specific spread revenue then enters the government budget.

B Rent Decomposition: Derivations and Additional Results

Section 4.3 decomposes the rent response by asking how each equilibrium input changes rental-market excess supply, the gap between the mass of landlords willing to supply a rental unit and the mass of households demanding one. This appendix formalizes that construction and reports the cumulative contributions. Recall that $h \in \{0,1,2\}$ indexes renters, owner-occupiers, and landlords. Given a path of household inputs $\mathbf{X}$, excess supply at $t+1$ is

$$\begin{aligned} \mathrm{ES}_{t+1}^r(\mathbf{X}) &= \mathcal{S}_{t+1}^r(\mathbf{X}) - \mathcal{D}_{t+1}^r(\mathbf{X}) \\ &= \sum_{s,h} \iint \left[\mathbb{P}_t(2 \mid a,h,y,s;\mathbf{X}) - \mathbb{P}_t(0 \mid a,h,y,s;\mathbf{X})\right] g_t(a,h,y,s;\mathbf{X})\, da\, dy. \end{aligned} \tag{B.1}$$

Thus, an input can change excess supply both by changing current tenure choices and by changing the distribution inherited from earlier choices. We keep both margins inside the economic channel generated by that input rather than reporting separate extensive and distribution channels.

The household block takes as inputs the paths

$$\mathbf{X}_t = \left\{ r_t^{\text{save}},\ r_t^{\text{debt,own}},\ r_t^{\text{debt,ll}},\ w_t N_t,\ \tau_t,\ p_t^r,\ \widetilde{p}_t^r,\ p_t^h,\ \xi_t \right\}. \tag{B.2}$$

Here τ_t denotes the path of taxes and transfers, $T_t(\cdot)$, $\widetilde{p}_t^r$ the pooled rent (the rents of younger leases are past values of p_t^r), and ξ_t the tenure-retention rates that govern lease survival. Retention rates move leases across lease states but not tenure choices, so they do not change excess supply. The three rates comove in equilibrium. We separate them only in the decomposition: r_t^{save} applies to positive financial positions, while $r_t^{\text{debt,own}}$ and $r_t^{\text{debt,ll}}$ apply to the mortgage debt of owner-occupiers and landlords, respectively. Let $\mathbf{X}^{ss}$ and $\mathbf{X}^{eq}$ denote the steady-state and equilibrium paths. For input j, let $\mathbb{1}_j$ select its path and define the one-input experiment

$$\mathbf{X}^j = \mathbf{X}^{ss} + \mathbb{1}_j(\mathbf{X}^{eq} - \mathbf{X}^{ss}), \qquad d^j\mathbf{ES}^r = \mathbf{ES}^r(\mathbf{X}^j) - \mathbf{ES}^r(\mathbf{X}^{ss}). \tag{B.3}$$

Result B.1 (Rental-market decomposition)**.** *To first order around the steady-state path, the equilibrium change in rental excess supply satisfies*

$$d\mathbf{ES}^r = d^{p^r,\widetilde{p}^r}\mathbf{ES}^r + \sum_{k=1}^{7} \chi^k + \epsilon^{ES}. \tag{B.4}$$

The seven indirect channels are

$$\chi^{\text{ll mortg.}} = d^{r^{\text{debt,ll}}}\mathbf{ES}^r, \qquad \chi^{\text{buyer mortg.}} = d^{r^{\text{debt,own}}}\mathbf{ES}^r,$$

$$\chi^{\text{portfolio},h} = d^{r^{\text{save}},h}\mathbf{ES}^r, \qquad h \in \{2,1,0\},$$

$$\chi^{\text{house price}} = d^{p^h}\mathbf{ES}^r, \qquad \chi^{\text{standard HANK}} = d^{wN}\mathbf{ES}^r + d^{\tau}\mathbf{ES}^r.$$

Here $d^{r^{\text{save}},h}\mathbf{ES}^r$ partitions the effect of the return on savings by the household's current tenure. The rental market clears in every quarter, so the market rent is pinned down date by date, and the rents of outstanding leases follow from it through (8) and (A.6). Let

$$J_p = \left.\frac{\partial \mathbf{ES}^r_{2:T}}{\partial \mathbf{p}^{r\prime}_{2:T}}\right|_{\mathbf{X}^{ss}}$$

denote the response of excess supply to the market-rent path, including its effect through the rents of outstanding leases. The rent contribution of channel k is therefore

$$d\mathbf{p}^{r,k}_{2:T} = -J_p^{-1}\chi^k_{2:T}, \qquad dp^{r,k}_1 = 0, \tag{B.5}$$

and $d\mathbf{p}^r = \sum_k d\mathbf{p}^{r,k} + \boldsymbol{\epsilon}^p$*, where* $\boldsymbol{\epsilon}^p$ *collects second-order terms.*

Derivation. Differentiating equation (B.1) attributes the first-order change in tenure probabilities and in the inherited distribution to the input path that generates them. First-order additivity across the one-input experiments gives equation (B.4). Rental-market clearing then requires the direct rent effect to offset the combined indirect pressure. Linearizing the clearing conditions with respect to the market-rent path gives $J_p\, d\mathbf{p}^{r,k}_{2:T} = -\boldsymbol{\chi}^k_{2:T}$; inverting J_p yields equation (B.5). □

The remaining discrepancy from exact additivity is reported as a first-order residual. The decomposition does not depend on the order in which the channels are introduced. Computed with the sequence-space Jacobian, as in Table B.1, each channel is linear in its input paths, so the channels add up exactly to the first-order response and coincide with a Shapley decomposition, which averages each channel's contribution over all orders of introduction. The one-input experiments in (B.3) evaluate each channel through the full nonlinear household block instead. Over quarters 1–24, they differ from the first-order channels by at most 0.04 percentage-point-quarters, and their joint residual, which collects all interactions among channels and hence any dependence on the order, is 0.9 percent of the gross excess-supply pressure over quarters 1–24 and 2.0 percent over quarters 1–100.

Cumulative contributions. Figure 5 plots the direct rent effect and the indirect channels, grouped into supply, demand, the house price, and standard HANK, in excess-supply units, and the cumulative contribution of each channel to the average rent. A negative bar in excess-supply units means that, at the steady-state rent, the channel contracts rental supply relative to demand and therefore places upward pressure on the clearing rent. Table B.1 reports the cumulative effects of each of the seven channels on excess supply and on the market rent. Because the rent mapping is dynamic, it uses the entire time profile of each excess-supply path, so the sign of a channel's cumulative excess-supply effect need not determine the sign of its cumulative rent contribution when the channel changes sign over the transition.

In rent-response units, which measure the market rent, the two landlord channels together contribute more than the standard-HANK channel over the first 24 quarters, 1.30 units against 0.96, and less over 100 quarters, 0.80 against 1.62; individually, the standard-HANK channel is the largest. The house-price response remains the main offset over both horizons. The realized cumulative rent response rises from 1.33 to 1.70 units.

C Numerical Solution

Household problem and stationary equilibrium. We solve the household problem on 287 asset nodes, evenly spaced between -20 and 45 times mean quarterly labor income, with a node at $a = 0$, and approximate the productivity process with a seven-state Rouwenhorst chain. Conditional on each tenure choice, we solve the consumption-saving problem with the endogenous grid method;

Channel	Notation	Excess supply (pp-quarters)	Rent contribution (rent-response units)
Rent (direct effect)	$d^{p^r}\mathbf{ES}^r$	1.06	—
Landlord mortgage cost	$\chi^{\text{ll mortg.}}$	−0.65	0.59
Homebuyer mortgage cost	$\chi^{\text{buyer mortg.}}$	−0.37	0.31
Portfolio effect on landlords	$\chi^{\text{portfolio,2}}$	−0.98	0.71
Portfolio effect on owners	$\chi^{\text{portfolio,1}}$	0.00	−0.06
Portfolio effect on renters	$\chi^{\text{portfolio,0}}$	−0.52	0.49
House price	$\chi^{\text{house price}}$	2.04	−1.68
Standard HANK	$\chi^{\text{standard HANK}}$	−0.58	0.96
First-order residual	ϵ^{ES}	0.00	0.00
Realized total	$d\mathbf{ES}^r$	0.00	1.33

Table B.1. Cumulative Rental-Market Decomposition, Quarters 1–24

Note. Excess supply is the landlord share minus the renter share. Its cumulative effects are measured in percentage-point-quarters (pp-quarters) of household mass. One rent-response unit is a one-percentage-point deviation of the market rent from p^r_{ss} for one quarter. The rent column reports the cumulative $d\mathbf{p}^{r,k}$ of (B.5); its first-order residual, ϵ^p, is 0.00 rent-response units, or 0.03 percent of the realized cumulative rent increase. In each column, the entries above the line sum to the realized total.

because the tenure choice and the borrowing constraints make the continuation value non-concave, we add the upper-envelope step of Iskhakov et al. (2017). The distribution is updated with a lottery on the asset grid (Young, 2010). No price depends on the lease state in the stationary equilibrium, so we solve the household problem on (a, h, y) and compute the split of renters and landlords across lease states as the fixed point of the lease transitions (A.5). The market-clearing conditions and the government budget hold in the stationary equilibrium up to residuals of order 10^{-12}.

Transitions. We compute the perfect-foresight transition after each shock over $T = 200$ quarters in sequence space. The unknowns are the paths of the house price, employment, nondurable inflation, the real interest rate, the market rent, the lump-sum transfer, the pooled rent, and the retention rates of renters and landlords in each lease state. The equations are the housing law of motion (18), the wage Phillips curve (13), the Taylor rule (24) with the Fisher equation (21), financial-market clearing (20), rental-market clearing, the fiscal rule (23), the pooled-rent recursion (A.6), and the consistency of the retention rates with household choices (A.3). Rental-market clearing holds exactly in every quarter: the market rent is an unknown at every date after the impact quarter, and no restriction is placed on the shape of its path.

We solve this system with a quasi-Newton method. The initial matrix is the sequence-space Jacobian at the stationary equilibrium, which combines the Jacobian of the household block, computed with the fake-news algorithm of Auclert et al. (2021) and central differences, with the derivatives of the remaining equilibrium conditions. Each accepted step updates the inverse matrix by Broyden's method, and a step is accepted only if it lowers the residual. The matrix affects only the speed of the solver: the equilibrium is the root of the nonlinear system, and we stop when the largest residual across all equations and dates falls below 10^{-10}. The monetary transitions of

Sections 4 and 5 converge in five iterations. The first-order decompositions of Appendix B and Section 5 use the same household Jacobian.

C.1 Welfare Computation

Consumption-equivalent units. We express welfare differences as consumption equivalents. Let Λ be the permanent proportional change in consumption under the benchmark rule ($\phi_r = 0$) that delivers the same aggregate utilitarian welfare (29) as the alternative rule. Split the benchmark objective into its consumption-utility part $\mathcal{W}_0^u = \sum_{t\geq 1} \widetilde{\beta}_t \int u(c_{it}, h_{it})\, di$ and the remainder $\mathcal{W}_0^n = \sum_{t\geq 1} \widetilde{\beta}_t \big[-\nu(N_t) - \frac{\theta_w}{2}(\pi_t^w)^2 \big]$, so that benchmark welfare is $\mathcal{W}_0 = \mathcal{W}_0^u + \mathcal{W}_0^n$, both evaluated at $\phi_r = 0$ and summed over the transition plus a steady-state continuation. Because $u(c, h)$ is CRRA in c, scaling every household's consumption by $(1 + \Lambda)$ multiplies $\mathcal{W}_0^u$ by $(1 + \Lambda)^{1-\gamma}$ and leaves $\mathcal{W}_0^n$ unchanged. The consumption equivalent of a rule with welfare $\mathcal{W}$ then solves $(1 + \Lambda)^{1-\gamma}\mathcal{W}_0^u + \mathcal{W}_0^n = \mathcal{W}$, that is, $\Lambda = \big[(\mathcal{W} - \mathcal{W}_0^n)/\mathcal{W}_0^u\big]^{1/(1-\gamma)} - 1$, and we report $10^4\Lambda$, in basis points.

The ϕ_r grids. We choose the rent coefficient by comparing welfare across policy rules that start from the same stationary equilibrium and differ only in ϕ_r. We evaluate $\phi_r \in \{0, 0.05, 0.10\}$ and continue in steps of 0.05 in the direction in which welfare rises, until it falls. We then search in steps of 0.01 around the best point, over $\phi_r \geq -0.50$. Figure C.1 reports the resulting welfare comparisons. Every transition used in Section 6 solves the equilibrium conditions with a largest residual below 10^{-10}.

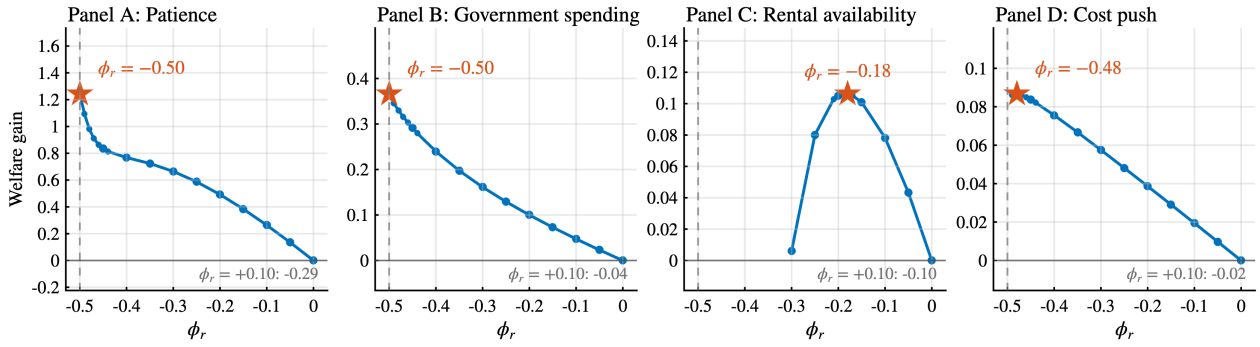


Figure C.1. Welfare Gains across Rent Coefficients

Note. Welfare gains relative to $\phi_r = 0$ under the same shock, with $\phi_\pi = 1.5$ throughout, in basis points of permanent consumption. Each point is a separately computed equilibrium transition, and the stars mark the optima. Welfare at $\phi_r = 0.10$ is reported by the annotations.

Distributional decomposition of welfare gains. Figure 7 splits the welfare change from moving to the optimized rule across initial cohorts and across the channels through which it reaches them. Consider a partition of the (a, h) state space into groups indexed by g, with masses m_g under the

distribution at the date of the shock. Tracking each group forward under the transition's own policies gives cohort welfare $\mathcal{W}_g(\phi_r)$, where the aggregate labor disutility and wage-adjustment cost in (29) are allocated across cohorts pro rata by mass and the steady-state continuation value by date-of-shock mass. The cohorts then add up, $\sum_g \mathcal{W}_g(\phi_r) = \mathcal{W}(\phi_r)$. Figure 7 reports the per-capita change $[\mathcal{W}_g(\phi_r^*) - \mathcal{W}_g(0)]/m_g$.

The channels are the one-input experiments of Appendix B, applied to welfare rather than to rental-market excess supply. Write $\mathcal{W}_g(\mathbf{X})$ for cohort g's welfare when the household block is solved at the input path $\mathbf{X}$ of (B.2), and let $\mathbf{X}^0$ and $\mathbf{X}^*$ be the equilibrium paths under $\phi_r = 0$ and $\phi_r = \phi_r^*$. With the selector $\mathbb{1}_j$ of (B.3), channel j's contribution to cohort g is

$$\Delta_g^j = \frac{\mathcal{W}_g(\mathbf{X}^0 + \mathbb{1}_j(\mathbf{X}^* - \mathbf{X}^0)) - \mathcal{W}_g(\mathbf{X}^0)}{m_g}.$$

The change in aggregate labor disutility and wage-adjustment costs is common to all cohorts and is reported as its own bar. The difference between the general-equilibrium total and the sum of the channel contributions is the interaction across channels: 16 percent of the total under the patience shock, and at most 1.3 percent under the other three shocks.

Welfare assessments (Dávila and Schaab, 2025). Figure 8 splits the marginal welfare effect of the rent coefficient into efficiency and redistribution. Let $V(a, h, y, s; \phi_r)$ denote the welfare of a household that starts the transition in state (a, h, y, s) under rule (28), measured as in (29): its expected discounted utility from consumption and housing, net of the labor disutility and wage-adjustment cost common to all households, plus a steady-state continuation. Aggregate welfare is $\mathcal{W}(\phi_r) = \sum_s \iint V(a, h, y, s; \phi_r)\, g_{ss}(a, h, y, s)\, da\, dy$. We express each household's gain in units of perpetual consumption, one unit of consumption at every date, which the household values at

$$\lambda(a, h, y, s) = \sum_{t \geq 1} \widetilde{\beta}_t \, \mathbb{E}_1 \left[u_c(c_{it}, h_{it}) \,\middle|\, (a_{i1}, h_{i1}, y_{i1}, s_{i1}) = (a, h, y, s) \right], \tag{C.1}$$

again with a steady-state continuation, evaluated along the transition at the assessed coefficient. Let $\bar{\lambda}$ be its population average and $\omega(a, h, y, s) = \lambda(a, h, y, s)/\bar{\lambda}$ the implied welfare weight, which averages to one. Writing $\mathrm{d}V^\lambda/\mathrm{d}\phi_r = \lambda^{-1}\, \mathrm{d}V/\mathrm{d}\phi_r$ for a household's willingness to pay for a higher coefficient, the marginal welfare effect in units of perpetual consumption splits into

$$\frac{1}{\bar{\lambda}} \frac{\mathrm{d}\mathcal{W}}{\mathrm{d}\phi_r} = \underbrace{\mathbb{E}_{ss}\left[\frac{\mathrm{d}V^\lambda(a, h, y, s; \phi_r)}{\mathrm{d}\phi_r} \right]}_{\text{efficiency}} + \underbrace{\mathbb{C}\mathrm{ov}_{ss}\left(\omega(a, h, y, s),\, \frac{\mathrm{d}V^\lambda(a, h, y, s; \phi_r)}{\mathrm{d}\phi_r} \right)}_{\text{redistribution}}, \tag{C.2}$$

where $\mathbb{E}_{ss}$ and $\mathbb{C}\mathrm{ov}_{ss}$ are taken over the stationary distribution g_{ss}. Efficiency adds up households' willingness to pay for a higher coefficient and does not depend on the planner's welfare weights. Redistribution is the covariance between the weights and that willingness to pay: it is positive when a higher coefficient favors the households that value consumption most at the margin, mostly

renters and levered owners.

We evaluate the derivatives in (C.2) by centered differences across neighboring points of the 0.05 grids, holding $\lambda(a, h, y, s)$ at its value at the assessed coefficient, and obtain household welfare and the numeraire (C.1) by integrating the stored household policies forward from $g_{ss}(a, h, y, s)$. The coefficients that efficiency alone favors, quoted in Section 6.3, are the zeros of its line in Figure 8, interpolated between grid points.

D Data

This appendix documents the data sources and construction of the calibration targets and untargeted moments. We use Spanish observations unless otherwise stated. Sample periods differ across sources according to availability; averages are computed over the observations specified below.

D.1 House Values and Rents

We construct the house-value-to-income ratio from the public statistical tables of the ECB Household Finance and Consumption Survey (HFCS), using the 2010, 2014, 2017, and 2021 waves. For each wave, we divide the mean value of the household's main residence among homeowners (HFCS statistical table B4) by mean annual gross household income across all households (HFCS statistical table I2). Both series are expressed in the same monetary units. The calibration target is the simple average of these four ratios, equal to 5.25. This is a ratio of group means, rather than an average of household-level value-to-income ratios.

In the model, the target applies to a reference dwelling of unit quality. The two housing qualities, 0.905 and 1.095, have midpoint one; this normalization does not impose a population-weighted mean quality of one. Because owner-occupiers live in the higher-quality dwelling (1.095), an owner-occupier's dwelling has a value of approximately 5.75 times mean annual income.

The level of rents is not targeted. We compare it with the data through the rental burden, the mean across renters of rent relative to the household's own gross income. In the EFF, 2014–2022 waves, we compute it for renters as defined in Appendix D.2, as annual rent paid (twelve times the monthly rent, `p2_31`) over gross household income (`renthog`), averaging the five implicates with cross-sectional weights and the four waves with equal weights. The mean burden is 35.0 percent. In the model, every renter's gross income is labor plus interest income, and the steady-state rent p^r_{ss} is 0.24 in units of mean quarterly labor income, which gives a mean burden of 30.4 percent. Renters' mean labor income is 0.83 of the population mean, against 0.74 for tenants in EU-SILC; the model does not target it.

The model reproduces the center of the burden distribution but not its upper tail. The median burden is 30.7 percent in the model and 24.1 percent in the EFF, where the mean exceeds the median by 11 points because of a tail of households whose rent is high relative to their income. The model's productivity process, whose lowest state is about two fifths of mean labor income,

does not generate that tail. The same compression makes aggregate rent a larger share of aggregate renter income in the model, 28.0 percent, than in the EFF, 20.4 percent, which is why the model's rent share of household spending exceeds Spain's HICP weight (Appendix D.8).

The EFF records the rent that tenants actually pay, as does EU-SILC, in which rent averaged 30.0 percent of tenant households' disposable income in Spain over 2004–2025 (Eurostat, `ilc_mded02`). Measures that assign each nonowner the local market rent are higher: Khametshin et al. (2024) report a mean burden of 38.7 percent in 2021 for households that do not own their main residence, valuing rent-free and reduced-rent accommodation at the average rent of its census tract. Our tenure classification excludes rent-free households.

D.2 Tenure Shares

We measure tenure shares in the 2014, 2017, 2020, and 2022 waves of the EFF, averaging the five implicates of each wave with cross-sectional weights and the four waves with equal weights. We classify a household as an owner-occupier if it owns its main residence (`p2_1`=2) and receives no rental income, as a landlord if it owns its main residence and receives income from renting real estate (`p7_1`=1 or `p7_2`>0), and as a renter if it does not own its main residence, receives no rental income, and pays rent (`p2_31`>0). Two groups have no counterpart in the model and are excluded: households that neither own their main residence nor pay rent, mostly in rent-free accommodation, which account for 7.2 percent of all households, and households that receive rental income without owning their main residence, which account for 1.7 percent.

Among the remaining 91.1 percent of households, renters account for 17.0 percent, owner-occupiers for 72.9 percent, and landlords for 10.0 percent. The renter share rises from 12.2 percent in 2014 to 20.6 percent in 2022. A landlord household may let more than one dwelling, so the landlord share counts households rather than rental units; Section 3.2 discusses why the model's landlord share equals its renter share.

D.3 Construction and Household Wealth

The residential investment share is gross fixed capital formation in dwellings divided by GDP at market prices. The numerator comes from Eurostat national accounts, dataset `nama_10_an6`, asset code `N111G`; the denominator comes from `nama_10_gdp`, item `B1GQ`. Both are annual current-price series in millions of euros. The average Spanish ratio over 1995–2025 is 6.9 percent. This measure covers investment in dwellings rather than total construction activity.

Using the same national accounts, we construct the nonresidential investment share as total gross fixed capital formation (`N11G`) less investment in dwellings (`N111G`), divided by GDP. The average over 1999–2025 is 15.6 percent. Comparing the residential and nonresidential shares helps assess the model's residential investment share when housing is its only produced illiquid asset.

Household wealth comes from the ECB Distributional Wealth Accounts (DWA). We sum the financial-asset components `F2M`, `F3`, `F5`, and `F62`, add housing wealth (`NUN`), and subtract liabilities.

This construction excludes nonfinancial business wealth and therefore differs from the DWA's broader published net-wealth aggregate. The income denominator is the four-quarter rolling sum of gross disposable income (B6G) from the ECB Quarterly Sector Accounts.

We average the quarterly wealth-to-income ratios over 2011Q4–2025Q4 for Spain and 2009Q1–2025Q4 for the euro area, obtaining 7.33 and 6.13, respectively. The corresponding housing-wealth-to-income ratios are 5.66 and 4.37. The wealth statistics cover households (S14), whereas the available income denominator also includes nonprofit institutions serving households (S1M).

The top-decile net-wealth share is taken from the HFCS 2021 wave. The Spanish hand-to-mouth benchmarks are taken from Kaplan et al. (2014), who distinguish households with little liquid wealth according to whether they hold illiquid assets.

Wealth by tenure. In the population we study, the average landlord holds 3.0 times mean net worth in Spain and 1.7 times in the model, and the average renter 0.14 and 0.27 times; relative to the median, landlords hold 3.4 against 1.7 times and renters 0.01 against 0.22 times. The model ranks the groups as the data do and places landlords even more at the top: 56 percent of them are in the top net-worth decile, against 40 percent in Spain, because its top decile consists mostly of landlords who own two dwellings. What the model compresses is the distance between landlords and tenants, which governs the gap in marginal utility that a utilitarian planner weighs.

Wealth concentration and hand-to-mouth households. Castellanos et al. (2024) show that allowing households to own a third property also improves the fit of the wealth distribution. Kaplan et al. (2014) document that 15.2 percent of Spanish households hold illiquid wealth but little liquid wealth, and only 4.4 percent hold neither. In the model, owner-occupiers and landlords who hold housing wealth but no positive net financial wealth are 16.0 percent of households, while renters holding no more than a quarter of mean annual income in financial wealth are 1.7 percent. The shortfall at the poor end is structural rather than a matter of calibration, since renters cannot borrow in our model. Essentially all wealthy hand-to-mouth households in Spain own housing, so the illiquid asset that defines the group is the one the model has.

D.4 Tenure-Transition Matrix

We estimate tenure transitions using the panel component of the EFF, linking households across the 2014–2017, 2017–2020, and 2020–2022 wave pairs. These pairs contain 3,634, 3,831, and 3,974 matched households, respectively, of which 3,300, 3,497, and 3,606 are renters, owner-occupiers, or landlords, as defined in Appendix D.2, in both waves. Households that live rent free or receive rental income without owning their main residence in either wave are dropped. We use all five imputed datasets supplied for each wave.

We build the matrix in three steps. First, for each wave pair j and each of the five imputed datasets, we weight matched households by the panel weight pesopan_1 (reported in the later wave and representative of the earlier wave's population), divide the weighted origin-to-destination

counts by the weighted count in each origin state, and average the five resulting matrices. This gives P_j, the transition matrix over the pair's horizon. Second, we convert each P_j into a quarterly matrix Q_j from its principal matrix root P_j^{1/q_j}, where q_j is the number of quarters between the two waves: $(q_1, q_2, q_3) = (12, 12, 8)$. The root has small negative entries, which we set to zero before renormalizing the rows. This step assumes a time-homogeneous quarterly Markov process between waves, so quarterly transitions are inferred rather than observed. Third, we average the three quarterly matrices row by row, weighting pair j's row h by n_{jh}, the unweighted number of matched households that start pair j in state h:

$$Q_{hk} = \frac{\sum_j n_{jh}(Q_j)_{hk}}{\sum_j n_{jh}}.$$

The survey observes tenure every two or three years. Converting each wave-pair matrix to quarterly frequency assumes that tenure follows a time-homogeneous Markov chain between waves. The principal quarterly root of each matrix has small negative entries, all in the cells linking renters and landlords and none larger than 0.04 percentage points in absolute value; we set them to zero and renormalize the rows. This step moves the targeted entries by less than a hundredth of a standard error, and the resulting quarterly matrices, raised back to the survey horizon, reproduce the observed matrices to within 0.16 standard errors outside the renter–landlord cells.

The resulting matrix describes transitions among continuing households. It excludes household formation and dissolution, so its implied stationary distribution need not reproduce cross-sectional tenure shares. Two of its entries, the flows from renting to owner-occupation and from landlord to owner-occupation, are calibration targets (Section 3.2); the remaining entries test the model's mobility across tenure states.

Table D.1 reports the nine entries of the pooled quarterly tenure-transition matrix in the data and in the model. Standard errors use the 1,000 replicate weights that the Banco de España distributes with the EFF panel weights. We recompute the full procedure, namely the matrix of each wave pair, its conversion to quarterly frequency, and the pooling, with each replicate weight, and combine the five implicates with Rubin's rules (Banco de España, 2021). The model reproduces every entry within sampling error except the flow from owner-occupation to renting, which it overstates, and hence the owner-occupier diagonal, which it understates. The flows between renting and being a landlord rest on 8 and 5 observed transitions; the model nearly rules them out and lies inside their percentile intervals.

A comparison at the survey horizon requires no conversion. We raise the model's quarterly matrix to the twelfth or eighth power and compare it with the observed matrix of each wave pair. Both sides then miss the moves that are reversed between waves, so this comparison is not affected by within-period churn. It confirms the misfit in the owner-occupier row. The flow from renting to owner-occupation fits in every wave pair, while the flow from owner-occupation to renting is overstated in every wave pair: over three years, 3.2 percent of owner-occupiers become renters in the model against 1.3 to 1.4 percent in the data. The flow from landlord to owner-occupation

is stable at about 1.8 percent per quarter between 2014 and 2020 but more than doubles, to 4.0 percent, between 2020 and 2022, a difference significant at the 1 percent level. The pooled target therefore averages over periods with different mobility.

Origin	Destination	Data	Model	Obs.
Renter	Renter	98.797 (0.144)	98.825	988
	Owner-occupier†	1.175 (0.145)	1.175	118
	Landlord	0.028 (0.022)	0.000	8
Owner-occupier	Renter	0.125 (0.021)	0.303	81
	Owner-occupier	99.265 (0.053)	99.038	6,218
	Landlord	0.610 (0.047)	0.659	643
Landlord	Renter	0.006 (0.011)	0.000	5
	Owner-occupier†	2.553 (0.240)	2.553	390
	Landlord	97.441 (0.240)	97.447	1,952

Notes: Quarterly transition probabilities in percent. Data: EFF panel 2014–2022 (three wave pairs), households that are renters, owner-occupiers, or owner-landlords in both waves of a pair, weighted by the longitudinal weight; principal matrix root of each pair with negative entries set to zero and rows renormalized, pooled by origin-state counts. Standard errors in parentheses from the 1,000 Banco de España panel replicate weights, combined across the five implicates with Rubin's rules. Obs. is the unweighted number of matched transitions summed over the three pairs. †Targeted in the calibration.

Table D.1. Quarterly Tenure Transitions: Data and Model

D.5 Lease Terms and Rental Turnover

Lease terms and rent updates. Article 9 of Ley 29/1994 de Arrendamientos Urbanos provides for a minimum term of five years when the landlord is an individual and seven years when it is a legal person, through mandatory extensions when the agreed term is shorter. These terms were introduced by Real Decreto-ley 7/2019. Article 18 permits rent updates at most annually when the contract provides for them. Temporary limits generally capped updates at 2 percent in 2022–2023 and 3 percent in 2024 (Real Decreto-ley 6/2022 and Ley 12/2023); from 2025, the INE reference index for annual rent updates, IRAV, applies to contracts within its scope. The model approximates these update provisions by consumer-price indexation, so a continuing lease retains its real signing rent; it does not reproduce each statutory cap.

EFF tenancy duration. We use the 2014, 2017, 2020, and 2022 waves of the EFF to measure how long market-rent tenants have lived in their current dwelling. The calculation selects households renting at market prices (`p2_1`=1 and `p2_30`=1), uses the reported year in which they began living in the dwelling (`p2_29`) and the interview year and month, and weights households by the cross-sectional weight `facine3`. Because the survey reports the start year rather than the start month, we approximate starts in earlier years by the middle of the year and starts in the interview year by the midpoint of the elapsed part of that year. We give each wave equal weight. This calculation uses the first of the five imputed datasets and gives a recent-mover share of 13.7 percent, rounded

to 14 percent in the main text, and a median tenancy duration of about four and a half years. Dwelling occupancy does not record a new contract signed with the same tenant, so this moment is a benchmark for lease turnover rather than an identical empirical counterpart.

Other turnover evidence. INE's *Índice de Precios de Vivienda en Alquiler*, table 59008, reports weights by contract age for dwellings let by individual landlords in the common tax territory. The new-contract category carries a weight of approximately 12 to 18 percent over 2021–2024. These are rent-expenditure weights, valued at the previous year's prices, rather than the fraction of dwellings with a new contract; we therefore do not treat them as a direct turnover target. The Observatori Metropolità de l'Habitatge de Barcelona's report *Els contractes de lloguer a la demarcació de Barcelona 2010–2023* (2025) finds that about a fifth of registered contracts ended each year in Barcelona city over 2010–2023. This local registry provides a higher-turnover benchmark.

Model counterpart. At the stationary equilibrium, an approximate quarterly lease-survival probability is $s \simeq \varphi\xi_{0,ss}\xi_{2,ss}$, implying a recent-lease share of approximately $1 - s^4$. With the calibrated retention rates, setting $\varphi = 1$ already implies approximately 14 percent of outstanding leases signed within the previous year, because tenure changes end leases. The choice $\varphi = 0.99$ also allows other terminations and generates a recent-lease share of 17.29 percent in the stationary lease-age distribution. We report this stock share, rather than the number of contracts signed during a year divided by the lease stock.

D.6 Mortgage Market

We measure the mortgage spread using monthly ECB interest-rate data. The mortgage rate covers new Spanish loans to households for house purchase with a floating rate or an initial rate-fixation period of up to one year (MIR series `M.ES.B.A2C.F.R.A.2250.EUR.N`). We subtract the monthly average one-year Euribor rate (FM series `M.U2.EUR.RT.MM.EURIBOR1YD_.HSTA`). Both rates are quoted in annual percent, so their difference is measured in percentage points.

The mean spread over May 2001–April 2026 is 1.38 percentage points. We set the mortgage spread κ to this mean at quarterly frequency, 0.00345, so that the mortgage rate exceeds the real rate on savings by 1.38 annualized percentage points at every date, as the margin over Euribor of a Spanish variable-rate mortgage does. The loan-to-value and loan-to-income limits and the amortization requirement are set as described in Section 3.1.

Owner-occupier debt. In the model, 9.9 percent of owner-occupiers are net debtors, against 40.7 percent in the EFF. Two thirds of the households that become owner-occupiers in the model are landlords who sell their rental unit and repay their mortgage when they sell, and buyers who were renters borrow about half the value of the house and often repay ahead of the minimum schedule.

D.7 Earnings Process Calibration

Persistence. We use the Swedish annual persistence estimate, $\rho_a = 0.8139$, of Flodén and Lindé (2001) as a benchmark for the time-varying productivity component. The wage autocorrelations reported by De la Roca (2014) for Spain decline slowly with the lag: 0.939, 0.908, 0.882, 0.861, 0.843, 0.828, 0.809 at one through seven years. To distinguish persistent individual heterogeneity from time-varying earnings risk, we fit two simple decompositions to these seven autocorrelations.

Without an independent transitory component, we fit

$$c_k = w + (1 - w)\rho^k,$$

obtaining $w = 0.782$ and $\rho = 0.765$. Allowing for an independent transitory component, we fit

$$c_k = a\left[w + (1 - w)\rho^k\right],$$

obtaining $a = 0.972$, $w = 0.739$, $\rho = 0.866$. In the second specification, the permanent share of total variance is $aw = 0.718$, while the independent transitory share is $1 - a = 0.028$. These illustrative fits place the annual persistence of the time-varying component between 0.77 and 0.87, around our benchmark. They are our calculations using the reported autocorrelations, rather than estimates reported by De la Roca or an independently identified decomposition.

Innovation dispersion and frequency conversion. Figure 4(c) of Arellano et al. (2022) reports a spread of approximately 0.52 between the 90th and 10th percentiles of men's residualized annual log-earnings changes in 2005–2006. We use this individual-earnings statistic as a benchmark for productivity risk.

For a stationary Gaussian AR(1), the corresponding spread of four-quarter endpoint changes is

$$2\Phi^{-1}(0.9)\sqrt{\frac{2\sigma_a^2}{1+\rho_a}},$$

where Φ is the standard normal distribution function. Equating this expression to 0.52 gives $\sigma_a^2 = 0.03733$. We convert to quarterly frequency by preserving persistence over a year and matching stationary variance:

$$\rho_y = \rho_a^{1/4}, \qquad \sigma_y^2 = \sigma_a^2 \frac{1 - \rho_y^2}{1 - \rho_a^2}.$$

These expressions give $\rho_y = 0.949823$ and $\sigma_y = 0.104016$. The calculation matches endpoint changes in the Gaussian process; it does not establish an exact match to changes in earnings aggregated over a year.

D.8 Actual Rents in the HICP

We obtain annual expenditure weights from Eurostat's HICP item-weight dataset (`prc_hicp_inw`), using actual rentals for housing (`CP041`). Published weights are expressed per thousand; we divide them by ten to report percentages. The euro-area series uses changing composition.

Spain's weight on actual rentals averages 3.1 percent over 2014–2022, the period of the EFF waves; Appendix F.7 reports the responses at this weight. Until 2022, Spain's HICP weights were built from the household budget survey, which places rents below their share in the national accounts, about 3.5 percent of household consumption excluding imputed rents (`nama_10_co3_p3`); since 2023, when the weights have been built from the national accounts, the weight has been about 3.6 percent. Figure D.1 reports the weight of actual rents in the euro-area HICP. Over 2000–2025, it averages approximately 6.3 percent, compared with 10.4 percent in Germany and 2.8 percent in Spain.

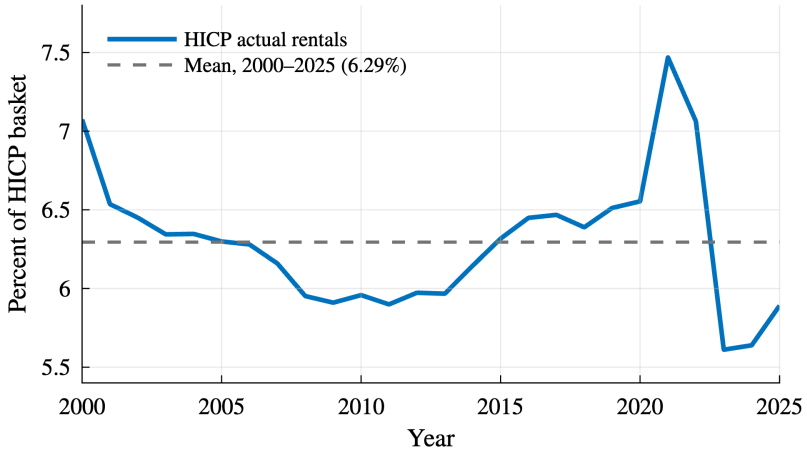


Figure D.1. Euro-Area HICP Weight of Actual Rents

Note. The figure reports the expenditure weight of actual rents paid by tenants in the euro-area HICP basket since 2000. The 2021 spike reflects pandemic reweighting, with no discrete change in rental-market coverage: an ECB Economic Bulletin box documents pandemic-induced shifts in HICP expenditure weights, with food and housing services increasing especially strongly (Gonçalves et al., 2021).

In the calibrated stationary equilibrium, rental expenditure represents

$$\omega_r^{\text{model}} = \frac{p_{ss}^r \mu_{ss}^0}{C_{ss} + p_{ss}^r \mu_{ss}^0} \simeq 5.3 \text{ percent}$$

of household spending on consumption and rent. This share follows from the calibrated rent and renter share, together with nondurable consumption, and it exceeds Spain's HICP weight because the model compresses the dispersion of rental burdens (Appendix D.1): at the EFF's aggregate burden of 20.4 percent, the model's share would be 3.9 percent. It describes the model's simplified consumption basket, whereas the empirical HICP weight reflects expenditure across a broader

basket. Section 5 uses the model's share, and Appendix F.7 also reports Spain's weight.

D.9 Shock Processes

Each shock in Section 6 is unanticipated and moves one exogenous variable along a first-order autoregressive path: its deviation from the steady-state value is $\sigma\rho^{t-1}$ in quarter $t \geq 1$. We set σ to one standard deviation of the innovation and ρ to its estimated persistence, and we give each shock the sign that raises rents. Table D.2 reports the values.

The patience shock is the domestic risk-premium shock of the ECB's New Area-Wide Model, which enters the bond Euler equation as our discount factor does, so its estimated standard deviation, 0.161 percent of β, and its persistence carry over without rescaling (Christoffel et al., 2008, Table 1). The government-spending process is an AR(1) on the log of Spanish real general-government consumption net of a quadratic trend, 1995Q1–2019Q4; one standard deviation is 0.72 percent of purchases, or 0.14 percent of GDP. The spread on landlords' debt follows the series to which we calibrate κ (Section 3.1): an AR(1) on its quarterly averages, 2003Q1–2026Q2, gives a persistence of 0.95 and a residual standard deviation of 22 annualized basis points. The cost-push shock lowers the elasticity across labor varieties, which raises the wage markup by 0.9 percent on impact. Its persistence is that of the New Area-Wide Model's wage-markup shock. Because that model normalizes the shock to its own wage Phillips curve, we set the size so that consumer-price inflation rises by 0.15 annualized percentage points on impact, the response to a one-standard-deviation wage-markup shock in Christoffel et al. (2008).

Shock	**Exogenous variable**	**Impact, σ**	**Persistence, ρ**
Patience	Discount factor, β_t	-0.0016	0.917
Government spending	Purchases, G_t / G_{ss}	$+0.72\%$	0.96
Rental availability	Spread on landlords' debt, κ_t^L (annualized)	$+22$ bp	0.95
Cost push	Log labor-variety elasticity, $\ln \epsilon_t$	-0.08	0.67

Table D.2. Shock Processes

Note. Each variable deviates from its steady-state value by $\sigma\rho^{t-1}$ in quarter $t \geq 1$. Sources: Christoffel et al. (2008), Table 1 (patience shock; persistence of the cost-push shock); Eurostat quarterly national accounts, `namq_10_gdp` (government spending); ECB Data Portal (spread); authors' calculations.

D.10 A Spanish Central Bank in a Monetary Union

Spain does not set its own interest rate. The ECB sets it in response to euro-area inflation, so the rule that Spanish households face is

$$i_t = r_{ss} + \phi_\pi \, \pi_t^{\mathrm{EA}} + \varepsilon_t^m, \tag{D.1}$$

whereas rule (24) responds to Spanish inflation. Euro-area inflation is a weighted average of inflation in Spain and in the rest of the euro area,

$$\pi_t^{\mathrm{EA}} = \vartheta\, \pi_t^{\mathrm{ES}} + (1-\vartheta)\, \pi_t^{\mathrm{xES}}, \tag{D.2}$$

where ϑ is Spain's weight in the euro-area HICP. The weight rose from 9.1 percent in 1999 to 12.8 percent in 2009 and stood at 11.9 percent in 2025, with an average of 11.4 percent over 1999–2025.[46] Substituting (D.2) into (D.1),

$$i_t = r_{ss} + \phi_\pi \vartheta\, \pi_t^{\mathrm{ES}} + \left[\phi_\pi (1-\vartheta)\, \pi_t^{\mathrm{xES}} + \varepsilon_t^m\right]. \tag{D.3}$$

Taken alone, the response to Spanish inflation is small: with $\phi_\pi = 1.5$, $\phi_\pi \vartheta \approx 0.17$. The bracketed term, however, moves with Spanish inflation whenever inflation in the rest of the euro area does. If $\pi_t^{\mathrm{xES}} = \pi_t^{\mathrm{ES}}$, the bracket folds into the response to Spanish inflation and (D.3) coincides with rule (24).

Inflation in Spain and in the rest of the euro area shares one cycle. Over 1999–2025, quarterly year-on-year HICP inflation in Spain and in the euro area excluding Spain has a correlation of 0.82. Writing the linear projection $\pi_t^{\mathrm{xES}} = a + b\, \pi_t^{\mathrm{ES}} + u_t$, rule (D.3) responds to Spanish inflation with the effective coefficient $\tilde{\phi}_\pi = \phi_\pi[\vartheta + (1-\vartheta)\, b]$; the constant a shifts the intercept, and the residual u_t, uncorrelated with Spanish inflation, acts as an additional policy shock. Over the same sample $b = 0.75$, so $\tilde{\phi}_\pi = 1.16$: seen from Spain, the ECB's rule satisfies the Taylor principle.

Figure D.2 shows the same co-movement in units of the policy rate. It plots the inflation term of the rule, $\phi_\pi \pi_t$ with $\phi_\pi = 1.5$, evaluated at Spanish and at euro-area inflation. The two prescriptions rise and fall together through the expansion before 2008, the disinflations of 2009 and 2014–2016, and the surge of 2021–2023, with a correlation of 0.86. The largest gap is in levels: before 2008, Spanish inflation averaged 3.2 percent against 2.2 percent in the euro area, which places the Spanish prescription about 1.6 percentage points higher, a difference absorbed by the intercept rather than by the dynamics. The argument concerns shocks common to the euro area, which move inflation in Spain and elsewhere together; the monetary shock of Section 4 is one by construction.

The MEDEA model of the Spanish economy faces the same modeling choice (Burriel et al., 2010). In a model of Spain alone in which the ECB responds to Spanish inflation only through Spain's weight, the indeterminacy region is "so large that the model becomes nearly useless." MEDEA therefore uses two limiting cases: a weight of one on Spanish aggregates in the rule, equivalent to an independent Spanish monetary policy, in its estimation over 1986–2007; and an

[46] Indices and country weights are from Eurostat (`prc_hicp_midx`, `prc_hicp_cow`) for 1999–2025, before the 2026 change in classification; the euro area has changing composition. We average monthly indices to quarters and compute year-on-year inflation. Inflation in the euro area excluding Spain follows from the chain-linked aggregation of the HICP: within each year, the euro-area index relative to the previous December is the weighted average of the member indices relative to that December, so the corresponding relative for the rest of the euro area is $(\mathrm{rel}_t^{\mathrm{EA}} - \vartheta\, \mathrm{rel}_t^{\mathrm{ES}})/(1-\vartheta)$. Rebuilding the official euro-area index from its members in the same way reproduces its inflation rate to within 0.014 percentage points on average.

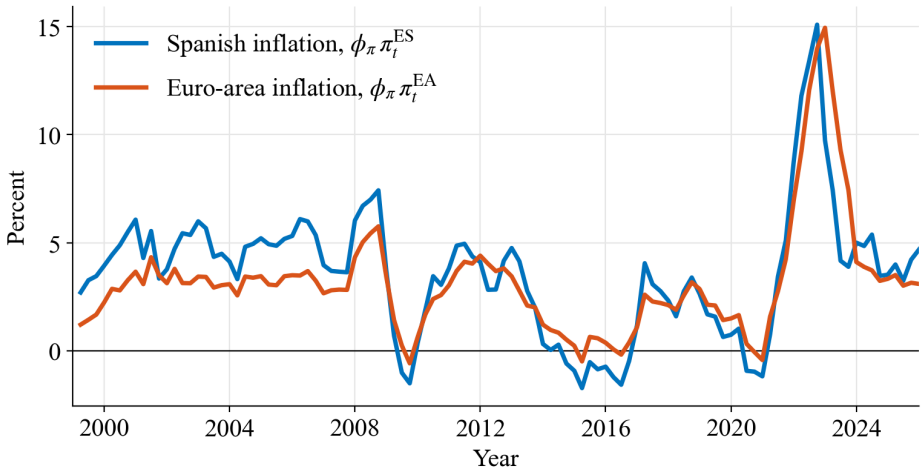


Figure D.2. Policy Prescriptions from Spanish and Euro-Area Inflation

Note. The figure plots $\phi_\pi \pi_t$ with $\phi_\pi = 1.5$, where π_t is year-on-year HICP inflation, in quarterly averages, in Spain and in the euro area (changing composition), 1999Q1–2025Q4. The correlation between the two series is 0.86. Source: Eurostat (`prc_hicp_midx`).

exogenous interest rate, on the grounds that Spain is too small to influence the ECB, in its estimation over 1997–2007. Rule (24) is the first case. The co-movement documented above connects the two: when inflation in the rest of the euro area moves with Spanish inflation, the euro-area rate that the second case takes as given is close to the rate a Spanish rule would set.

E Monetary Policy and Rents in Spain: Empirical Evidence

This appendix describes the vector autoregression behind Figure 4: the data, the identification of the monetary shock, the full set of responses, and their robustness.

E.1 Data

Table E.1 lists the series. The estimation sample is 2005:3–2019:12 ($T = 178$ months after six lags). The HICP excluding rents is built by chain-linking the all-items index and the index of actual rentals with Eurostat's item weights, and all price indices are seasonally adjusted. The sample ends in December 2019, as in Dias and Duarte (2019) and Corsetti et al. (2022): after 2020, most of the surprises around the hiking cycle are information shocks in the decomposition of Jarociński and Karadi (2020), and the period contains the pandemic, the energy shock, and the 2022–2024 cap on updates of existing Spanish leases.

Variable	Source	Units
ECB monetary surprise	EA-MPD (Altavilla et al., 2019)	basis points
One-year rate	ECB, euro-area AAA yield curve	percent
Industrial production	Eurostat	100 log, seasonally adjusted
HICP excluding rents	Eurostat	100 log, seasonally adjusted
HICP actual rentals	Eurostat	100 log, seasonally adjusted
Spain–Germany ten-year spread	ECB	percentage points

Table E.1. Data for the Spanish VAR

E.2 Identification and Estimation

The surprise is the change in the one-year overnight index swap rate in a window around each ECB announcement, aggregated to monthly frequency as in Gertler and Karadi (2015). Because the ECB sets its rate for the euro area as a whole, the shock is plausibly exogenous to Spanish conditions (Appendix D.10). We estimate the monthly VAR in levels

$$y_t = c + \sum_{\ell=1}^{6} A_\ell \, y_{t-\ell} + u_t, \qquad y_t = \left(z_t,\ i_t,\ ip_t,\ p_t,\ q_t,\ s_t\right)', \qquad u_t = B\,\varepsilon_t, \tag{E.1}$$

where z_t is the surprise, i_t the one-year rate, ip_t industrial production, p_t the HICP excluding rents, q_t the HICP index of actual rentals, s_t the spread, and B the lower-triangular Cholesky factor of the residual covariance matrix. The monetary shock is the innovation in the surprise, ordered first; using the surprise as an internal instrument identifies the effects of the shock without requiring the VAR to be invertible (Plagborg-Møller and Wolf, 2021). Responses are scaled so that the one-year rate rises by 25 basis points on impact, and real rents are $q_t - p_t$. The first-stage F-statistic of the one-year-rate residual on the surprise is 11.9. Information criteria select between one and eleven lags; we use six and report twelve below. Bands come from a residual moving-block bootstrap with 24-month blocks and 2,000 draws (Jentsch and Lunsford, 2019), and the figures report the median of the draws.

E.3 Responses

Figure E.1 reports every variable in the VAR. The one-year rate rises in a hump, peaking at 39 basis points after three months, and returns to zero after about seven. Industrial production falls by 1.3 percent after a year, with the 68 percent band below zero, and the spread does not move significantly. Prices excluding rents fall by 0.7 percent after a year, with the 90 percent band below zero from month 9 to month 35. The rents that tenants pay do not respond significantly at any horizon, so real rents rise by 0.6 percent after a year, with the 90 percent band above zero from month 9 to month 18. The shock accounts for a modest share of fluctuations, as monetary shocks typically do: about a tenth of the forecast-error variance of prices excluding rents and a twentieth of that of real rents at two years (Table E.2).

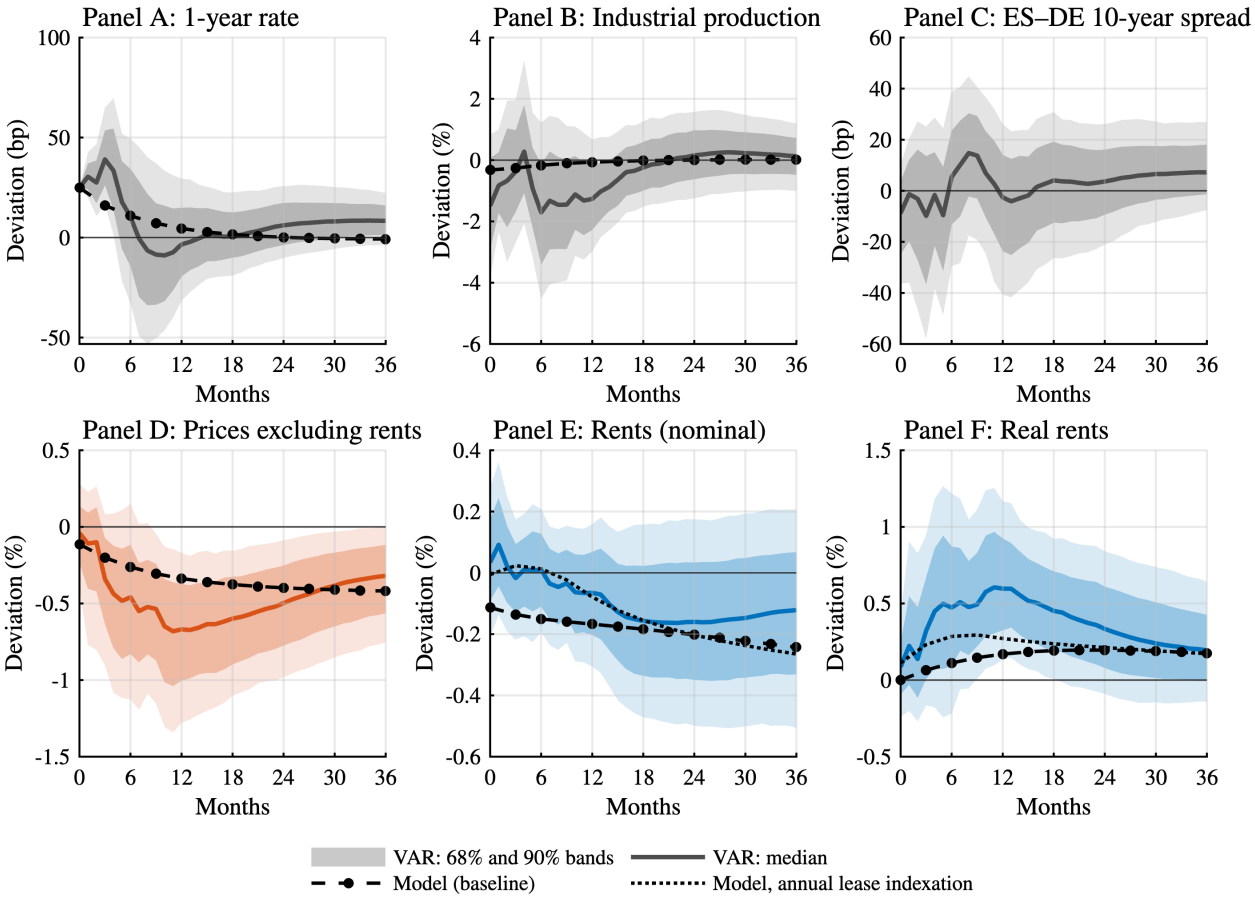


Figure E.1. Spanish VAR: All Responses

Note. Responses to a monetary tightening that raises the one-year rate by 25 basis points on impact; median of 2,000 bootstrap draws with 68 and 90 percent bands. Model lines as in Figure 4. The model's counterpart of industrial production is nondurable output Y_t^c, and that of the HICP rent index is the nominal average rent $P_t^c \overline{p}_t^r$. The spread has no model counterpart.

E.4 Robustness

Figure E.2 reports the median responses of alternative specifications against the baseline bands. The first row varies the specification: twelve lags, INE's CPI rent instead of the HICP index, no spread, and the spread together with oil prices. Real rents are higher after a year in all 24 specifications that combine six or twelve lags, the two rent indices, two price indices, and three sets of controls, and the results are unchanged when the HICP excluding rents is seasonally adjusted with month dummies instead.

The second row varies the identification and the sample. When output, prices, and rents are restricted not to respond within the month of the shock, prices fall by 0.5 percent and real rents rise by 0.5 percent after a year. Press-conference surprises can carry information about the economic outlook as well as about policy (Jarociński and Karadi, 2020); keeping only the announcements at which the swap rate and stock prices move in opposite directions gives larger responses of the

	12 months	24 months
One-year rate	0.04 [0.02, 0.21]	0.03 [0.02, 0.19]
Industrial production	0.02 [0.01, 0.15]	0.03 [0.01, 0.14]
Prices excluding rents	0.05 [0.01, 0.19]	0.11 [0.02, 0.23]
HICP rents	0.02 [0.01, 0.08]	0.03 [0.00, 0.12]
Real rents	0.03 [0.01, 0.18]	0.05 [0.02, 0.19]

Table E.2. Share of Forecast-Error Variance Due to the Monetary Shock

Note. Point estimates and bootstrap 90 percent bands.

same sign, but a weaker instrument ($F = 6.0$). The surprises of 2008 and 2011, the two tightening cycles in the sample, account for 70 percent of the variance of the instrument. Setting the surprises of any single year to zero leaves the signs unchanged, with real rents between 0.16 and 0.72 percent after a year; setting both to zero leaves too little variation to estimate the responses precisely ($F = 5.3$). Extending the sample to 2025, with monthly dummies for 2020:3–2021:7, gives the same signs with wider bands ($F = 8.9$).

E.5 Model and Evidence

In Figures 4 and E.1, the model's counterpart of the one-year rate is its one-year nominal yield, the average of the policy rates over the next four quarters, whose annualized deviation is $\sum_{k=0}^{3}(i_{t+k} - r_{ss})$. We scale the model so that this yield rises by 25 basis points in the impact quarter, as the estimated one-year rate does in the impact month, and plot model quarter t at month $3(t-1)$. The dotted line solves the model with the rents of continuing leases updated once a year by the previous year's inflation (Appendix F.10). Under this indexation the model's nominal average rent stays flat for three quarters and then declines with the price level, as the HICP rent index does. Appendix F.1 compares the model with estimates for the United States and the euro area on the same nominal scale.

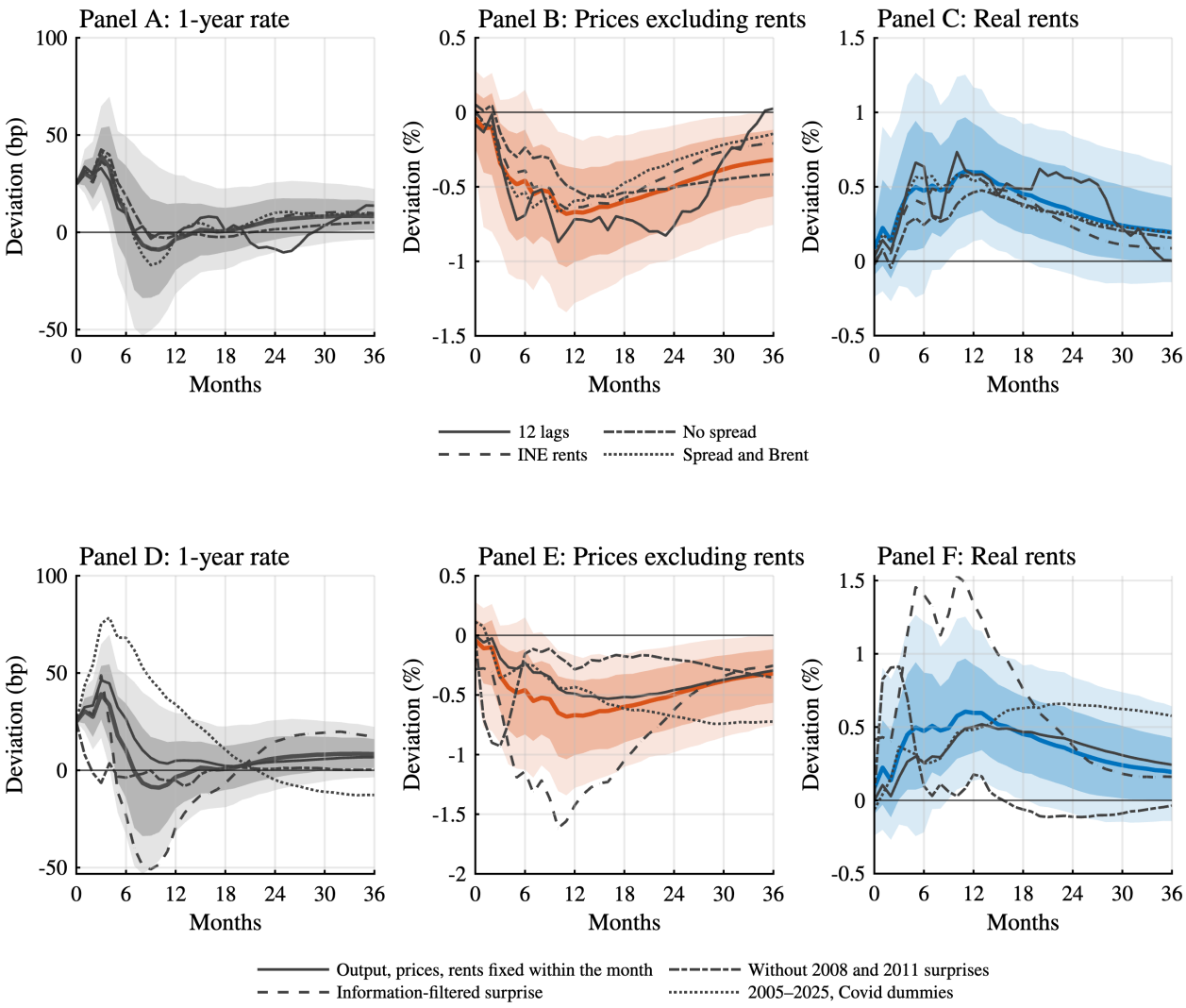


Figure E.2. Spanish VAR: Robustness

Note. Bands: baseline specification, 68 and 90 percent. Lines: bootstrap medians of the alternative specifications, each normalized to raise the one-year rate by 25 basis points on impact. Information-filtered surprise: announcements at which the one-year swap rate and the EURO STOXX 50 move in the same direction are set to zero.

F Additional Figures and Tables

F.1 Model and Evidence on a Common Scale

Table F.1 compares the model's responses to a monetary tightening with estimates for the United States and the euro area. Each study normalizes its shock on a nominal interest rate, and we rescale the model to the same rise in the corresponding nominal rate: its one-year nominal yield for Dias and Duarte (2019) and Abramson et al. (2025), as in Figure 4, and its policy rate on impact for Dias and Duarte (2022) and Corsetti et al. (2022). The one-year-yield targets are the first-quarter averages of the estimated one-year-rate responses, 0.31 and 0.34 percentage points; the latter converts Abramson et al.'s 25-basis-point mortgage-rate shock into a one-year-yield change using our estimate from local projections with instrumental variables.[47] The policy-rate target is 25 basis points. The model's 25-basis-point innovation raises the nominal one-year yield by 0.056 percentage points and the policy rate by 10 basis points on impact, so the scaling factors are 5.5, 6.1, and 2.6.

On this scale, both rents rise by less than the point estimates but lie within their 68 percent bands. Part of the remaining gap reflects lease contracts: the estimates are for the United States, where leases typically run twelve months, whereas a lease keeps its signing rent for five years in our calibration, so the market rent rises by less and passes into the average rent more slowly. The decline in homeownership is about half of the estimates and lies within both of their 68 percent bands. The homeownership response is limited by the housing stock: because each landlord lets a single, smaller unit, the renter share can rise only as the aggregate housing stock falls (equation (19)), and the stock falls by less than 0.003 percent over the transition. House prices fall on impact by slightly more than in the estimates, within both 68 percent bands, and recover within two years, whereas the estimated decline persists. Consumption falls by more than in the Spanish estimates of Corsetti et al. (2022) on impact and recovers within two years, whereas in the data it keeps falling for three to four years.

[47] The conversion estimates the response of the one-year Treasury yield, averaged over the first quarter, at 1.37 times the impact response of the 30-year fixed mortgage rate, using the surprises of Bauer and Swanson (2023) over 2011–2022. The first stage is weak ($F = 2.8$), so the reported bands do not capture uncertainty in this conversion.

	Horizon	Evidence	Model
Homeownership rate (pp)	quarters 8–12	-0.058 to -0.070	-0.035
Nominal house price (%)	impact	-0.18 to -0.28	-0.35
Real average rent (%)	mean, months 0–24	0.27 [0.12, 0.43]	0.17
Real market rent, new leases (%)	mean, months 0–24	1.04 [0.52, 1.56]	0.85
Private consumption, Spain (%)	impact	-0.08	-0.17
Private consumption, Spain (%)	quarters 12–15	about -0.55	0.01

Table F.1. Model and Evidence on a Common Scale

Note. Evidence: homeownership and house prices from Dias and Duarte (2022), point estimates under their two instruments; average rent from Dias and Duarte (2019), converted from their CPI deflator to prices excluding shelter, as in the model; new-lease rent from Abramson et al. (2025), built from listings; consumption from Corsetti et al. (2022), the Spanish response to a euro-area shock, read from their figure, which reports no bands for individual countries. Brackets give the averages over months 0–24 of the pointwise 68 percent bands. Model responses are rescaled as described in the text. The model's new-lease rent is the market rent p_t^r, the rent at which leases are signed.

F.2 Rents and Lease Length

Figure F.1 repeats the monetary experiment of Section 4.1 for lengths J of the period at the signing rent between one quarter and five years, holding φ and all other parameters at their calibrated values. Every lease pays the same rent in the stationary equilibrium, so J does not affect the steady state. It shapes the market rent. With $J = 1$, a new lease pays the market rent for a single quarter and the pooled rent afterward, so the market rent must move by more to clear the rental market: it rises by 1.68 percent in the second quarter and falls below its steady state from the seventh, and the average rent peaks at 0.15 percent in the fifth quarter and is back to 43 percent of its peak in the twelfth quarter. As J rises, a new lease commits both parties to its rent for longer, and the market rent rises by less and declines more slowly; its fall below steady state shrinks to 3.4 percent of its peak with $J = 8$ and disappears with $J = 20$. The average rent peaks later and lower, and its deviation in the twelfth quarter rises from 43 percent of its peak with $J = 1$ to 85 percent with $J = 8$ and 93 percent with $J = 20$ (Panel C). The aggregate responses are nearly identical: consumption falls by between 0.062 and 0.066 percent on impact for every J. The mean age of outstanding leases, and hence the mean lag of the rent index, changes by less than two quarters across J, from 20.5 quarters with $J = 1$ to 22.3 with $J = 20$: the persistence of the average rent comes from the path of the market rent.

The length J matters more for the rent-inclusive rule of Section 5, which responds to the path of the average rent. With $J = 1$, the average rent rises faster and returns sooner, so under the model's rent share the rule adds about seven times as much tightening over the first two years, a cumulative 10.4 against 1.4 annualized basis points on the policy rate, and gives it back from the eleventh quarter. Consumption then falls by about as much as under the benchmark rule over two years and by 11 percent less over five, whereas with $J = 20$ it falls by 3 percent more over two years and 1 percent more over five. For every J, the rule delivers less nondurable disinflation per unit of output lost.

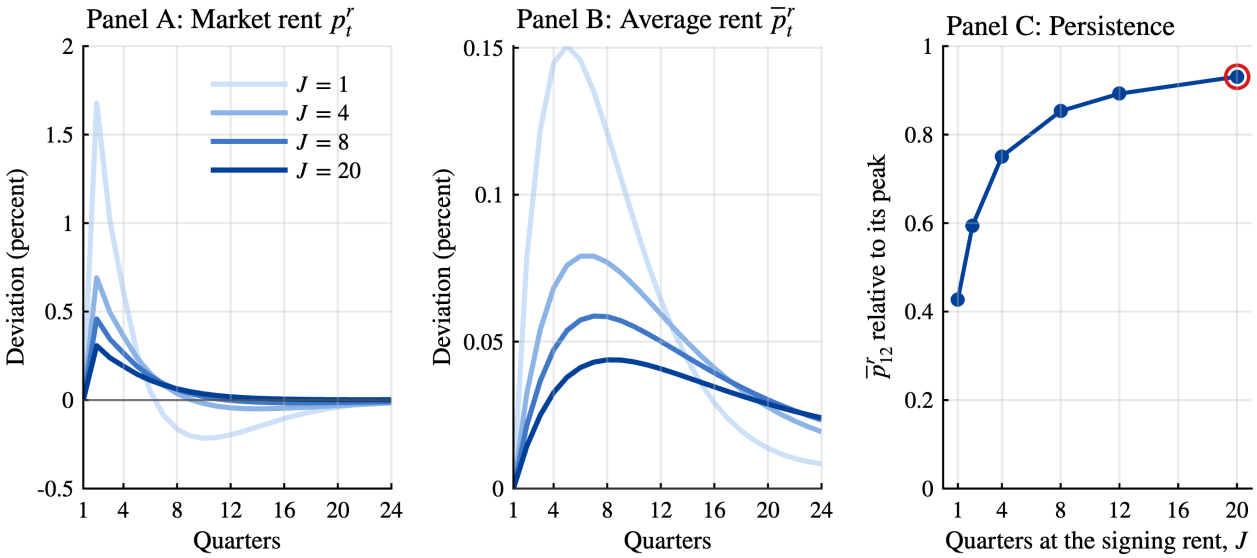


Figure F.1. Rents and the Length of the Period at the Signing Rent

Note. Responses of the real market rent p_t^r (Panel A) and the real average rent $\overline{p}_t^r$ (Panel B) to the monetary shock of Section 4.1, in percent deviations from steady state, for four values of J, the number of quarters during which a lease keeps the real rent at which it was signed. Panel C reports the average rent in quarter 12 relative to its peak for $J \in \{1, 2, 4, 8, 12, 20\}$; the circled point is the baseline, $J = 20$. All other parameters are held at their calibrated values.

F.3 Rents and the Rental Yield

Figure F.2 repeats the monetary experiment of Section 4.1 in economies calibrated to different gross rental yields. We replace the residential-investment target of Section 3.2 with a target for the annual yield, $4p_{ss}^r/(p_{ss}^h \tilde{h}_1)$, between 3.5 and 6.3 percent, and recalibrate all internal parameters to the remaining targets. Because the house-value-to-income ratio pins down the house price, the yield fixes the steady-state market rent. Depreciation falls from 0.0073 to 0.0011 per quarter as the yield falls from 6.3 to 3.5 percent, and residential investment from 10.9 to 1.6 percent of GDP. Lower yields also move the rental burden away from its Spanish value: at a yield of 3.8 percent it is 22.7 percent, against 30.4 percent in our calibration and 35 percent in the data.

The sign of the rent response changes with the yield. Above a yield of about 3.8 percent, a tightening raises both the market rent and the average rent, as in the baseline; below it, both fall. Over the first three years, the real average rent is on average 3.6 basis points above its steady state at a yield of 6.3 percent and 1.6 basis points below it at 3.5 percent. In the second quarter, the market rent goes from 40 basis points above steady state to 14 basis points below (Panels A and B).

The switch follows the response of the house price. Its fall on impact is approximately inversely proportional to the yield, from 8.6 basis points at a yield of 6.3 percent to 15.4 basis points at 3.5 percent (Panel C). The price–rent ratio is the inverse of the yield, so at low yields more of a dwelling's value comes from rents in the distant future, and the same rise in the real rate lowers its price by more. A larger fall makes housing cheaper for prospective landlords and homebuyers,

which strengthens the channel that offsets the rise in rents in Section 4.3. The adjustment runs through the price rather than through quantities. At lower yields fewer households move into renting: 0.4 basis points of households in the eighth quarter at a yield of 3.5 percent, against 1.4 at 6.3 percent. The housing stock is within 0.2 basis points of its steady state ten years after the shock in every calibration.

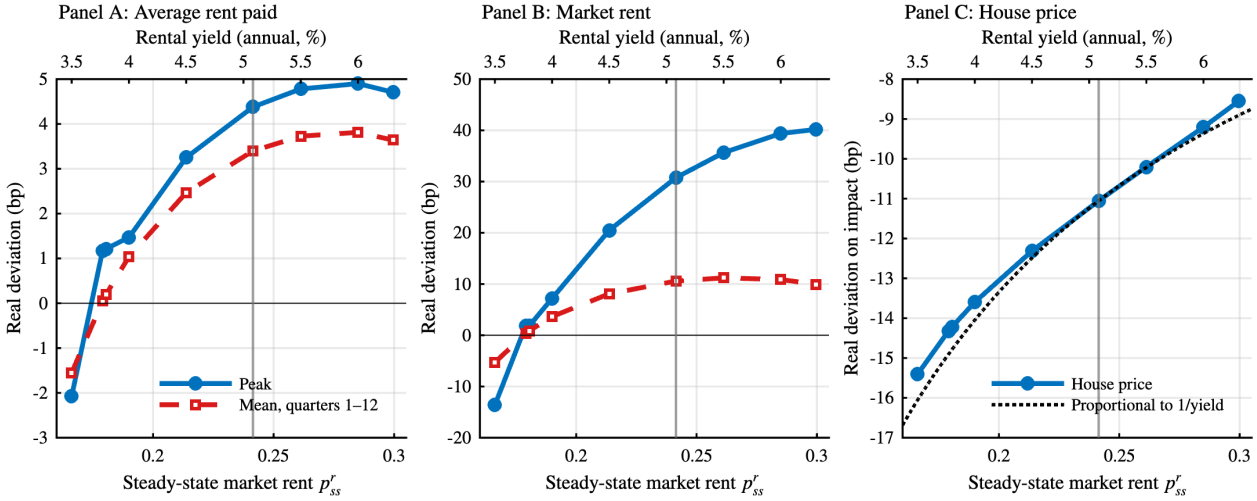


Figure F.2. Rents and the Rental Yield

Note. Responses to the monetary shock of Section 4.1 in economies recalibrated to the annual gross rental yield on the top axis, with residential investment untargeted. Panels A and B report the real average rent $\overline{p}^r_t$ and the real market rent p^r_t: the peak is the deviation largest in absolute value over the first ten years, and the mean is the average over quarters 1 to 12, in basis points. Panel C reports the real house price on impact, in basis points; the dotted line is proportional to the inverse of the yield and passes through the baseline. The vertical line marks the baseline calibration.

F.4 Rental-Supply Elasticity

When we perturb the market-rent path around the calibrated steady state, the model's rental-supply elasticity is 0.37 over the impact quarter and the subsequent three quarters, just below the short-run range of 0.4 to 0.7 discussed below, and 1.06 over the following four quarters, between the short- and long-run benchmarks and above the 0.81 implied by the estimated model of De Llanos (2026).[48]

The elasticity rises with the rental yield. In the recalibrated economies of Appendix F.3, it is 0.80 in quarters 5 to 8 at a yield of 3.5 percent and 1.22 at 6.3 percent. Because the renter share is a target, the number of additional landlords per unit increase in the rent falls slightly across these economies. The elasticity rises because the same increase is a smaller percentage of a higher rent.

Rotberg and Steinberg (2024) infer a long-run rental-supply elasticity of 1.4 from the share

[48]The elasticity is a partial-equilibrium object: we move the market rent with persistence 0.97, close to the 0.98 that De Llanos (2026) estimates for his housing-demand shifter, let the rents of outstanding leases follow (8) and (A.6), and hold other prices, labor income, taxes, and lease-retention rates at steady state. Quarters 5–8 are his year-one horizon.

of property taxes borne by renters in the United States, and obtain a short-run elasticity of 0.4 to 0.7 from the rise in rents in Warsaw and Kraków after the 2022 inflow of Ukrainian refugees documented by Trojanek and Głuszak (2022). The long-run estimate allows the aggregate housing stock to adjust; the short-run one refers to a horizon at which the stock is nearly fixed (their footnote 11). Our partial-equilibrium experiment measures the willingness to supply rental housing behind both estimates: it moves the market rent and records how many households choose to let a dwelling, holding other prices and incomes fixed.

A related object is the tenure-supply curve of Greenwald and Guren (2025), the price–rent ratios at which landlords supply different amounts of rental housing, which they estimate to be steep. In the model, this curve is also steep. The landlord margin gives it its slope, but the number of rental units can change only as construction changes the housing stock (equation (19)): after the monetary tightening of Section 4.1, the stock moves by less than 0.3 basis points while the house price falls by 11. A shift in rental demand therefore moves rents and house prices more than the number of rental units. The two objects play complementary roles. The partial-equilibrium elasticity measures how much the rent must rise to induce entry when holding a rental unit becomes more costly; the slow adjustment of the stock limits how much tenure reallocation the economy can absorb in the short run.

F.5 Tenure-Transition Responses

The tenure flows behind the rent channels of Section 4.3 are visible in the transition probabilities. Figure F.3 plots, for each current tenure state $h \in \{0, 1, 2\}$, the deviation $\mathbb{P}_t(h' \mid h) - \mathbb{P}_{ss}(h' \mid h)$ for next-period choices h', where $\mathbb{P}_t(h' \mid h)$ is the share of households in tenure state h at date t that choose h'. The homebuyer mortgage-cost channel appears in the renter panel: on impact, current renters are 0.03 percentage points more likely to remain renters and as much less likely to become owner-occupiers, so the tightening raises rental demand by keeping marginal renters in the rental market.[49]

The landlord channels appear in the owner and landlord panels. On impact, the probability that an owner-occupier becomes a landlord rises from 0.66 to 0.68 percent per quarter, and the probability that a landlord sells its rental unit rises from 2.55 to 2.60 percent. Owner-occupiers outnumber landlords almost four to one, so the additional entry more than offsets the additional exit, and the landlord share rises with the renter share, as clearing requires. Exits recede after the first year, and from the seventh quarter landlords are less likely to sell than in the stationary equilibrium.[50]

[49] The model holds the household unit fixed, so all adjustment runs through tenure choices. In the data, downturns can also reshape household composition: household size is countercyclical because young adults move back into, or delay leaving, the parental home (Dyrda et al., 2024).

[50] The discrete housing ladder shapes these flows: households cannot adjust the scale of a rental portfolio, so a landlord who reduces rental-property holdings must exit landlord status, and rental supply that existing landlords would otherwise provide by expanding their portfolios comes here through entry by new landlord households.

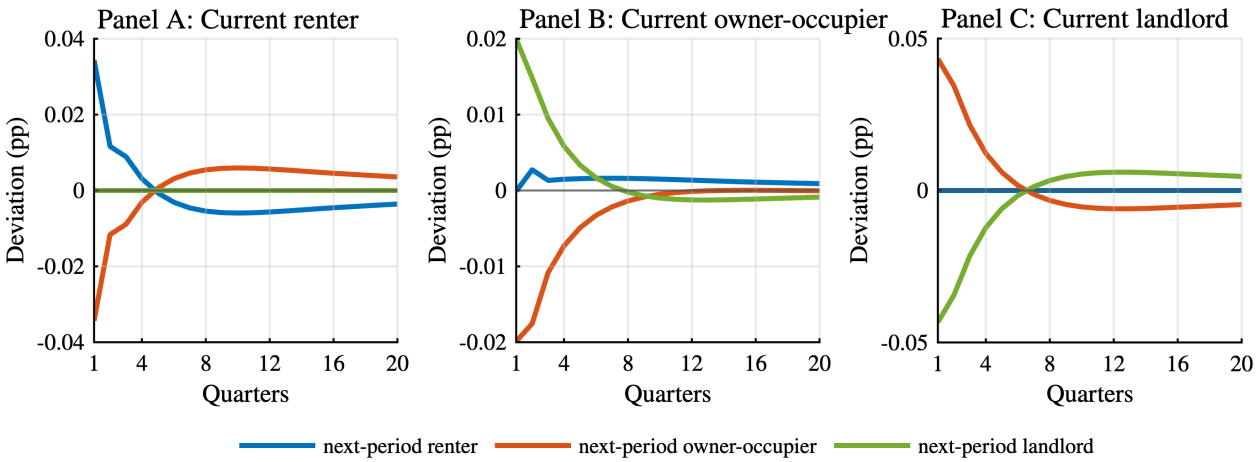


Figure F.3. Tenure-Transition Response to a Contractionary Monetary Shock

Note. Each panel conditions on the household's current tenure state and reports deviations in next-period tenure-choice probabilities from the stationary equilibrium, in percentage points. Conditional probabilities are computed by normalizing expected transition masses within each current tenure state.

F.6 Consumption Incidence of a Monetary Shock across Households

We fix each household group, defined below, at the date of the shock and follow its members forward as they move across tenure and balance-sheet states. We compare each initial cohort's per-capita consumption along the transition with that of the identical initial population propagated under steady-state policies. This comparison avoids conflating the consumption response of a given cohort with changes in the composition of contemporaneous tenure groups.

We report two partitions: borrowers and savers, defined by net financial assets; and renters, mortgagor-owners, unlevered owners, mortgagor-landlords, and unlevered landlords, defined by tenure and leverage. Table F.2 reports, for both partitions, the impact response and the discounted sum of all consumption deviations over the transition. Figure F.4 reports impact responses across net-financial-asset bins in Panel A and the consumption paths of three representative initial cohorts in Panel B.

Balance-sheet exposure. Mortgagor-owners experience the largest impact consumption contraction, about a third larger than that of unlevered owners, and their consumption falls further in the second quarter, when the higher debt service comes due; mortgagor-landlords contract slightly more than unlevered landlords.[51] Tenure also matters within net-financial-asset bins: renters holding small positive balances contract by more than the average borrower.

Mortgage repricing connects these consumption responses to the landlord mortgage-cost and homebuyer mortgage-cost channels of Section 4.3. Higher borrowing costs discourage the provision

[51]The impact response is anticipatory rather than contemporaneous. The real return on assets carried into the first period is set before the shock, so debt service at $t = 1$ remains at its steady-state rate. Indebted households adjust consumption in anticipation of higher debt service from the following quarter onward.

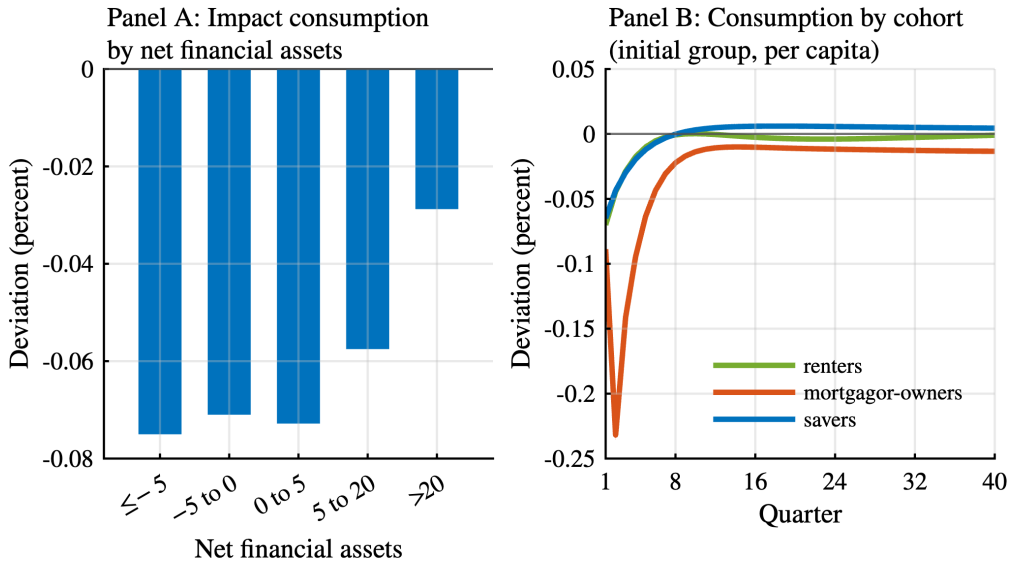


Figure F.4. Consumption Incidence of the Monetary Shock

Note. Panel A reports impact consumption responses by date-of-shock net-financial-asset bin (in units of quarterly steady-state income), in percent deviations from steady state. Panel B reports per-capita consumption paths for three representative initial cohorts (renters, mortgagor-owners, and savers), measured as percent deviations from each cohort's no-shock counterfactual; these span the exposure range but are not a partition, since savers ($a \geq 0$) include the renters and the unlevered owners and landlords.

of rental housing and entry into homeownership; they also reduce the resources available for consumption among households already carrying mortgage debt. The cohort responses reported here incorporate all equilibrium channels, rather than isolating these mortgage effects.

Rent exposure. Renters experience a smaller impact contraction than mortgagor-owners but a larger one than unlevered owners and landlords, and, unlike them, they consume less over the transition as a whole. Higher rental payments reduce the resources available for consumption and offset part of the benefit from higher returns on financial savings.

Lease stickiness determines when the higher rental bill arrives. Households signing new leases face the market rent, while renters on older leases keep paying the rent at which they signed (Section 2.2). The would-be buyers who remain renters therefore generally absorb the increase gradually. All rents remain at their steady-state values on impact, so the initial consumption response reflects changes in other equilibrium conditions and anticipation of future payments. As rents subsequently rise, rental exposure adds a burden that a classification based only on financial assets does not capture.

	SS share	Impact	Discounted sum
By net financial assets			
Borrowers ($a < 0$)	0.121	−0.073	−1.250
Savers ($a \geq 0$)	0.879	−0.065	0.117
By housing tenure			
Renters	0.170	−0.070	−0.134
Mortgagor-owners	0.066	−0.089	−1.593
Unlevered owners	0.594	−0.066	0.155
Mortgagor-landlords	0.056	−0.061	−0.968
Unlevered landlords	0.114	−0.057	0.177

Table F.2. Consumption Incidence of the Monetary Shock across Initial Cohorts

Note. Each cohort is fixed at the date of the shock and tracked forward; entries are per-capita consumption deviations from the cohort's no-shock counterfactual, in percent. *Impact* is the t=1 response. *Discounted sum* adds all deviations over the first 100 quarters, discounted at the steady-state real rate; a positive value means that the cohort consumes more over the transition than in the counterfactual. Owner-occupiers and landlords are split by net financial assets into mortgagor ($a < 0$) and unlevered ($a \geq 0$) groups. Renters are not split by net financial assets: they cannot borrow (the third case of (5) imposes $a' \geq 0$ whenever $h' = 0$), so every renter is a saver. SS share is the steady-state population share.

F.7 Rent-Inclusive Targets: Responses

Figure F.5 compares the responses to the monetary shock of Section 4.1 under the benchmark rule and under the model's rent share. Figure F.6 does the same for the composites with owner-occupied housing of Section 5.3. Table F.4 reports the impact, cumulative, and root-mean-square responses under the benchmark rule, the model's rent share, and the two composites; Table 4 in Section 5.2 summarizes them relative to the benchmark rule.

		Actual rentals		Owner-occupied housing	
	Benchmark (no rents)	Spain's weight	Model's share	Rental equivalence	Net acquisitions
First two years (quarters 1–8)					
Real interest rate	+	+1.3	+2.3	+8.3	−0.2
Real house price	−	−1.8	−3.1	−11.5	−8.6
Real average rent	+	+1.9	+3.4	+12.5	+14.4
Consumption	−	−1.6	−2.7	−9.9	−9.4
Nondurable inflation	−	0.0	0.0	+0.3	−5.6
First five years (quarters 1–20)					
Real interest rate	+	+1.3	+2.3	+8.4	+0.6
Real house price	−	−1.4	−2.4	−8.8	−5.5
Real average rent	+	+1.8	+3.1	+11.5	+15.5
Consumption	−	−0.8	−1.4	−5.0	−8.6
Nondurable inflation	−	+1.5	+2.7	+10.7	−1.0

Table F.3. Rent-Inclusive Targets: Responses Relative to the Benchmark Rule, Both Weights and Horizons

Note. Responses to the monetary policy shock of Section 4.1, cumulated over quarters 1–8 and 1–20. The first column gives the direction of the cumulative deviation from steady state under the benchmark rule, whose index excludes rents ($\omega_r = 0$), the same at both horizons. Each other entry is the difference between the cumulative deviation under the rule in the column and under the benchmark rule, in percent of the benchmark's absolute cumulative deviation; a negative entry means the variable is lower than under the benchmark rule. Actual rentals enter the index at Spain's HICP weight, 3.1 percent, or at the model's rent share, 5.3 percent. Rental equivalence weighs owner-occupied housing by 13 percent and net acquisitions by 9 percent, with actual rentals at the model's share scaled by $1 - \omega_o$. Table 4 in Section 5.2 reports the first two years.

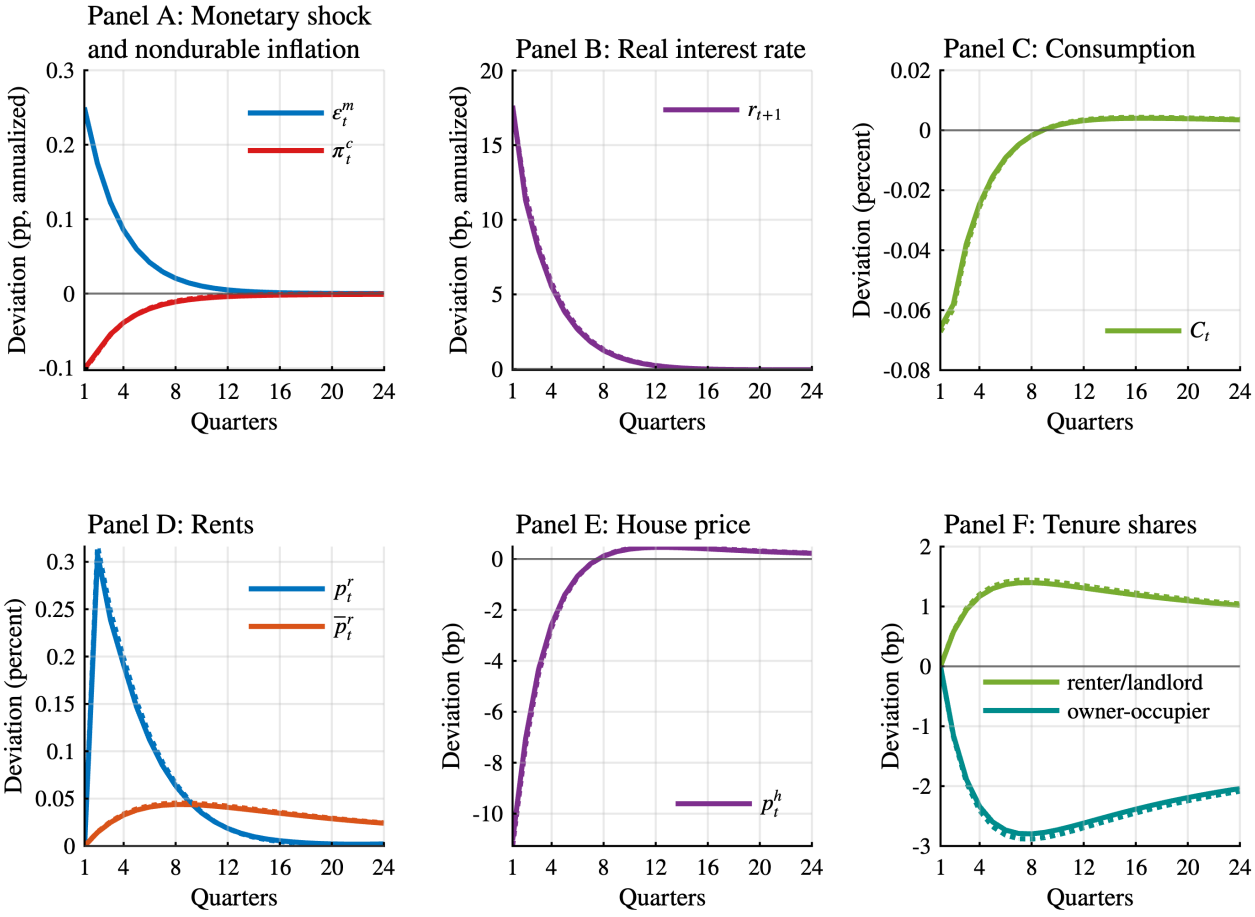


Figure F.5. Monetary Policy Shock under Alternative Policy Rules: Model's Rent Share

Note. Responses under $\omega_r = 0$ (solid lines) and $\omega_r = 0.053$, the model's rent share (dotted lines). Panel A reports the monetary policy innovation and nondurable inflation in annualized percentage points; Panel B reports the real interest rate in annualized basis points; Panel C reports consumption in percent deviations from steady state. Panel D reports real market and average rents in percent deviations; Panel E reports house-price deviations in basis points; Panel F reports tenure-share deviations in basis points.

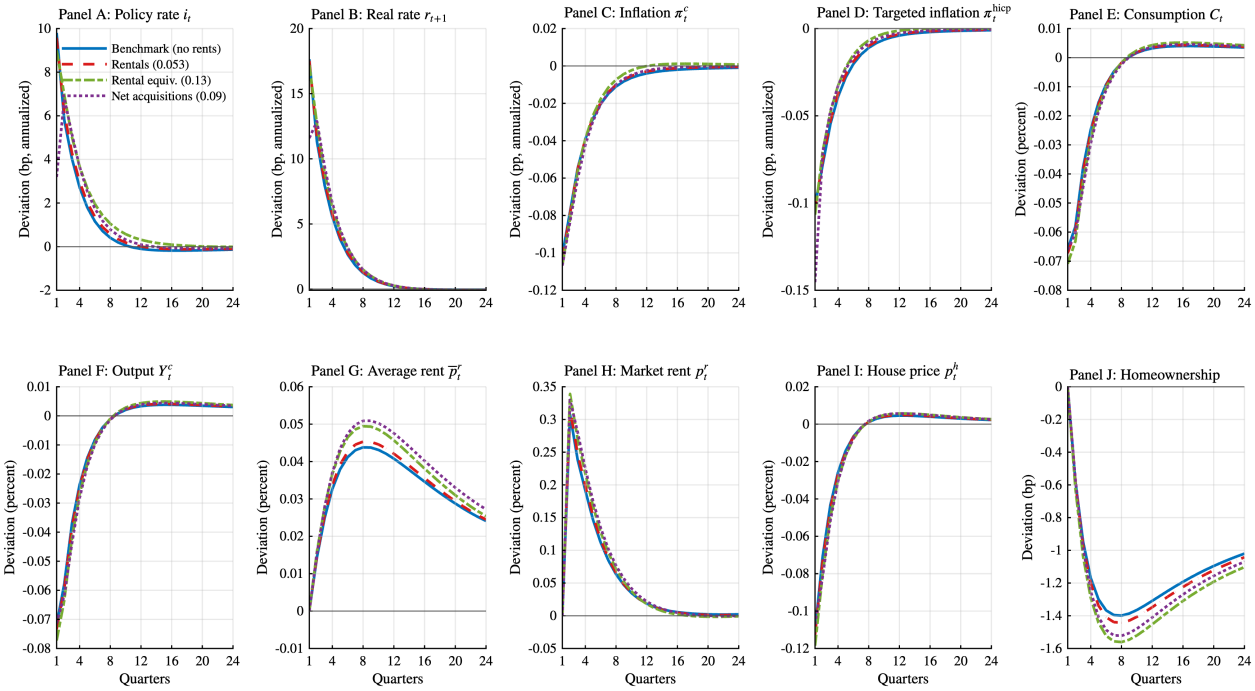


Figure F.6. Owner-Occupied Housing in the Inflation Target

Note. Responses to a 25-basis-point annualized monetary shock under the benchmark rule, the model's rent share on actual rentals ($\omega_r = 0.053$), rental equivalence ($\omega_o = 0.13$), and net acquisitions ($\omega_o = 0.09$), with the composite index of Section 5.3 and actual rentals at the model's share scaled by $1 - \omega_o$. Interest and inflation rates in annualized units; quantities and prices in percent deviations from steady state; homeownership in basis points.

	Rule	Impact	Cumulative		Root mean square	
			Q1–8	Q1–20	Q1–8	Q1–20
Consumption C (%)	Benchmark (no rents)	−0.066	−0.218	−0.179	0.036	0.023
	Rentals (0.053)	−0.067	−0.224	−0.182	0.036	0.023
	Rental equiv. (0.13)	−0.071	−0.239	−0.188	0.039	0.025
	Net acquisitions (0.09)	−0.066	−0.238	−0.194	0.038	0.024
Nondurable output Y^c (%)	Benchmark (no rents)	−0.072	−0.218	−0.180	0.037	0.023
	Rentals (0.053)	−0.073	−0.224	−0.183	0.038	0.024
	Rental equiv. (0.13)	−0.077	−0.241	−0.191	0.040	0.026
	Net acquisitions (0.09)	−0.071	−0.238	−0.194	0.039	0.025
Nondurable inflation π^c (pp, annualized)	Benchmark (no rents)	−0.101	−0.347	−0.387	0.053	0.034
	Rentals (0.053)	−0.103	−0.348	−0.377	0.053	0.034
	Rental equiv. (0.13)	−0.107	−0.346	−0.346	0.055	0.035
	Net acquisitions (0.09)	−0.107	−0.367	−0.391	0.056	0.036
Real rate r_{t+1} (bp, annualized)	Benchmark (no rents)	17.6	52.0	54.0	8.4	5.3
	Rentals (0.053)	17.5	53.2	55.3	8.5	5.4
	Rental equiv. (0.13)	17.2	56.3	58.6	8.7	5.5
	Net acquisitions (0.09)	11.6	51.9	54.4	7.7	4.9
Average rent $\overline{p}^r$ (%)	Benchmark (no rents)	0.000	0.238	0.679	0.033	0.036
	Rentals (0.053)	0.000	0.246	0.700	0.034	0.037
	Rental equiv. (0.13)	0.000	0.268	0.757	0.037	0.040
	Net acquisitions (0.09)	0.000	0.272	0.785	0.038	0.041

Table F.4. Owner-Occupied Housing in the Inflation Target

Note. Responses to a 25-basis-point annualized monetary shock, in deviations from steady state, under the benchmark rule, a rule on actual rentals at the model's rent share (0.053), and HICP-plus-owner-occupied-housing composites in which owner-occupied housing has weight ω_o and every other item, actual rentals at the model's share included, is scaled by $1 - \omega_o$: rental equivalence ($\omega_o = 0.13$, priced with the average rent) and net acquisitions ($\omega_o = 0.09$, the nominal house price). Cumulative entries sum the quarterly deviations; the root mean square is taken over deviations from steady state.

F.8 The Cost of the Rent-Inclusive Rule across Weights

Figure F.7 extends Table 4 to rent weights $\omega_r \in \{0, 0.01, \ldots, 0.15\}$, holding the economy and the monetary policy shock fixed. The responses of the real interest rate, the house price, the average rent and consumption move monotonically with the weight, and almost in proportion to it. At $\omega_r = 0.15$, over the first two years the rise in the real interest rate, the fall in house prices, the rise in the average rent and the fall in consumption are 7, 10, 10 and 8 percent larger than under the benchmark rule. Nondurable inflation over the same two years is essentially unchanged at every weight. Over five years, the rule delivers less disinflation as the weight rises, 9 percent less at $\omega_r = 0.15$. The costs of responding to rents therefore scale with the weight on rents, and the disinflation they buy does not.

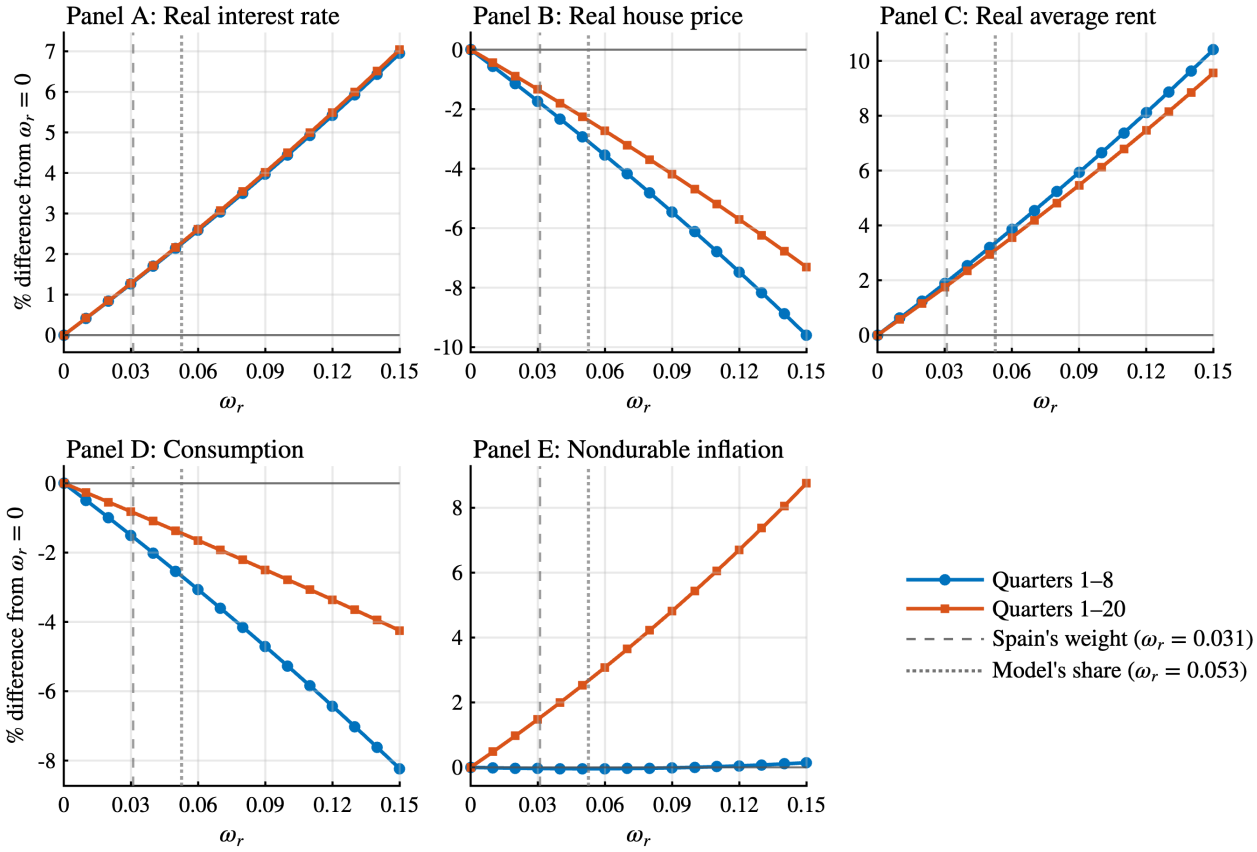


Figure F.7. Rent-Inclusive Targets across Rent Weights

Note. Responses to the monetary policy shock of Section 4.1. Each panel reports the difference between the cumulative deviation from steady state under weight ω_r and under the benchmark rule ($\omega_r = 0$), in percent of the benchmark's absolute cumulative deviation, over quarters 1–8 and 1–20, as in Table 4; a negative value means the variable is lower than under the benchmark rule. The dashed and dotted lines mark Spain's HICP weight and the model's rent share.

		Benchmark rule		Additional, $\omega_r = 0.031$		Additional, $\omega_r = 0.053$	
	SS share	Impact	Cumulative	Impact	Cumulative	Impact	Cumulative
Borrowers ($a < 0$)	0.121	−0.073	−1.474	−0.0009	−0.0162	−0.0016	−0.0278
Savers ($a \geq 0$)	0.879	−0.065	−0.180	−0.0008	−0.0030	−0.0013	−0.0052
Renters	0.170	−0.070	−0.258	−0.0008	−0.0017	−0.0014	−0.0028
Mortgagor-owners	0.066	−0.089	−1.862	−0.0011	−0.0208	−0.0019	−0.0357
Unlevered owners	0.594	−0.066	−0.194	−0.0008	−0.0033	−0.0014	−0.0058
Mortgagor-landlords	0.056	−0.061	−1.151	−0.0008	−0.0124	−0.0013	−0.0213
Unlevered landlords	0.114	−0.057	−0.133	−0.0007	−0.0023	−0.0011	−0.0040

Table F.5. Consumption Incidence of Rent-Inclusive Rules across Initial Cohorts

Note. Cohorts are fixed at the date of the 25-basis-point monetary shock, as in Table F.2. Entries are per-capita consumption deviations from the cohort's no-shock counterfactual, in percent. *Cumulative* sums negative deviations over the first 100 quarters. *Additional* columns report the difference between the rent-inclusive rule and the benchmark rule; a negative entry is a larger loss.

F.9 Distributional Incidence of the Rent-Inclusive Rule

Table F.5 applies the fixed-cohort method of Appendix F.6 to the two monetary policy rules. The last four columns measure the additional consumption loss generated by including rent inflation in the policy index, at each of the two weights. The impact contraction increases for every cohort. Although the real rate chosen on impact is slightly lower under the rent-inclusive rule, the return on assets carried into that period is predetermined. Households instead adjust to the anticipated path: indebted households face higher future debt service, while renters anticipate higher rental payments.

The additional cumulative contraction is largest for mortgagor-owners. Measured relative to each cohort's no-shock consumption, it is about 12 times that of renters. Mortgage exposure therefore remains central to the incidence of a rule that responds to rent inflation. Renters also lose through higher rental payments and the broader contraction. The cumulative loss increases for every tenure cohort. These consumption comparisons describe exposure to the policy-induced equilibrium adjustment; welfare additionally depends on households' marginal utilities.

F.10 Nominal Lease Indexation

In the baseline, a lease keeps the real rent at which it was signed: its nominal rent is updated every quarter by current consumer-price inflation (Section 2.2). Spanish leases are instead updated once a year, on the anniversary of the contract, by the change in consumer prices over the previous year. We consider three alternatives that index continuing leases to past inflation: a quarterly update by the previous quarter's inflation; an annual update, on each anniversary, by the change in consumer prices over the four quarters ending in the previous quarter; and nominal rents fixed for the life of the lease. A new lease pays the market rent in every case, and the stationary equilibrium is unchanged, since steady-state inflation is zero.

Under the three alternatives, the rents of continuing leases do not respond to inflation in the impact quarter. The tightening lowers prices, so their real value rises by the fall in the price level, and measured rent inflation reflects only the new leases, 4.7 percent of the total: it is -0.005 annualized percentage points on impact, against -0.10 for nondurable inflation. Remark 1 therefore no longer holds. The policy index exceeds nondurable inflation on impact by 0.3 basis points under Spain's weight and 0.5 under the model's rent share, and the rent term raises the policy rate already on impact, by 0.6 to 0.7 basis points under the model's rent share, rather than from the second quarter.

Figure F.8 shows that the real average rent rises on impact and peaks higher and earlier than in the baseline. Under annual updates the increase is shorter-lived, because each anniversary passes the lower price level into the nominal rent. With nominal rents fixed for the life of the lease, the market rent rises by half as much, because a lease whose nominal rent is fixed gains real value as prices keep falling, so a lower rent at signing delivers the same real rent over the life of the contract. Table F.6 repeats Table 4 under each indexation. Under both weights on actual rentals, the additional tightening stays below one basis point, house prices fall by more than under the benchmark rule, and the rent term buys no additional disinflation over two years. The consumption differences are smaller than in the baseline when leases are indexed to past inflation, and larger over two years when nominal rents are fixed; under annual updates, consumption falls slightly less than under the benchmark rule over five years.

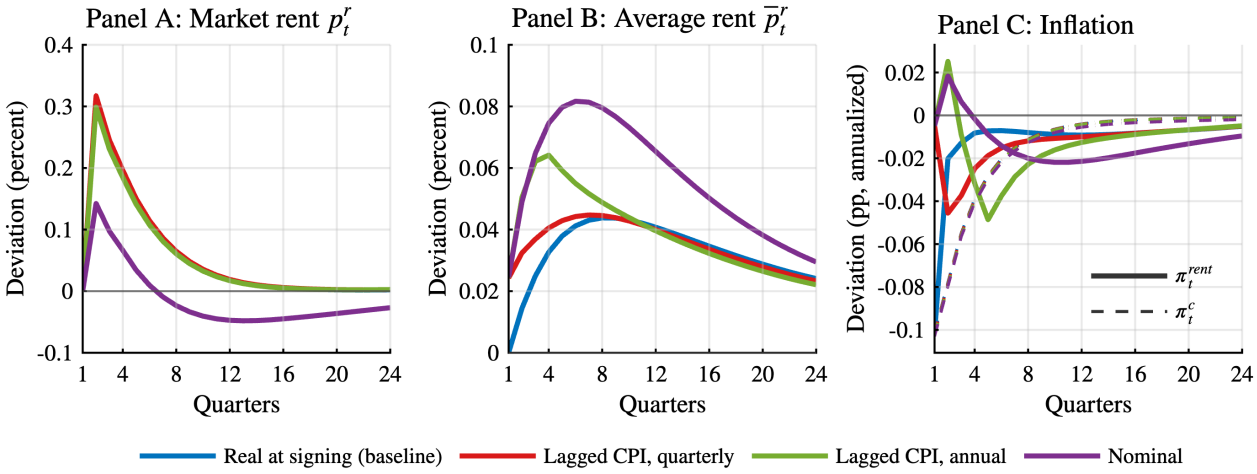


Figure F.8. Rents under Alternative Lease Indexations

Note. Responses to the monetary policy shock of Section 4.1 under the benchmark rule. In the legend, CPI denotes the model's consumer-price inflation, π_t^c, which excludes rents and approximates the update provisions of Spanish leases (Appendix D.5). Panels A and B report percent deviations from steady state; Panel C reports annualized percentage points.

	Benchmark		Actual rentals				Owner-occupied housing			
	(no rents)		Spain's weight		Model's share		Rental equivalence		Net acquisitions	
	Q1–8	Q1–20	Q1–8	Q1–20	Q1–8	Q1–20	Q1–8	Q1–20	Q1–8	Q1–20
Real rents fixed at signing (baseline)										
Real interest rate	+	+	+1.3	+1.3	+2.3	+2.3	+8.3	+8.4	−0.2	+0.6
Real house price	−	−	−1.8	−1.4	−3.1	−2.4	−11.5	−8.8	−8.6	−5.5
Real average rent	+	+	+1.9	+1.8	+3.4	+3.1	+12.5	+11.5	+14.4	+15.5
Consumption	−	−	−1.6	−0.8	−2.7	−1.4	−9.9	−5.0	−9.4	−8.6
Nondurable inflation	−	−	0.0	+1.5	0.0	+2.7	+0.3	+10.7	−5.6	−1.0
Indexed every quarter to the previous quarter's CPI inflation										
Real interest rate	+	+	+1.5	+1.5	+2.6	+2.5	+9.2	+8.9	+0.2	+0.8
Real house price	−	−	−1.4	−1.1	−2.4	−1.9	−8.4	−6.7	−7.8	−5.1
Real average rent	+	+	+0.8	+0.6	+1.4	+1.1	+4.7	+3.7	+11.1	+12.8
Consumption	−	−	−1.1	−0.3	−1.8	−0.5	−6.3	−1.3	−8.5	−7.7
Nondurable inflation	−	−	+0.5	+1.9	+0.8	+3.3	+3.2	+11.9	−4.8	−0.4
Indexed on each anniversary to the CPI change over the previous year										
Real interest rate	+	+	+2.0	+1.8	+3.5	+3.0	+11.4	+10.0	+1.0	+1.3
Real house price	−	−	−1.1	−0.9	−1.9	−1.6	−5.9	−4.8	−7.3	−4.6
Real average rent	+	+	−0.1	−0.5	−0.2	−0.9	−1.4	−3.9	+8.6	+9.9
Consumption	−	−	−0.6	+0.5	−1.0	+0.8	−2.7	+3.4	−7.6	−6.5
Nondurable inflation	−	−	+1.4	+2.7	+2.3	+4.6	+8.3	+15.3	−3.2	+0.8
Nominal rents fixed for the life of the lease										
Real interest rate	+	+	+3.1	+2.8	+5.3	+4.8	+19.5	+16.9	+2.6	+2.7
Real house price	−	−	−2.4	−1.0	−4.2	−1.6	−14.0	−3.8	−9.4	−4.6
Real average rent	+	+	+1.7	+0.8	+2.9	+1.2	+7.7	−0.3	+11.3	+10.5
Consumption	−	−	−2.3	−0.7	−3.8	−1.1	−12.0	+0.4	−10.5	−8.9
Nondurable inflation	−	−	+1.6	+4.4	+2.9	+7.7	+13.8	+29.5	−2.6	+3.7

Table F.6. Rent-Inclusive Targets under Nominal Lease Rigidity: Responses Relative to the Benchmark Rule

Note. Responses to the monetary policy shock of Section 4.1 under four indexations of the rents of continuing leases; a new lease pays the market rent p_t^r in every case. CPI denotes the model's consumer-price inflation, π_t^c, which excludes rents and approximates the update provisions of Spanish leases (Appendix D.5). The first two columns give the direction of the cumulative deviation from steady state under the benchmark rule, whose index excludes rents ($\omega_r = 0$), with the same indexation, over quarters 1–8 and 1–20. Each other entry is the difference between the cumulative deviation under the rule in the column and under that benchmark rule, in percent of the benchmark's absolute cumulative deviation; a negative entry means the variable is lower than under the benchmark rule. Under the benchmark rule, the cumulative real average rent over quarters 1–8 is +0.24, +0.31, +0.42 and +0.54 percent-quarters under the four indexations, in the order of the panels, and the cumulative fall in consumption 0.22, 0.22, 0.21 and 0.20 percent-quarters. Actual rentals enter the index at Spain's HICP weight, 3.1 percent, or at the model's rent share, 5.3 percent. Rental equivalence weighs owner-occupied housing by 13 percent and net acquisitions by 9 percent, with actual rentals at the model's share scaled by $1 - \omega_o$.

F.11 Patience Shock

Figure F.9 compares the responses to the patience shock under the benchmark (solid lines) and the optimized rule (dotted lines). The optimized rule removes most of the rise in the real rate after the first quarter. Rents, the house price, and the tenure shares move much less, while consumption, output, and nondurable inflation rise by more.

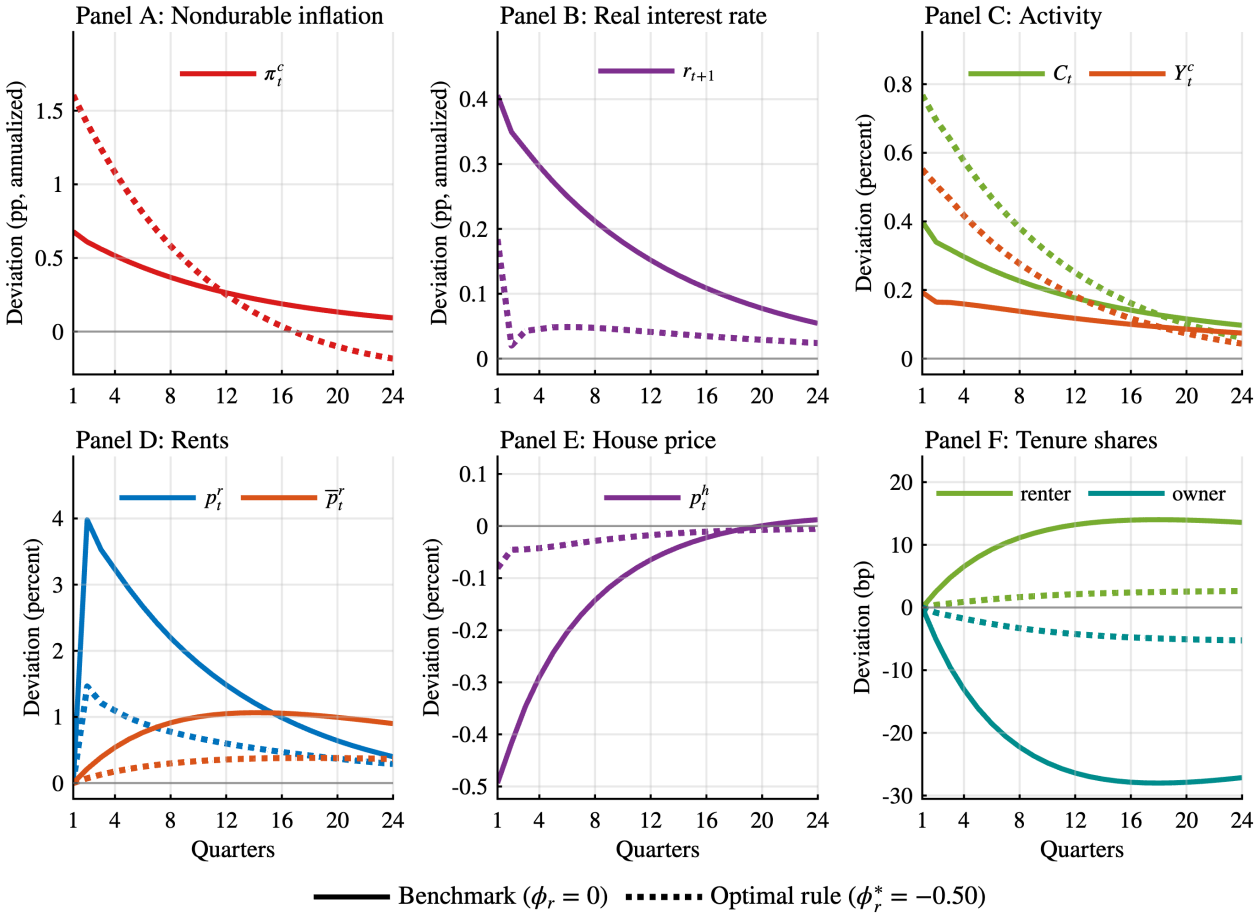


Figure F.9. Patience Shock under Alternative Policy Rules

Note. Responses under $\phi_r = 0$ (solid lines) and the optimized rule, $\phi_r = -0.50$ (dotted lines), with $\phi_\pi = 1.5$.

F.12 Government Spending Shock

Figure F.10 compares the responses to the government-spending shock. As under the patience shock, the optimized rule removes most of the rise in the real rate. Consumption stays close to its steady state, rents rise by half as much, and nondurable inflation is higher. Figure F.11 reports the distributional decomposition.

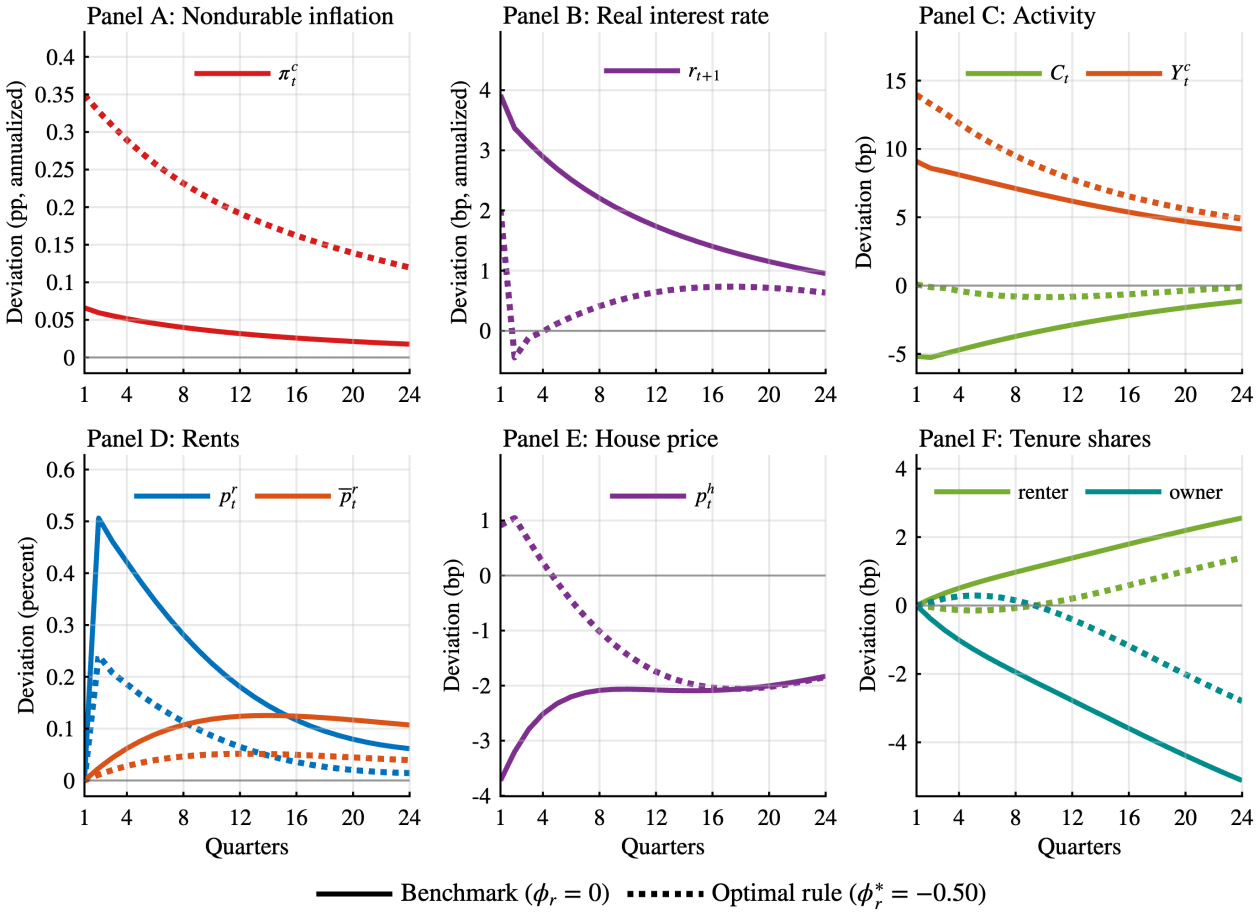


Figure F.10. Government Spending Shock under Alternative Policy Rules

Note. Responses under $\phi_r = 0$ (solid lines) and the optimized rule, $\phi_r = -0.50$ (dotted lines), with $\phi_\pi = 1.5$.

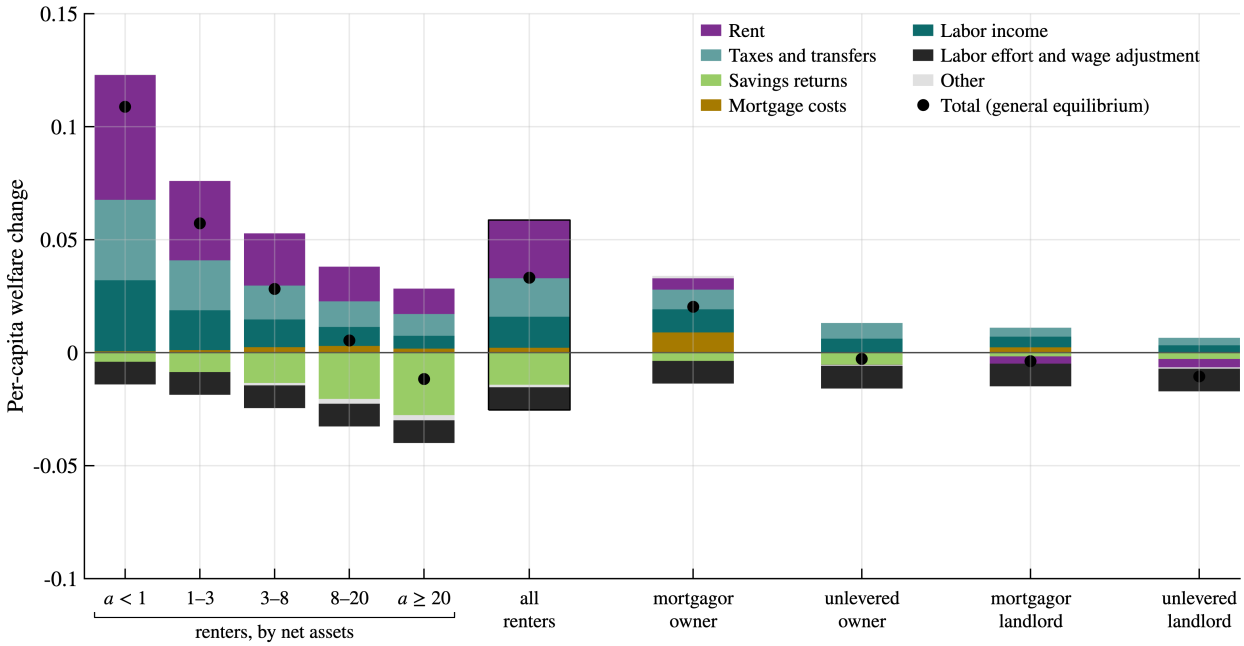


Figure F.11. Distributional Decomposition of the Welfare Gain: Government Spending Shock

Note. Per-capita welfare change from moving ϕ_r from 0 to -0.50, by date-of-shock cohort, in utility units. Cohort definitions, channels, colors, and decomposition conventions follow Figure 7.

F.13 Rental Availability Shock

Figure F.12 compares the responses to the rental supply shock. The shock raises rents with almost no movement in nondurable inflation, so the rule responds only through its rent term. The optimized rule lowers the real rate by up to five basis points; rents rise less, the house price rises more, and more households become owners. Figure F.13 reports the distributional decomposition.

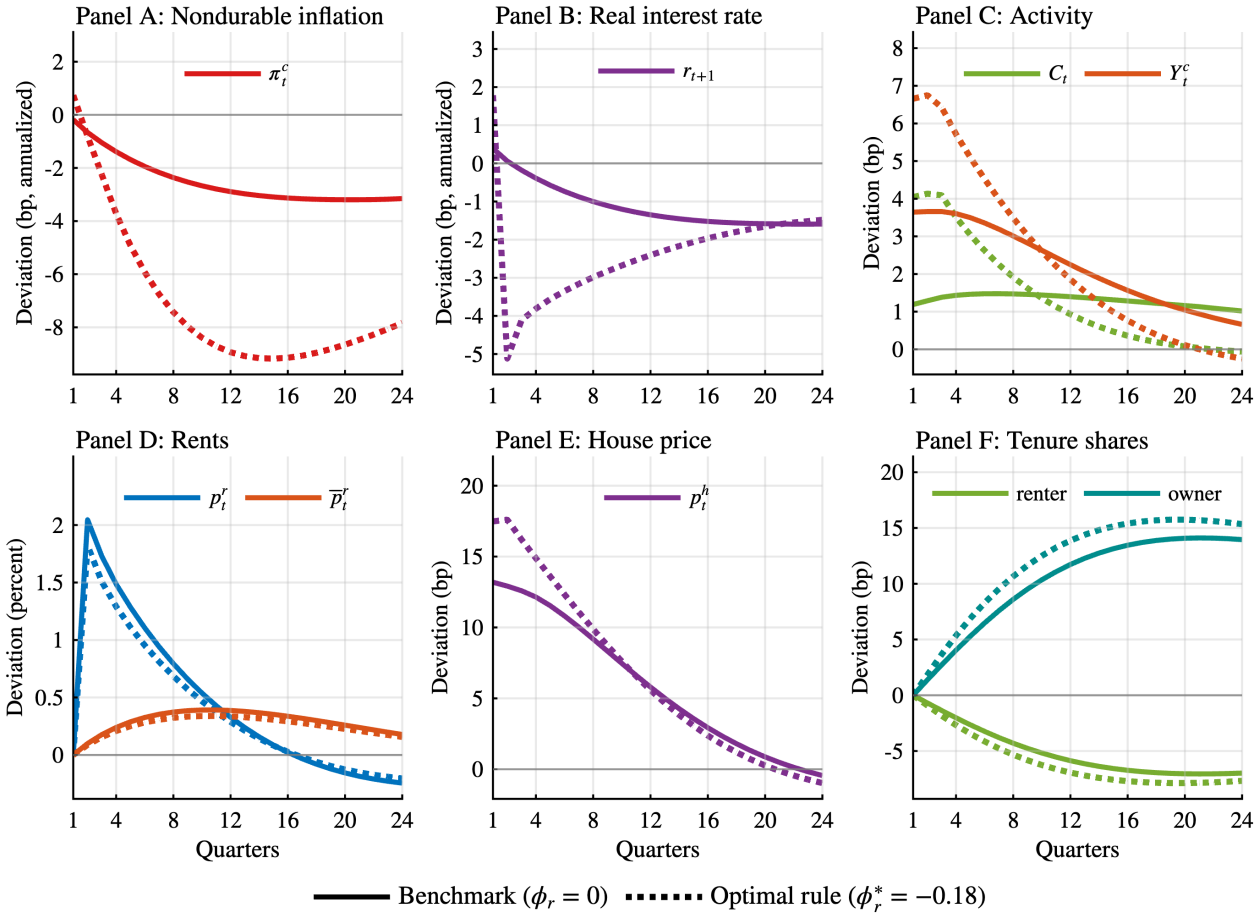


Figure F.12. Rental Supply Shock under Alternative Policy Rules

Note. Responses under $\phi_r = 0$ (solid lines) and the optimized rule, $\phi_r = -0.18$ (dotted lines), with $\phi_\pi = 1.5$.

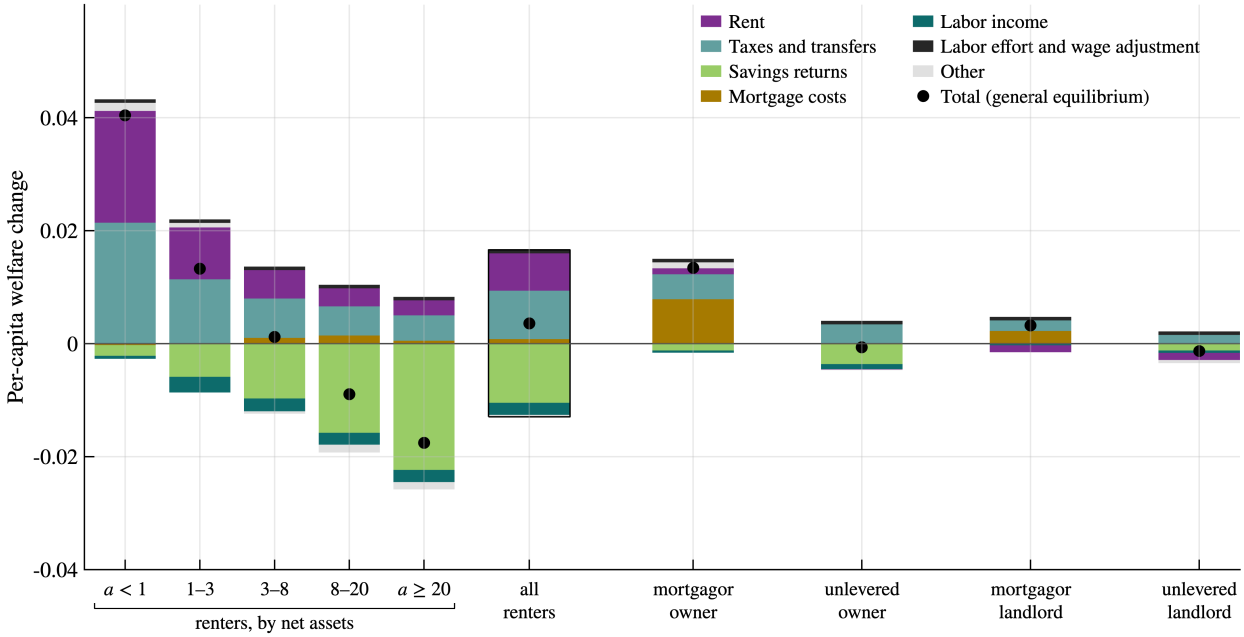


Figure F.13. Distributional Decomposition of the Welfare Gain: Rental Supply Shock

Note. Per-capita welfare change from moving ϕ_r from 0 to -0.18, by date-of-shock cohort, in utility units. Cohort definitions, channels, colors, and decomposition conventions follow Figure 7.

F.14 Cost-Push Shock

Figure F.14 compares the responses to the cost-push shock. The optimized rule halves the rise in the real rate on impact. Rents rise by half as much, the house price falls by less, and consumption falls by half as much on impact; nondurable inflation is slightly higher on impact and slightly lower afterward. Figure F.15 reports the distributional decomposition.

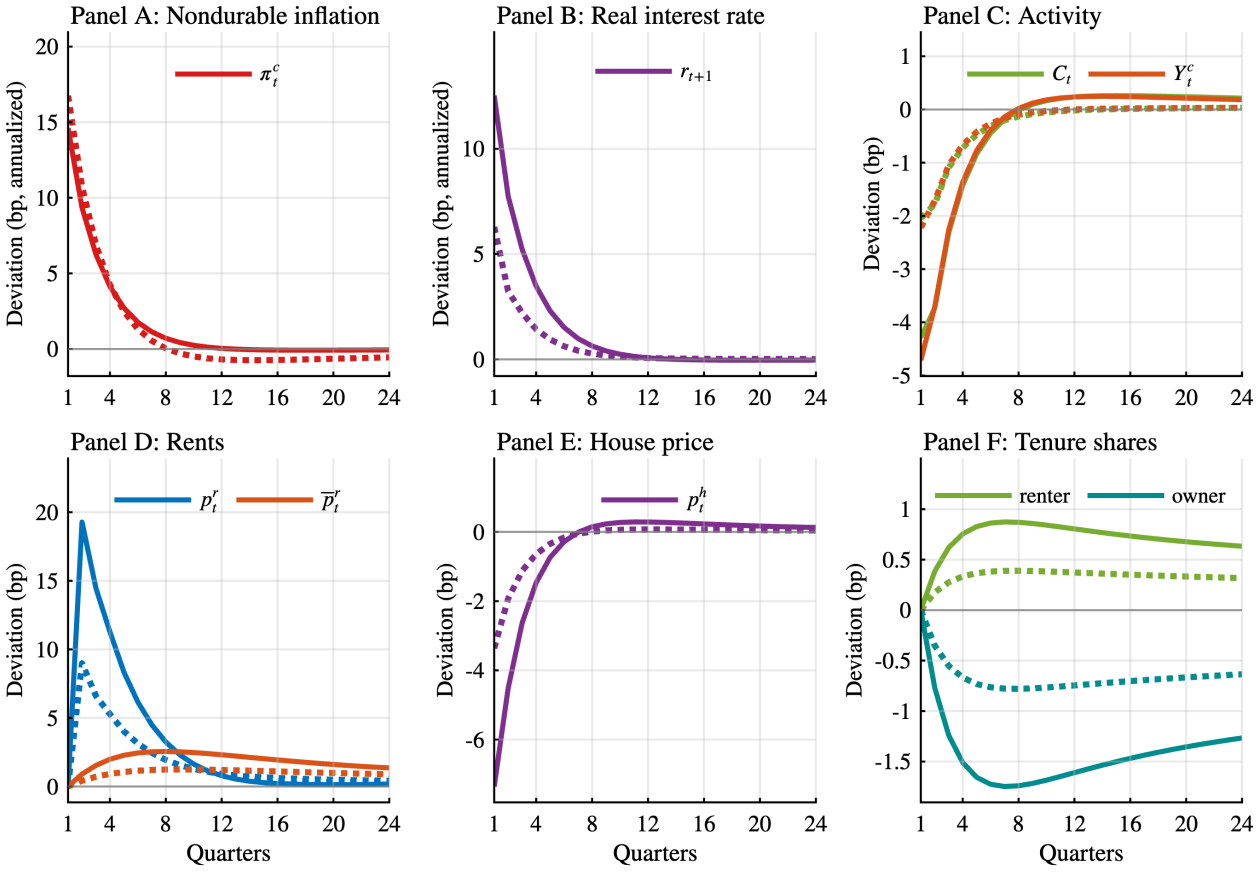


Figure F.14. Cost-Push Shock under Alternative Policy Rules

Note. Responses under $\phi_r = 0$ (solid lines) and the optimized rule, $\phi_r = -0.48$ (dotted lines), with $\phi_\pi = 1.5$.

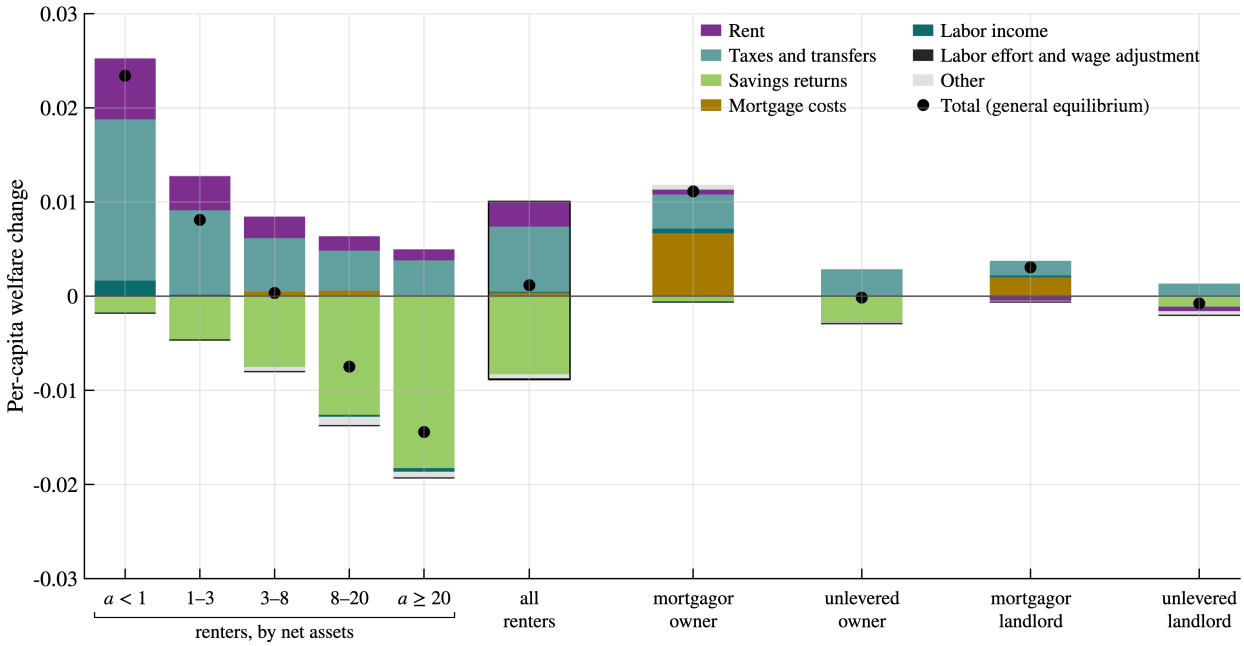


Figure F.15. Distributional Decomposition of the Welfare Gain: Cost-Push Shock

Note. Per-capita welfare change from moving ϕ_r from 0 to -0.48, by date-of-shock cohort, in utility units. Cohort definitions, channels, colors, and decomposition conventions follow Figure 7.

F.15 Alternative Rules and Fiscal Closures

Table F.8 reports the welfare gain of rules that vary ϕ_π and ϕ_r jointly. Each row holds the effective coefficient on nondurable inflation in (30), $\phi_\pi + \phi_r$, fixed, so that along a row only the response to real rent growth changes. Table F.9 reports the gains of the optimized rules by tenure group, in consumption equivalents.

Table F.7 reports the gain from moving ϕ_r from 0 to -0.20 under alternative rules, fiscal closures and lease blocks, each against its own $\phi_r = 0$ benchmark. The gain is positive in every case. With interest-rate smoothing, the gain under the rental supply shock peaks at a smaller coefficient, near -0.13, where it is about 30 percent of the baseline gain.

	Demand		Supply	
	Patience	Government spending	Rental supply	Cost push
Baseline	0.492	0.100	0.105	0.039
Interest-rate smoothing, $\rho_i = 0.8$	0.536 (1.09)	0.090 (0.90)	0.018 (0.17)	0.032 (0.82)
Output gap in the rule, $\phi_y = 0.125$	0.528 (1.07)	0.034 (0.34)	0.055 (0.53)	0.034 (0.87)
Transfer proportional to productivity	0.395 (0.80)	0.119 (1.19)	0.056 (0.53)	0.034 (0.89)
Shorter leases, $J = 4$	0.774 (1.57)	0.119 (1.19)	0.109 (1.04)	0.041 (1.06)
Lease survival, $\varphi = 0.95$	0.728 (1.48)	0.117 (1.17)	0.115 (1.10)	0.041 (1.06)

Table F.7. Robustness of the Negative Response

Note. Welfare gain from moving ϕ_r from 0 to -0.20, with $\phi_\pi = 1.5$, after a one-standard-deviation shock, in basis points of permanent consumption, each against its own $\phi_r = 0$ benchmark. In parentheses, the ratio to the baseline gain. With smoothing, the rule is $i_t = r_{ss} + \rho_i(i_{t-1} - r_{ss}) + (1-\rho_i)(\phi_\pi \pi_t^c + \phi_r \pi_t^{\text{rent}})$; the output-gap term adds $\phi_y(Y_t^c/Y_{ss}^c - 1)$ to rule (28).

Figure F.16 decomposes the change in the transfer under the four optimized rules. The lower real rate cuts the government's debt service, debt falls, and the fiscal rule returns the saving to households as a transfer that builds up over several years. Summed over the transition, lower interest payments account for almost all of the change in the transfer; tax and spread revenues move much less.

F.16 A Rule on the Real Rent Level

Section 6 studies a rule on rent inflation, the object a price index measures. Here the rule reads instead the gap of the real average rent from its steady state,

$$i_t = r_{ss} + \phi_\pi \pi_t^c + \phi_r \frac{\overline{p}_t^r - p_{ss}^r}{p_{ss}^r},$$

with $\phi_\pi = 1.5$. By (27), the real-rent component of the inflation rule sums to the rent gap: with a negative coefficient, it eases while real rents rise and tightens by the same cumulative amount as they fall back, so it shifts the timing of policy. A level term instead responds to the gap for as

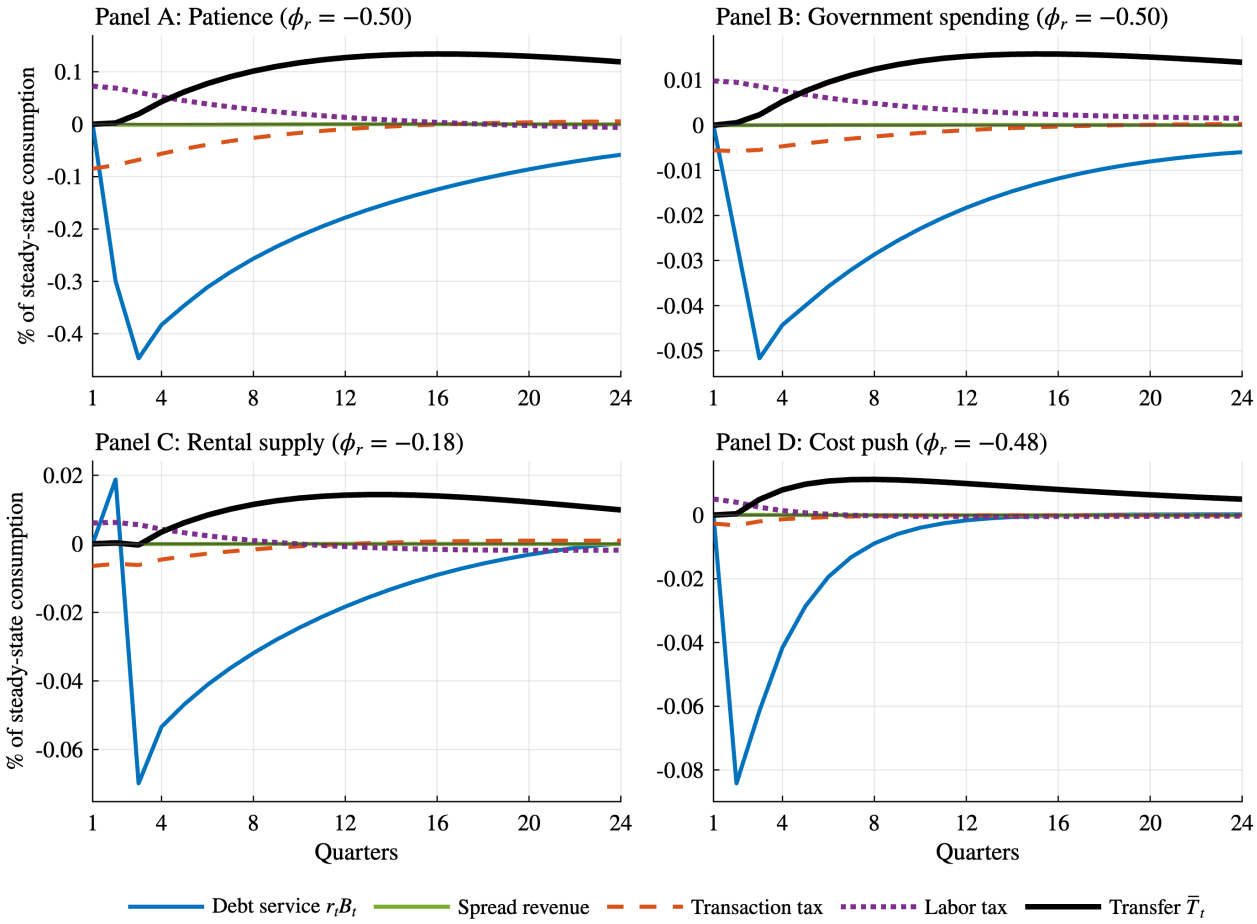


Figure F.16. Decomposition of the Fiscal Adjustment under the Optimized Rules

Note. Differences between each optimized rule and the benchmark $\phi_r = 0$ under the same shock, in percent of steady-state consumption. Higher revenues and lower debt service raise the transfer $\overline{T}_t$.

long as the gap lasts. The two coefficients are on different scales and are not directly comparable. Table F.10 reports the optimal coefficients, found on the grids of Appendix C.1 over $\phi_r \geq -2$.

Under the rental supply shock, the rent gap is created by the higher landlord spread, and a negative coefficient keeps the policy rate low for as long as rents stay high: in the fourth and fifth years, the real interest rate is lower than under the benchmark rule by about half of the remaining increase in the spread, whereas the inflation rule eases only while rents rise. Under the government-spending shock, the gain is negligible. Under the patience shock, real rents peak in quarters 12 to 16, after the expansion, so easing on the level would arrive once the boom has passed, and the optimal coefficient is slightly positive. Under the cost-push shock, welfare keeps rising up to the edge of the grid, -2.00, but flattens: -1.00 already delivers 84 percent of the gain.

$\phi_\pi + \phi_r$	$\phi_r = 0$	-0.1	-0.2	-0.3	-0.4
Panel A: Patience shock					
1.1	-0.201	0.154	0.420	0.619	0.768
1.2	0.081	0.329	0.518	0.665	**0.779**
1.3	0.112	0.325	0.492	0.626	0.733
1.4	0.067				
1.5	0.000	0.187	0.341	0.469	0.577
1.6	-0.070				
1.7	-0.138				
Panel B: Government spending shock					
1.1	0.160	0.181	0.202	0.221	**0.240**
1.2	0.093	0.118	0.140	0.161	0.180
1.3	0.050	0.077	0.100	0.121	0.141
1.4	0.021				
1.5	0.000	0.026	0.048	0.069	0.088
1.6	-0.016				
1.7	-0.029				
Panel C: Rental supply shock					
1.1	-0.295	-0.352	-0.410	-0.467	-0.522
1.2	-0.080	-0.044	-0.016	0.006	0.023
1.3	-0.023	0.047	0.105	0.153	0.194
1.4	-0.005				
1.5	0.000	0.088	0.162	0.226	**0.281**
1.6	0.001				
1.7	0.000				
Panel D: Cost-push shock					
1.1	0.069	0.071	0.072	0.074	**0.076**
1.2	0.050	0.053	0.055	0.058	0.060
1.3	0.033	0.036	0.039	0.041	0.044
1.4	0.016				
1.5	0.000	0.004	0.008	0.012	0.015
1.6	-0.015				
1.7	-0.029				

Table F.8. Rent Term or a Less Hawkish Rule?

Note. Welfare gain of the rule (ϕ_π, ϕ_r) over the benchmark $(1.5, 0)$, in basis points of permanent consumption, after a one-standard-deviation shock. Each row holds $\phi_\pi + \phi_r$ fixed; the column $\phi_r = 0$ holds the rules with no rent term. Bold: the best rule in each panel.

	Share	Demand		Supply	
		Patience $\phi_r^* = -0.50$	Government spending $\phi_r^* = -0.50$	Rental supply $\phi_r^* = -0.18$	Cost push $\phi_r^* = -0.48$
All households	1.000	1.242	0.366	0.106	0.087
Renters	0.170	9.839	2.154	0.232	0.076
Mortgagor-owners	0.066	12.724	1.770	1.171	0.971
Unlevered owners	0.594	−2.592	−0.274	−0.064	−0.014
Mortgagor-landlords	0.056	−0.493	−0.459	0.385	0.365
Unlevered landlords	0.114	−7.126	−1.355	−0.165	−0.099

Table F.9. Welfare Gains of the Optimized Rules by Tenure

Note. Consumption-equivalent gain of the optimized rule over the benchmark $\phi_r = 0$ after a one-standard-deviation shock, in basis points of permanent consumption, for all households and for cohorts fixed at the date of the shock. Share: the cohort's mass in the stationary distribution. The all-households gain weights the cohort gains by marginal utility times consumption, not by share.

	Demand		Supply	
	Patience	Government spending	Rental supply	Cost push
Rule on the real average-rent gap, $(\overline{p}_t^r - p_{ss}^r)/p_{ss}^r$				
Optimal weight, ϕ_r^*	0.03	−0.05	−0.16	−2.00
Welfare gain at ϕ_r^*	0.270	0.004	0.681	0.122
Rule on nominal average rent inflation, π_t^{rent}				
Optimal weight, ϕ_r^*	−0.50	−0.50	−0.18	−0.48
Welfare gain at ϕ_r^*	1.242	0.366	0.106	0.087

Table F.10. A Rule on the Real Rent Level

Note. Each column optimizes the rule against a one-standard-deviation shock, holding $\phi_\pi = 1.5$. Welfare gains are consumption equivalents relative to the $\phi_r = 0$ benchmark under the same shock, in basis points of permanent consumption. Under the cost-push shock, the optimum lies at the edge of the grid.